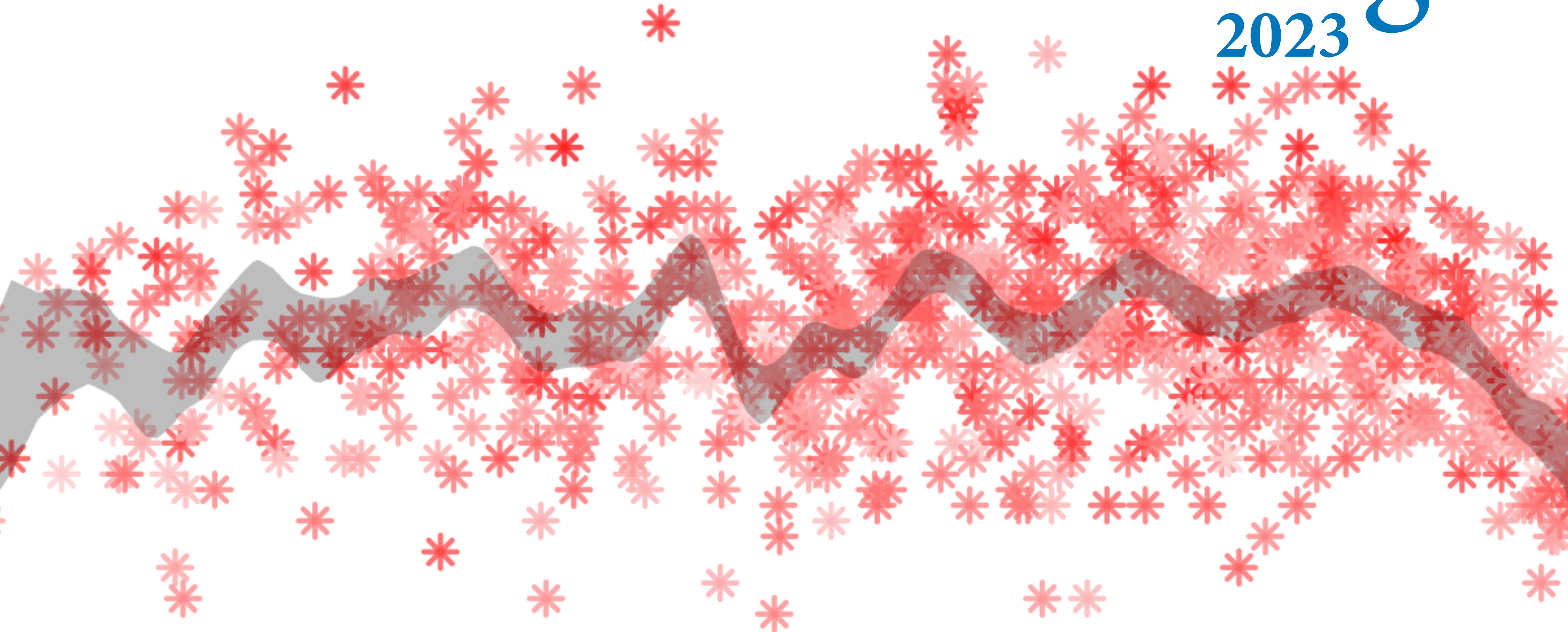


Statistical Rethinking

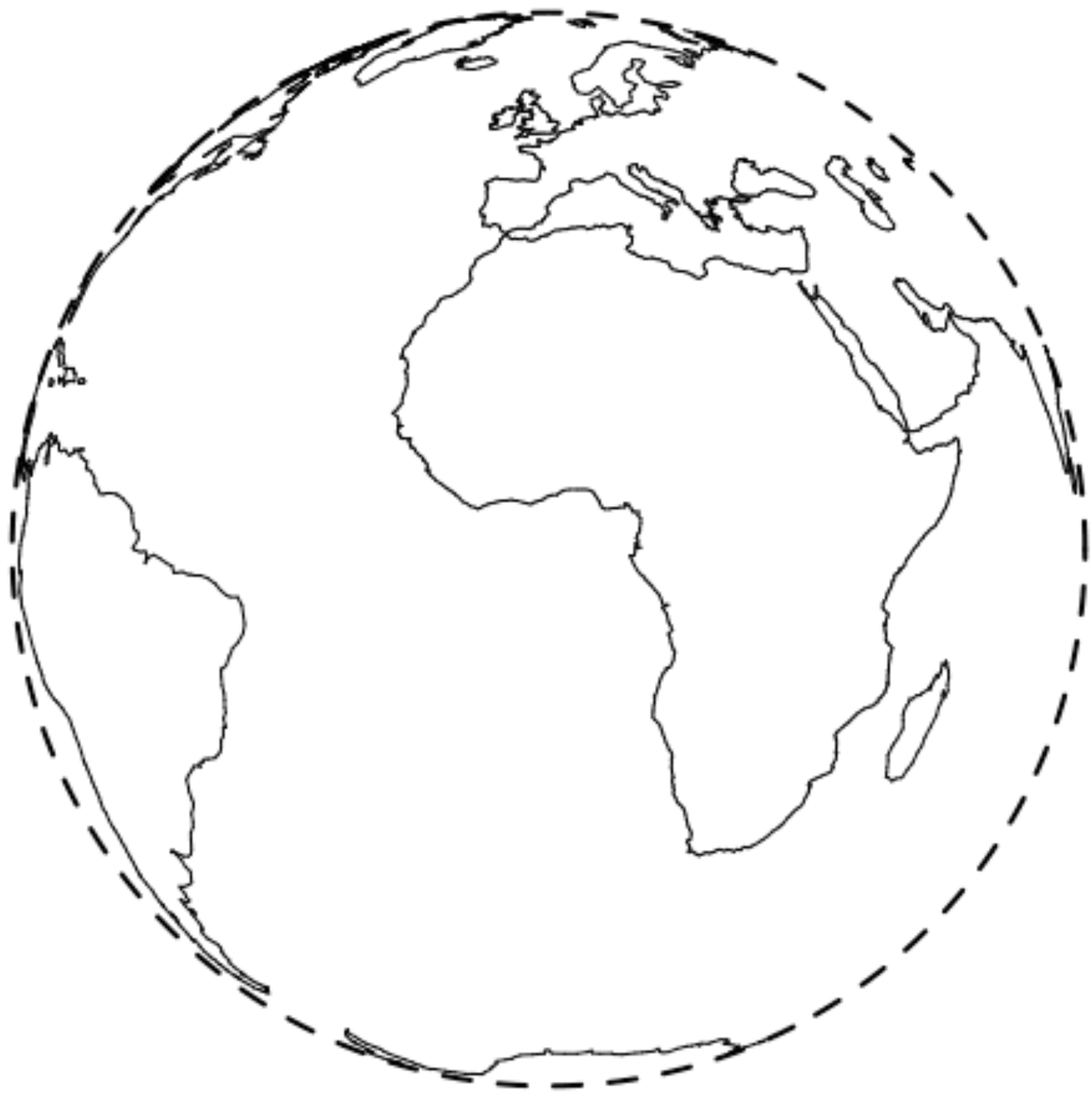
2023



2. The Garden of Forking Data



What
proportion of
the surface is
covered with
water?



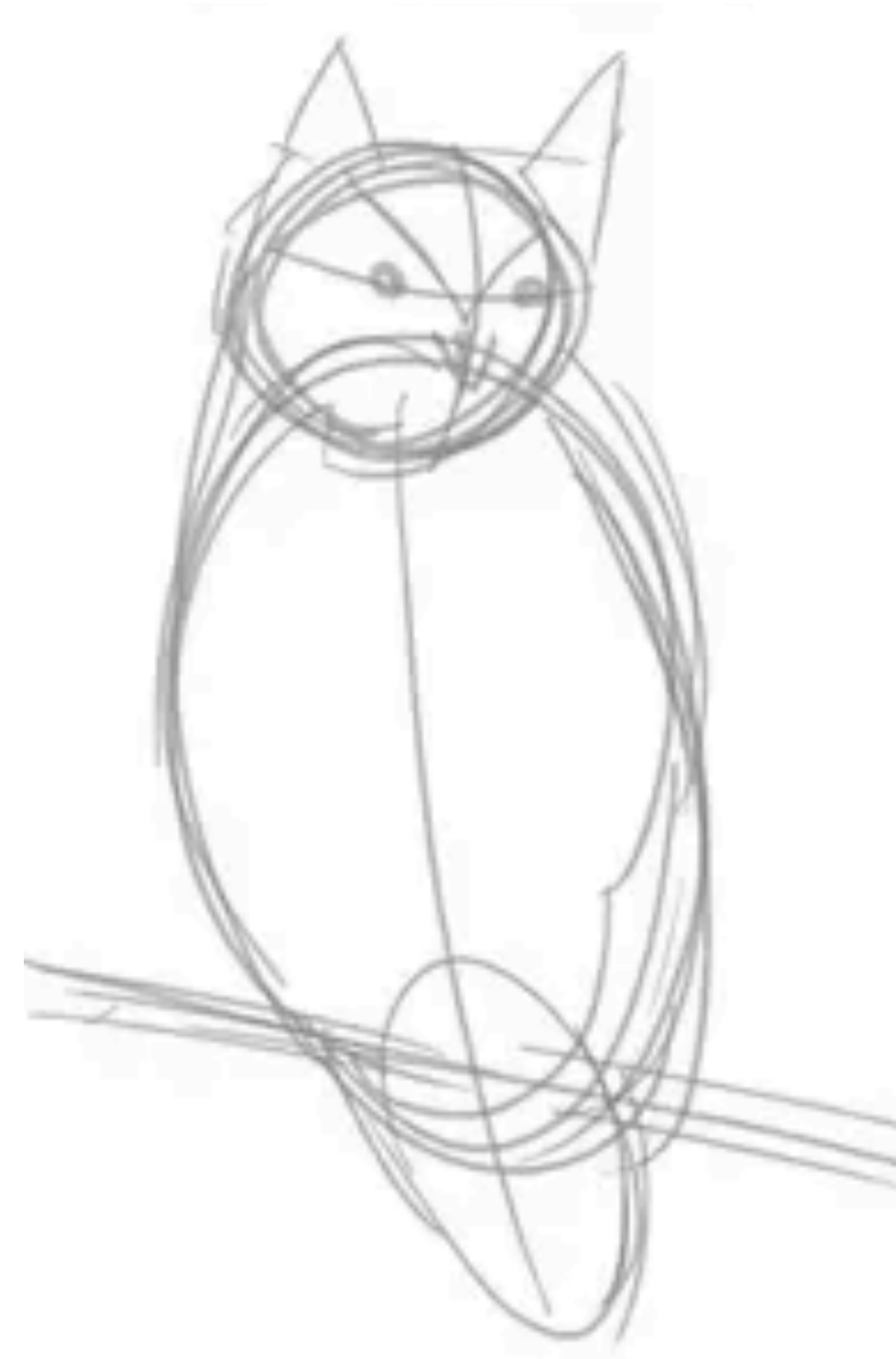
How should we use the sample?

How to produce a summary?

How to represent uncertainty?

Workflow

- (1) Define generative model of the sample
- (2) Define a specific estimand
- (3) Design a statistical way to produce estimate
- (4) Test (3) using (1)
- (5) Analyze sample, summarize



Generative model of the globe

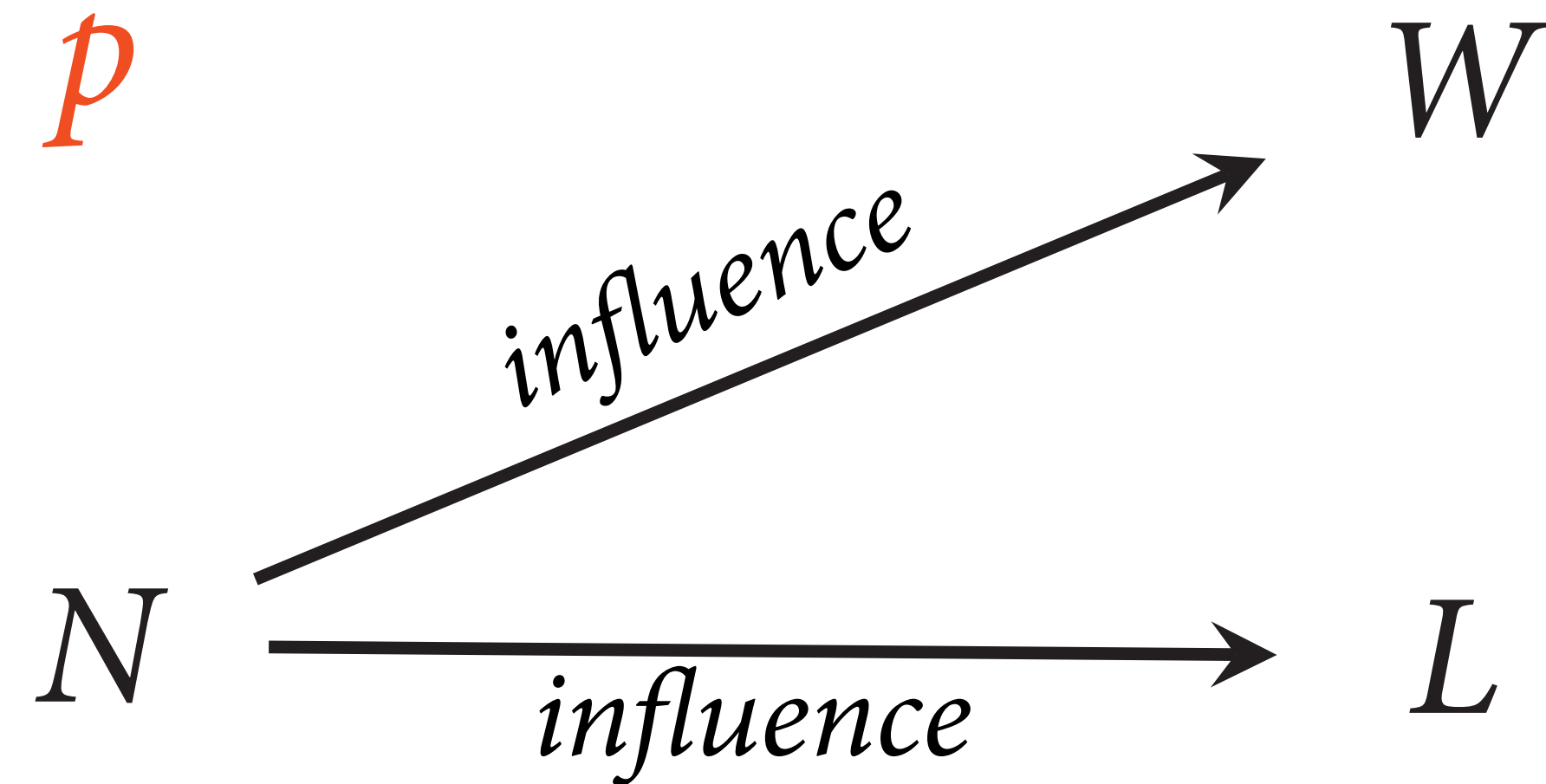
Begin conceptually: How do the variables influence one another?

proportion of water p W water observations

number of tosses N L land observations

Generative model of the globe

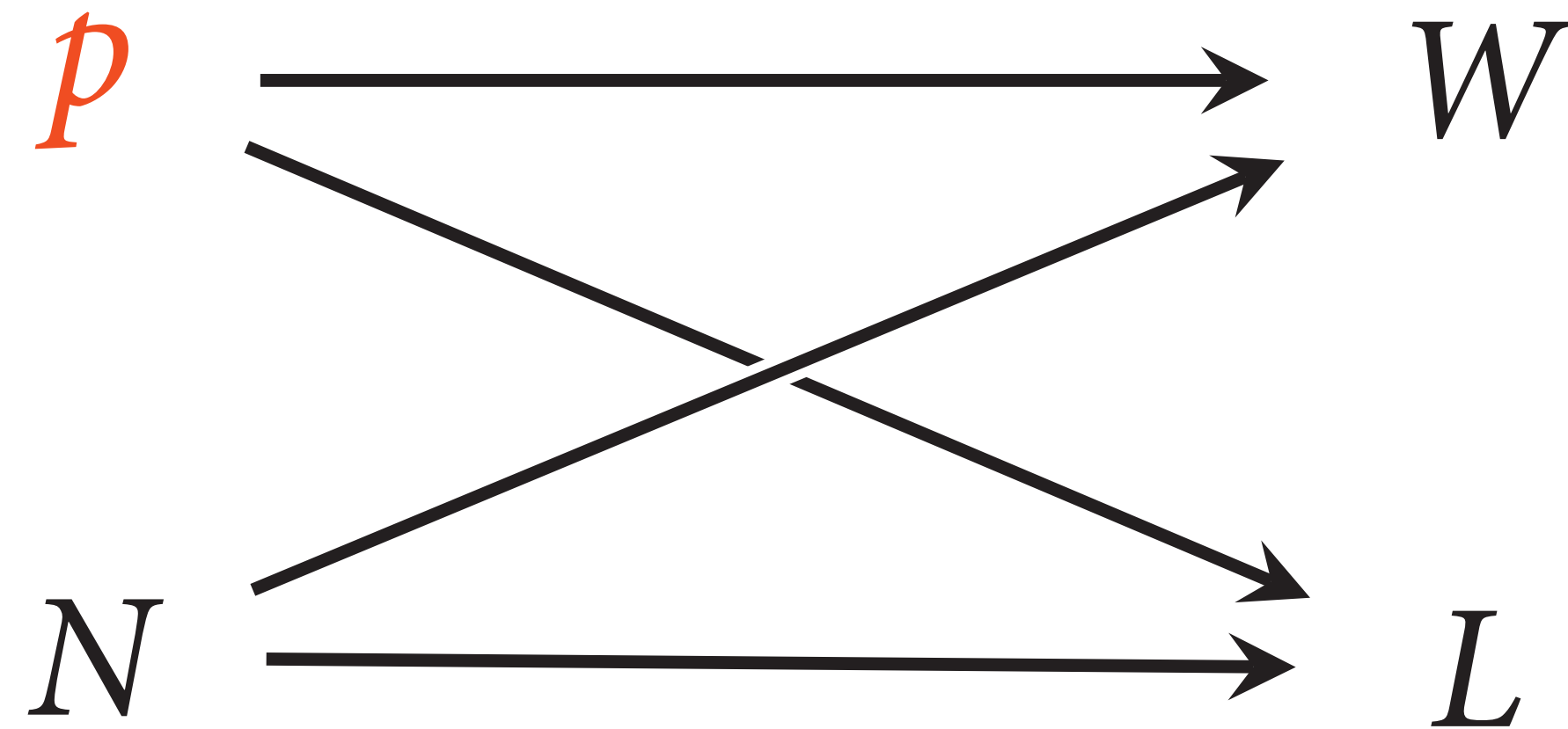
Begin conceptually: How do the variables influence one another?



N influences W and L

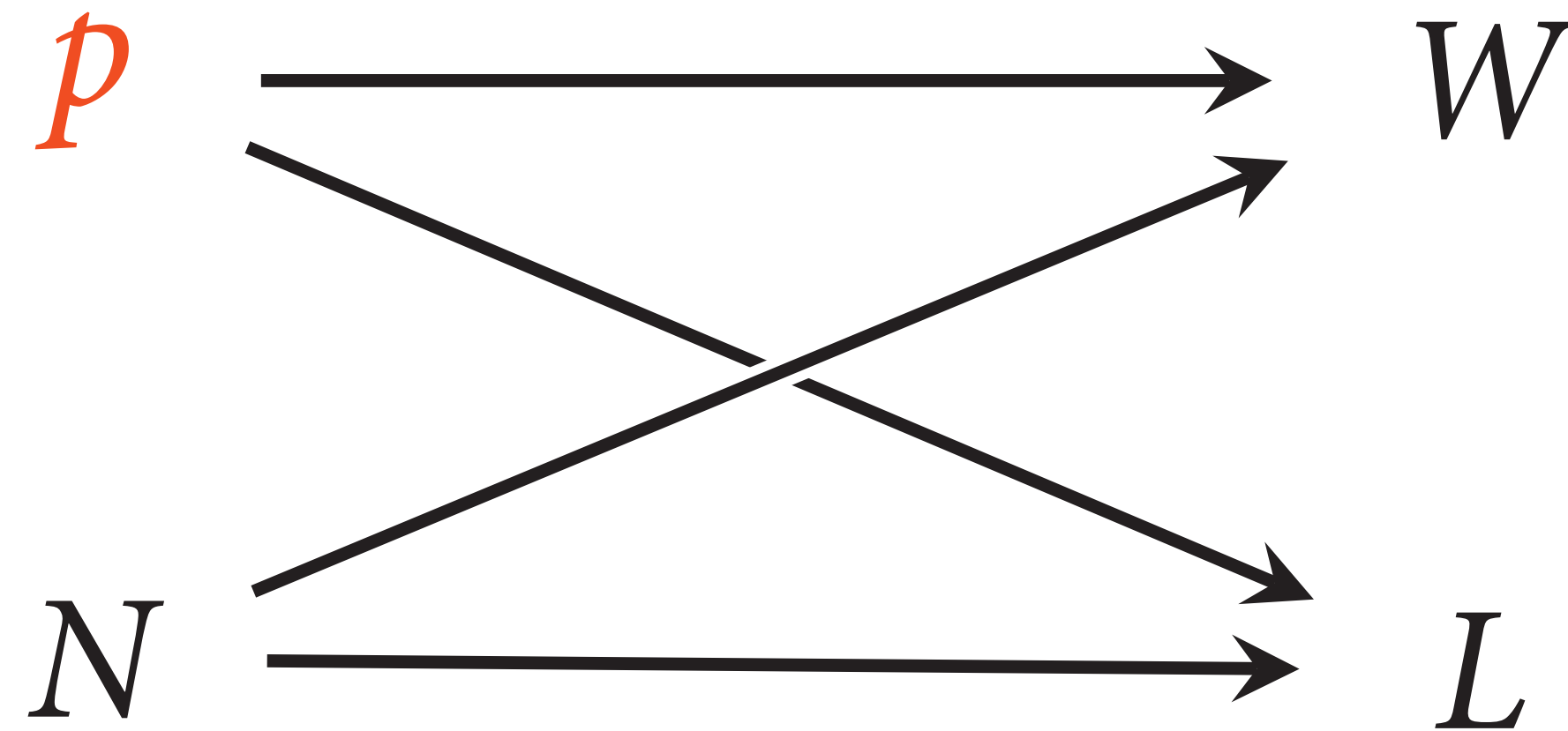
Generative model of the globe

Begin conceptually: How do the variables influence one another?



Generative model of the globe

Generative assumptions: What do the arrows mean exactly?



$$W, L = f(p, N)$$

Workflow

- (1) Define generative model of the sample
- (2) Define a specific estimand
- (3) Design a statistical way to produce estimate
- (4) Test (3) using (1)
- (5) Analyze sample, summarize

Bayesian data analysis

For each possible explanation of the sample,

Count all the ways the sample could happen.

Explanations with more ways to produce the sample are more plausible.

The Garden of Forking Data

El jardín de los datos que se bifurcan





*For each possible **proportion**
of water on the globe,*

*Count all the ways the sample
of tosses could happen.*

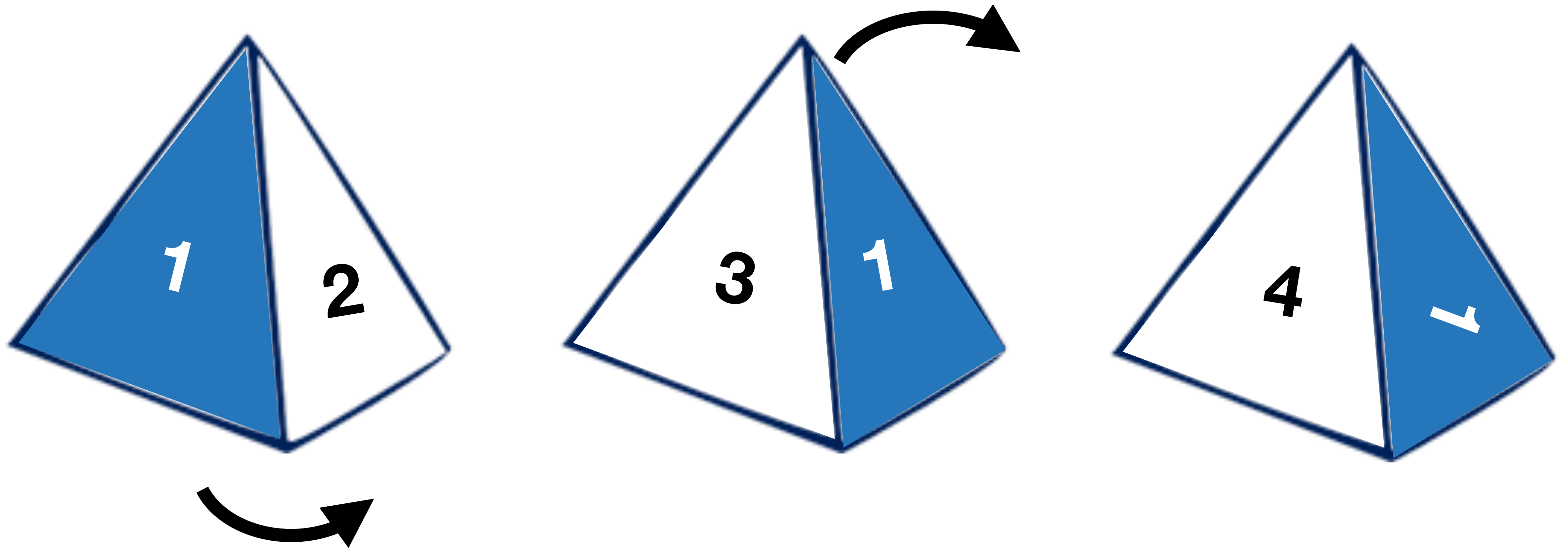
*Proportions with more ways
to produce the sample are
more plausible.*

Linking the following example to Bayes' Rule



A Four-sided Globe

covered 25% by water



Other (more) interesting proportions:

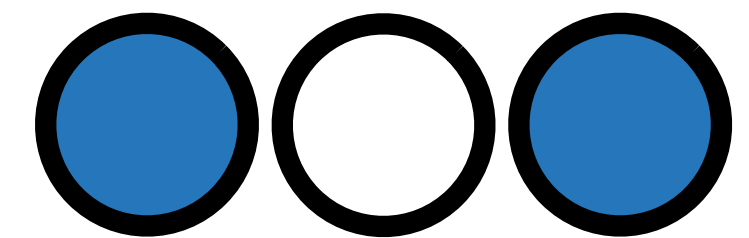
- Fraction of people that respond to an intervention.
 - Drug treatment
 - Policy change
 - Acute stress
 - Etc.
- Fraction of people that show a given preference.
 - Accept a 30/70 Ultimatum game offer
 - Wait for 1 month to get an extra 50 CHF
 - Chose a given risky option over a safe option
 - Etc.

Garden of Forking Data

Possible d4 globes:

- (1) ○ ○ ○ ○
- (2) ● ○ ○ ○
- (3) ● ● ○ ○
- (4) ● ● ● ○
- (5) ● ● ● ●

Observe:

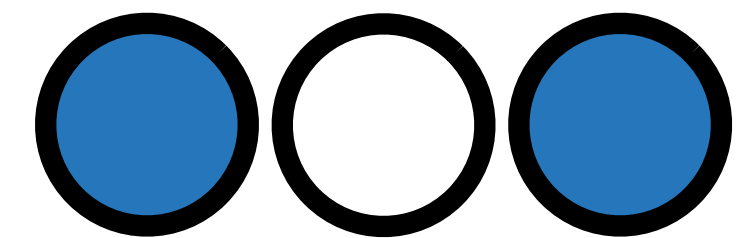


Garden of Forking Data

Possible d4 globes:

(2) ● ○ ○ ○ 25%

Observe:



Possible outcomes of the first toss given 25% water



Figure 2.2

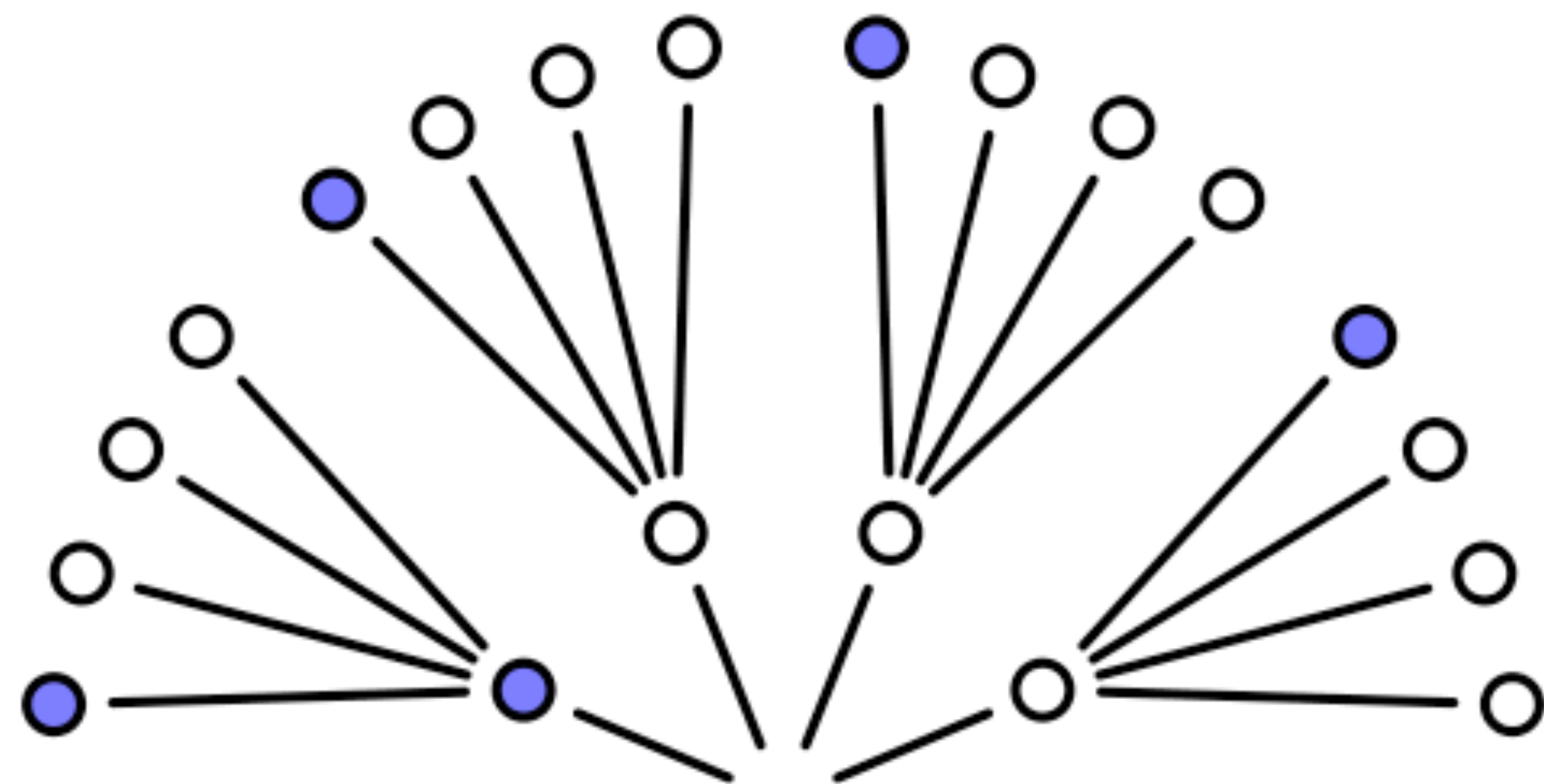


Figure 2.2

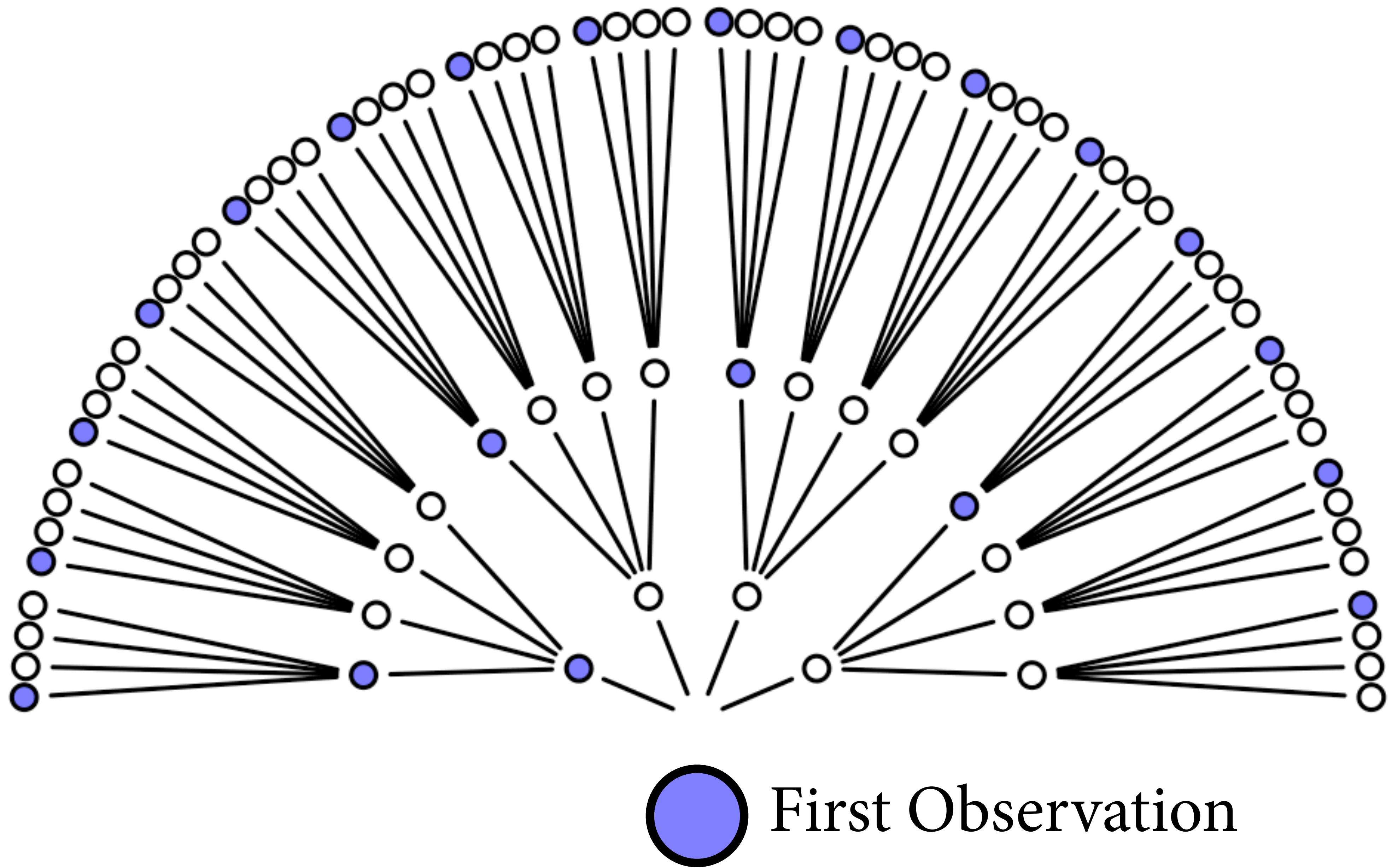


Figure 2.2

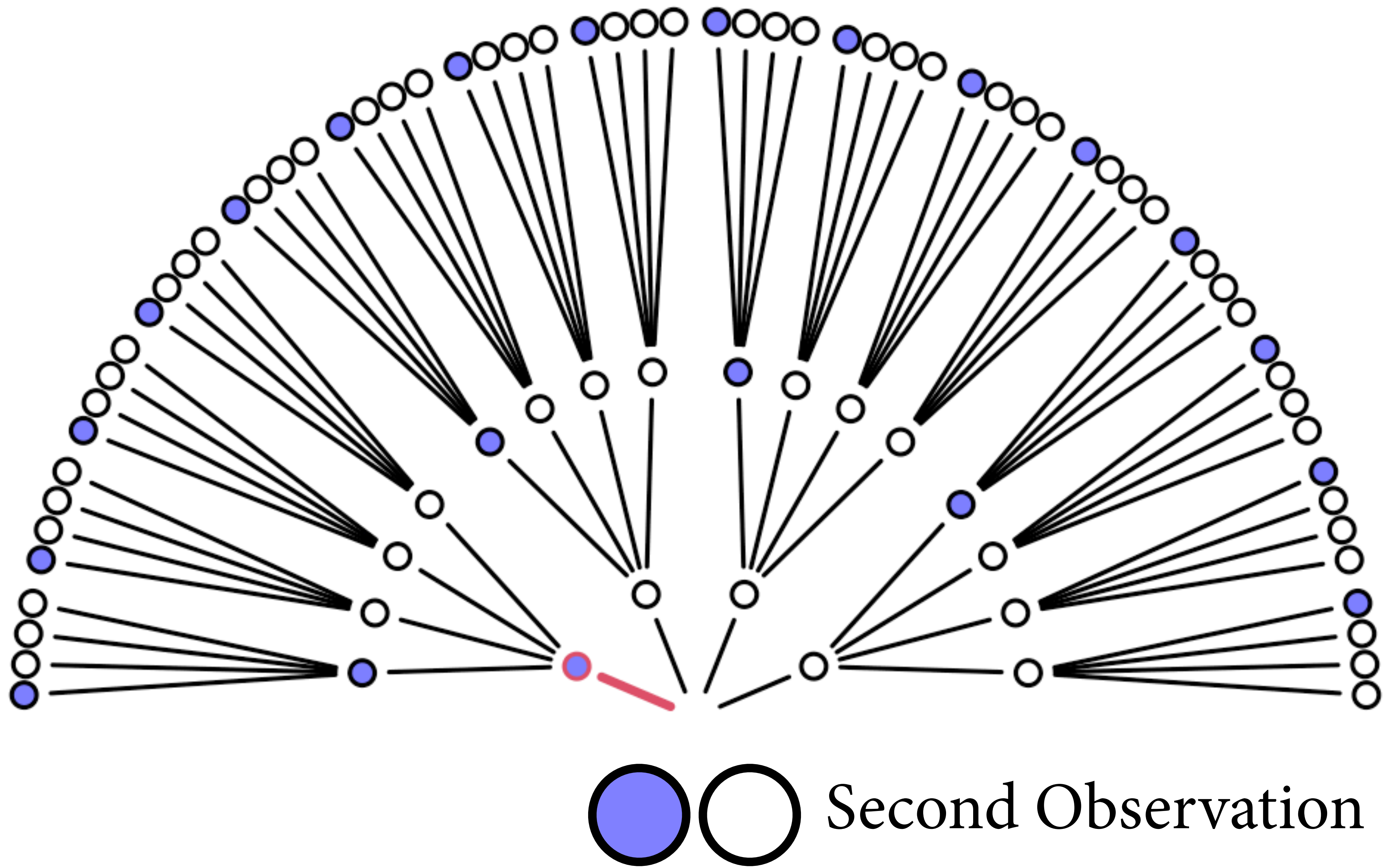


Figure 2.2

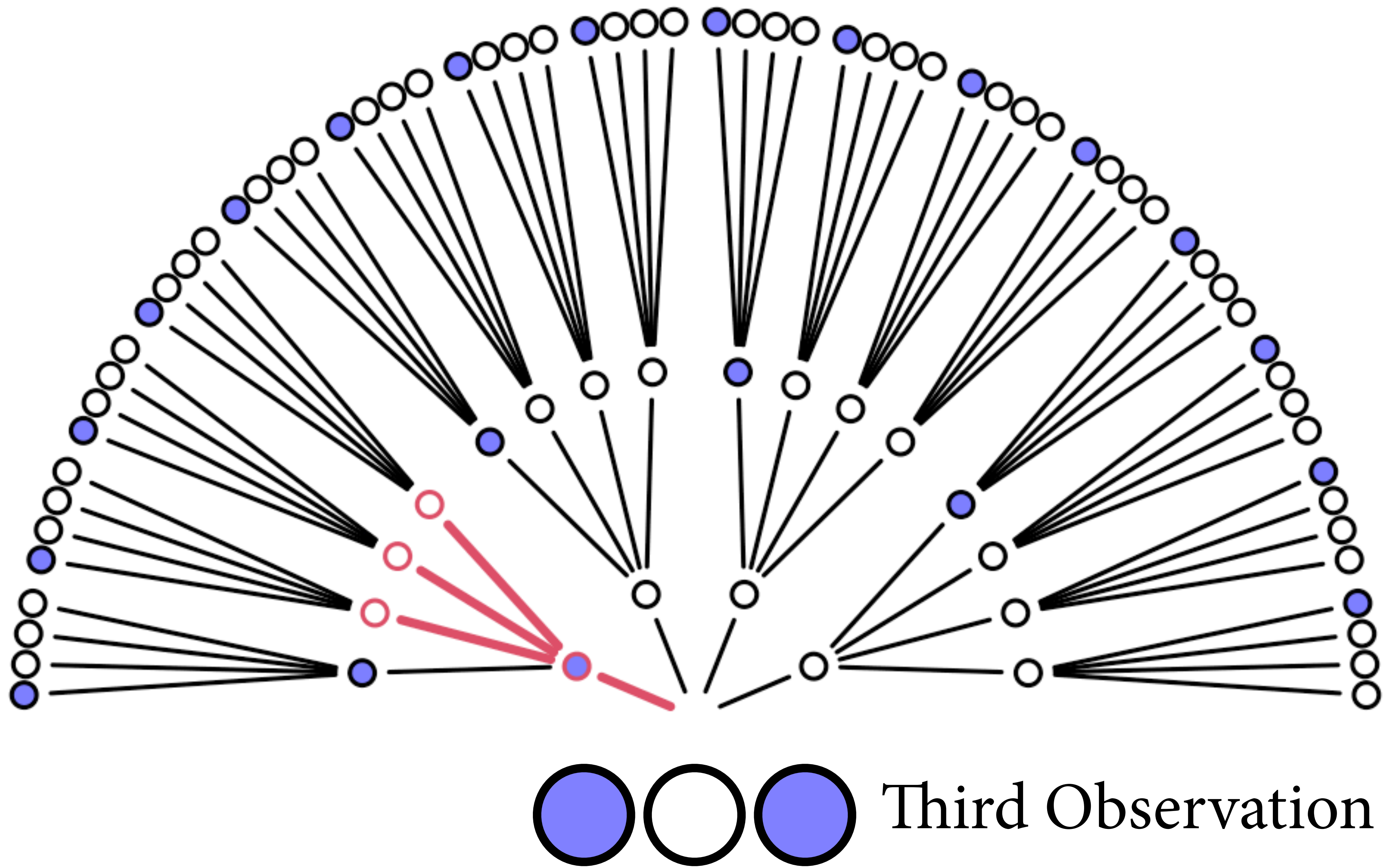
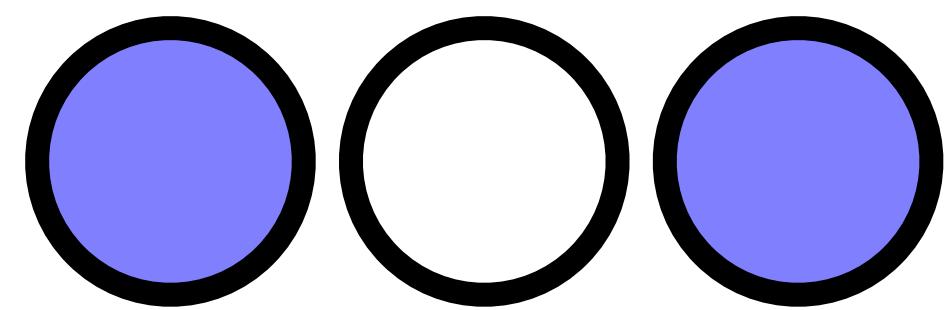


Figure 2.2

3 Ways to see



for 25% water

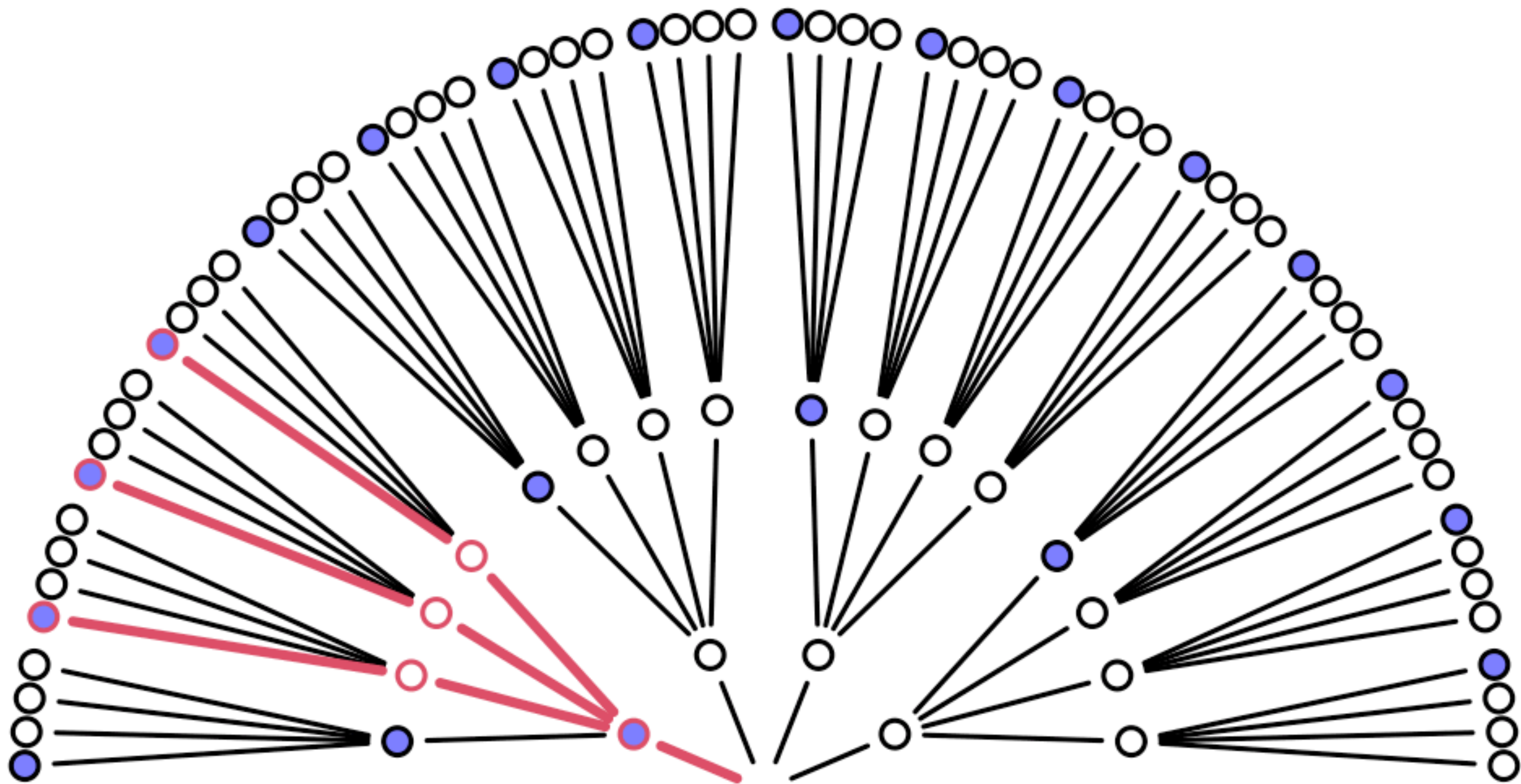
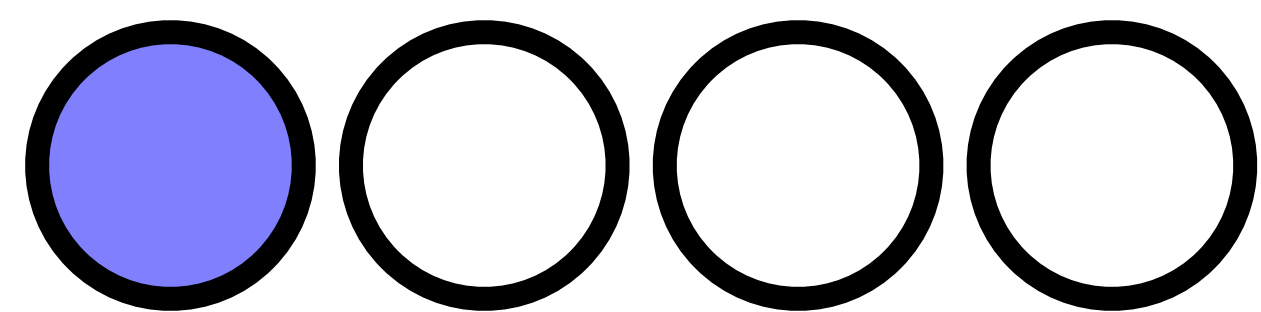
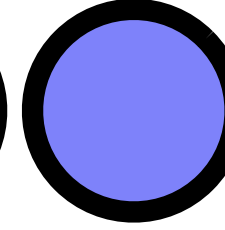
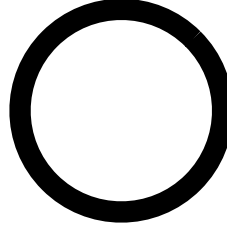
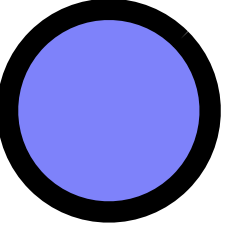
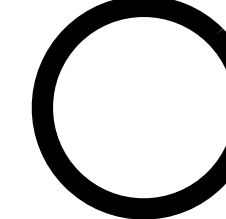
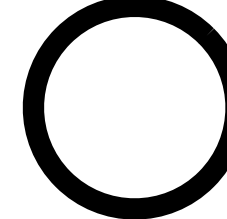
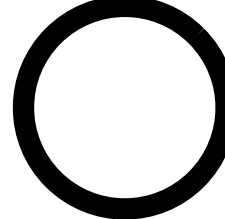
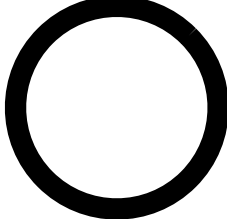


Figure 2.2

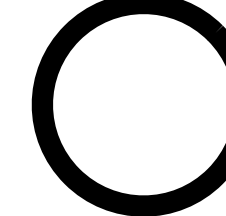
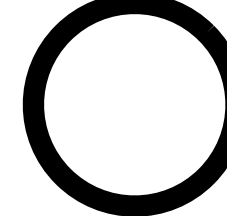
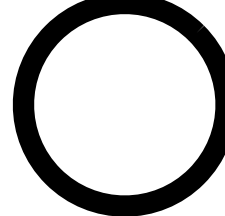
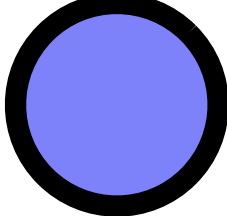
Garden of Forking Data

Possible globes:

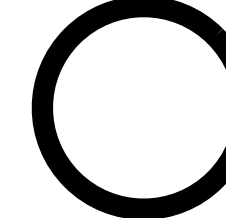
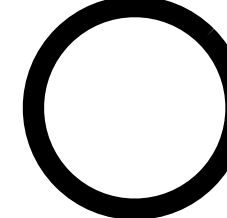
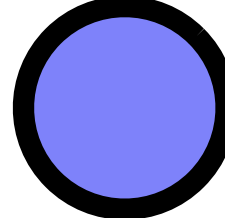
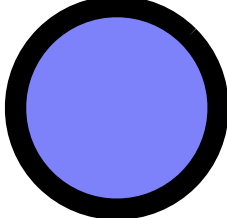
Ways to produce 

(1) 

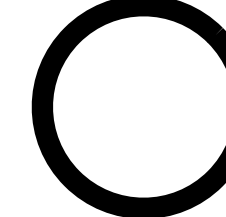
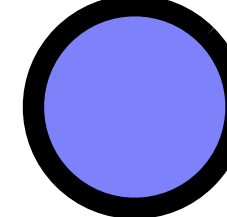
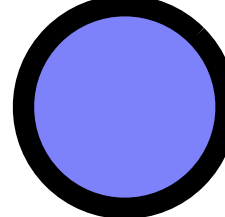
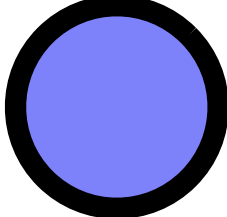
?

(2) 

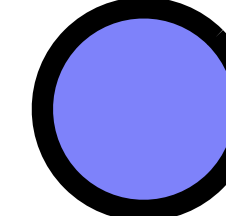
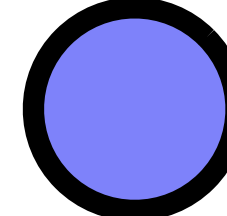
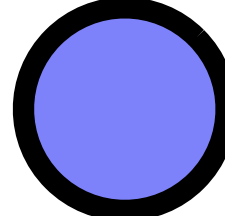
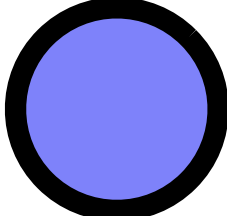
3

(3) 

?

(4) 

?

(5) 

?

Garden of Forking Data

Possible globes:

- (1) ○ ○ ○ ○
- (2) ● ○ ○ ○
- (3) ● ● ○ ○
- (4) ● ● ● ○
- (5) ● ● ● ●

Ways to produce ●○●

0

3

?

?

?

Garden of Forking Data

Possible globes:

- (1) ○ ○ ○ ○
- (2) ● ○ ○ ○
- (3) ● ● ○ ○
- (4) ● ● ● ○
- (5) ● ● ● ●

Ways to produce ●○●

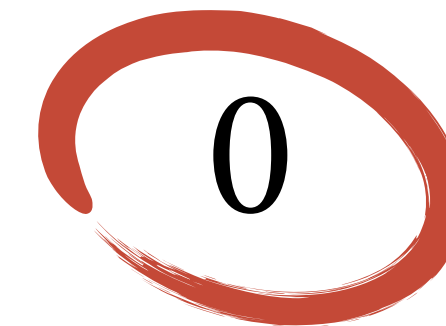
0

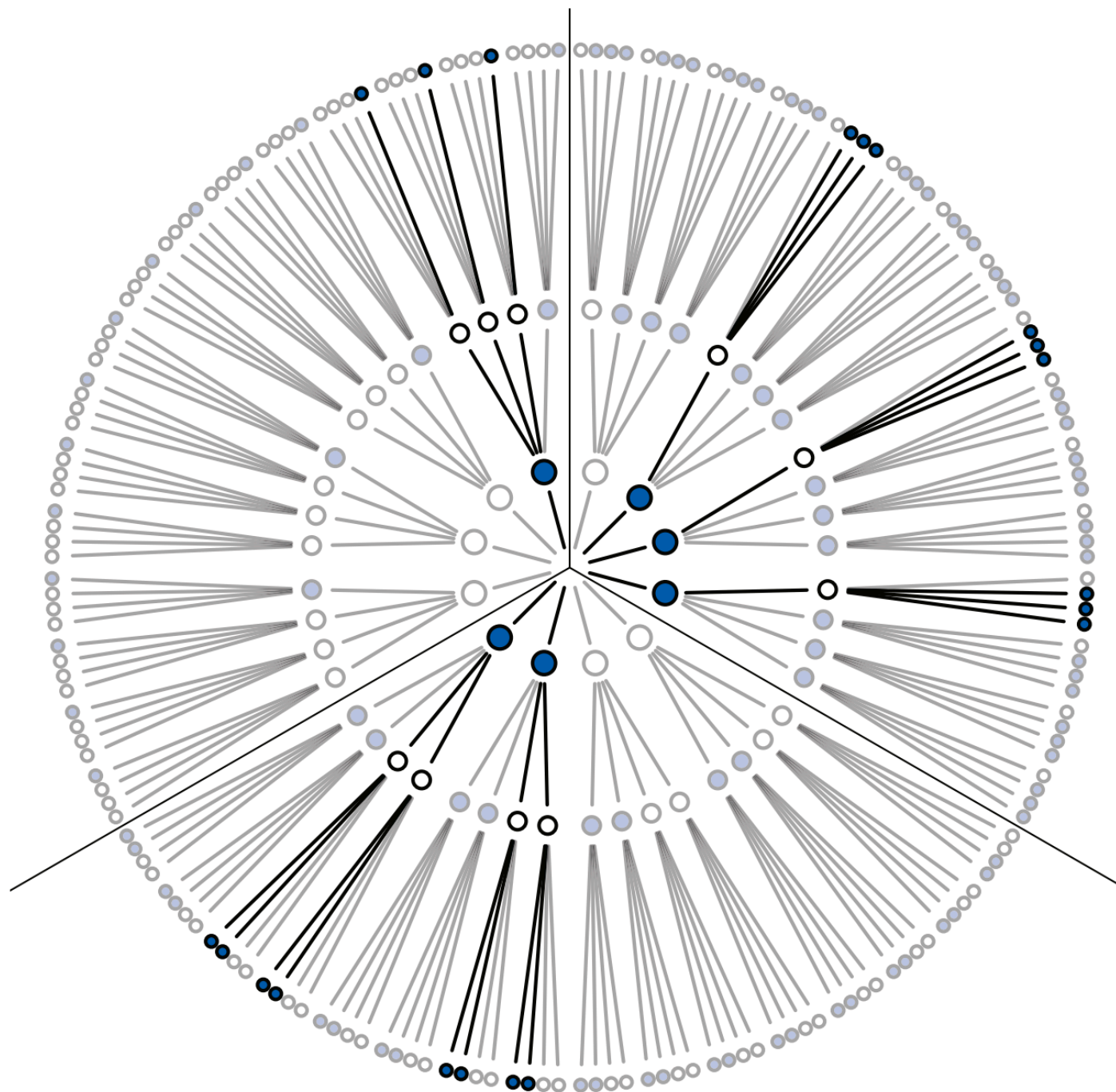
3

?

?

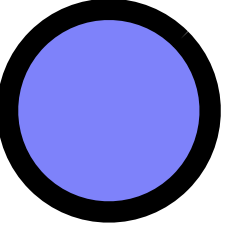
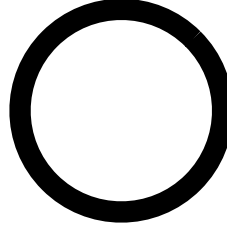
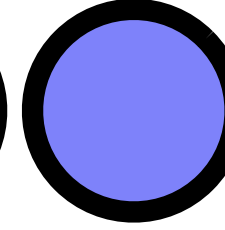
0

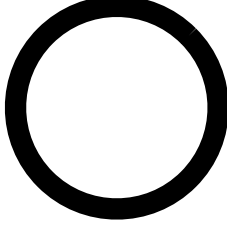
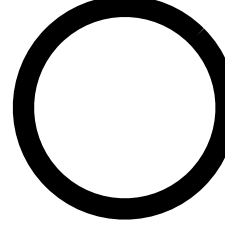
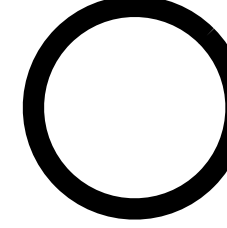
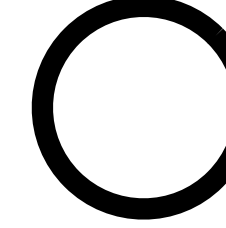
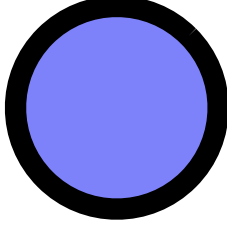
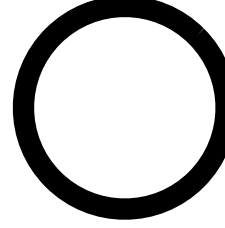
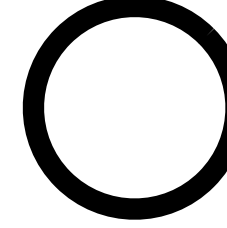
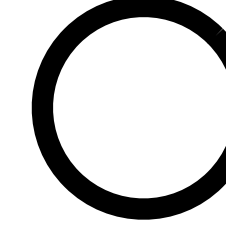
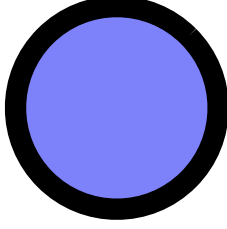
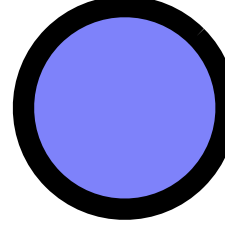
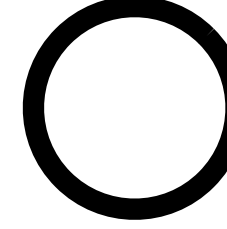
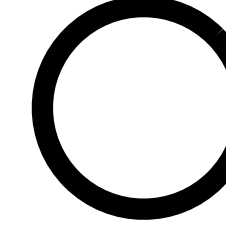
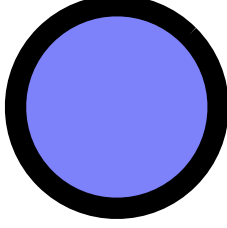
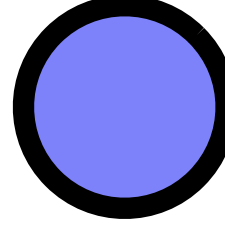
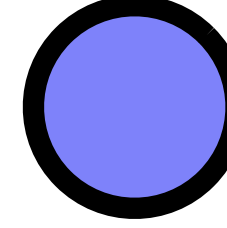
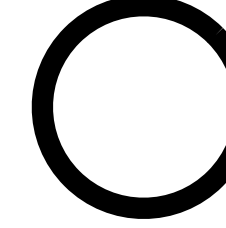
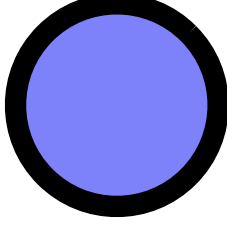
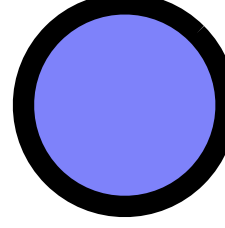
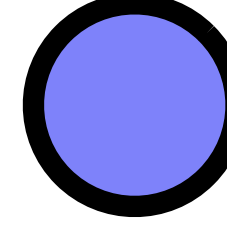
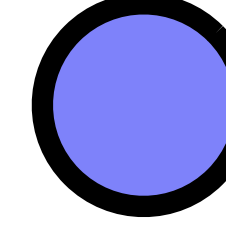




Garden of Forking Data

Possible globes:

Ways to produce   

- (1)    
- (2)    
- (3)    
- (4)    
- (5)    

0

3

8

9

0

Counts to plausibility

Unglamorous basis of applied probability:

Things that can happen more ways are more plausible.

Possibility	
[○ ○ ○ ○]	0%
[● ○ ○ ○]	25%
[● ● ○ ○]	50%
[● ● ● ○]	75%
[● ● ● ●]	100%

Counts to plausibility

Unglamorous basis of applied probability:

Things that can happen more ways are more plausible.

Possibility	Ways to produce ●○●
[○○○○]	0%
[●○○○]	25%
[●●○○]	50%
[●●●○]	75%
[●●●●]	100%

Counts to plausibility

Unglamorous basis of applied probability:

Things that can happen more ways are more plausible.

Possibility		Ways to produce ●○○●
[○○○○]	0%	$0 \times 4 \times 0 = 0$
[●○○○]	25%	$1 \times 3 \times 1 = 3$
[●●○○]	50%	
[●●●○]	75%	
[●●●●]	100%	

Counts to plausibility

Unglamorous basis of applied probability:

Things that can happen more ways are more plausible.

Possibility		Ways to produce ●○○●
[○○○○○]	0%	$0 \times 4 \times 0 = 0$
[●○○○○]	25%	$1 \times 3 \times 1 = 3$
[●●○○○]	50%	$2 \times 2 \times 2 = 8$
[●●●○○]	75%	
[●●●●○]	100%	

Counts to plausibility

Unglamorous basis of applied probability:

Things that can happen more ways are more plausible.

Possibility		Ways to produce ●○○●
[○○○○○]	0%	$0 \times 4 \times 0 = 0$
[●○○○○]	25%	$1 \times 3 \times 1 = 3$
[●●○○○]	50%	$2 \times 2 \times 2 = 8$
[●●●○○]	75%	$3 \times 1 \times 3 = 9$
[●●●●○]	100%	$4 \times 0 \times 4 = 0$

Updating

Another draw from the bag: ●
(AKA: Toss the globe again)

Possibility

[○○○○○]

[●○○○○]

[●●○○○]

[●●●○○]

[●●●●○]

[●●●●●]

Updating

Another draw from the bag: ●

Possibility	Ways to produce ●○●
[○○○○○]	$0 \times 4 \times 0 = 0$
[●○○○○]	$1 \times 3 \times 1 = 3$
[●●○○○]	$2 \times 2 \times 2 = 8$
[●●●○○]	$3 \times 1 \times 3 = 9$
[●●●●○]	$4 \times 0 \times 4 = 0$

Updating

Another draw from the bag: ●

Possibility	Ways to produce ●○●	Ways to produce ●
[○○○○○]	$0 \times 4 \times 0 = 0$	0
[●○○○○]	$1 \times 3 \times 1 = 3$	1
[●●○○○]	$2 \times 2 \times 2 = 8$	2
[●●●○○]	$3 \times 1 \times 3 = 9$	3
[●●●●○]	$4 \times 0 \times 4 = 0$	4

Updating

Another draw from the bag: ●

Possibility	Ways to produce ●○●	Ways to produce ●	Ways to produce ●○●●
[○○○○○]	$0 \times 4 \times 0 = 0$	0	$0 \times 0 = 0$
[●○○○○]	$1 \times 3 \times 1 = 3$	1	
[●●○○○]	$2 \times 2 \times 2 = 8$	2	
[●●●○○]	$3 \times 1 \times 3 = 9$	3	
[●●●●○]	$4 \times 0 \times 4 = 0$	4	

Updating

Another draw from the bag: ●

Possibility	Ways to produce ●○●	Ways to produce ●	Ways to produce ●○●●
[○○○○○]	$0 \times 4 \times 0 = 0$	0	$0 \times 0 = 0$
[●○○○○]	$1 \times 3 \times 1 = 3$	1	$3 \times 1 = 3$
[●●○○○]	$2 \times 2 \times 2 = 8$	2	$8 \times 2 = 16$
[●●●○○]	$3 \times 1 \times 3 = 9$	3	$9 \times 3 = 27$
[●●●●○]	$4 \times 0 \times 4 = 0$	4	$0 \times 4 = 0$

The whole sample

[illegible]

The whole sample

[illegible]

The whole sample

Possibility	Observations:	●	○	●	●	●	○	●	○	●
[○○○○○]		0	0	0	0	0	0	0	0	$0 = 0^6 \times 4^3$
[●○○○○]		1	3	3	3	3	9	9	27	$27 = 1^6 \times 3^3$
[●●○○○]		2	4	8	16	32	64	128	256	$512 = 2^6 \times 2^3$
[●●●○○]		3	3	9	27	81	81	243	243	$729 = 3^6 \times 1^3$
[●●●●○]		4	0	0	0	0	0	0	0	$0 = 4^6 \times 0^3$

The whole sample

Possibility	Observations:	●	○	●	●	●	○	●	○	●
[○○○○○]		0	0	0	0	0	0	0	0	$0 = 0^6 \times 4^3$
[●○○○○]		1	3	3	3	3	9	9	27	$27 = 1^6 \times 3^3$
[●●○○○]		2	4	8	16	32	64	128	256	$512 = 2^6 \times 2^3$
[●●●○○]		3	3	9	27	81	81	243	243	$729 = 3^6 \times 1^3$
[●●●●○]		4	0	0	0	0	0	0	0	$0 = 4^6 \times 0^3$

Ways for p to produce $W, L = (4p)^W \times (4-4p)^L$

Probability

Probability: Non-negative values that sum to one

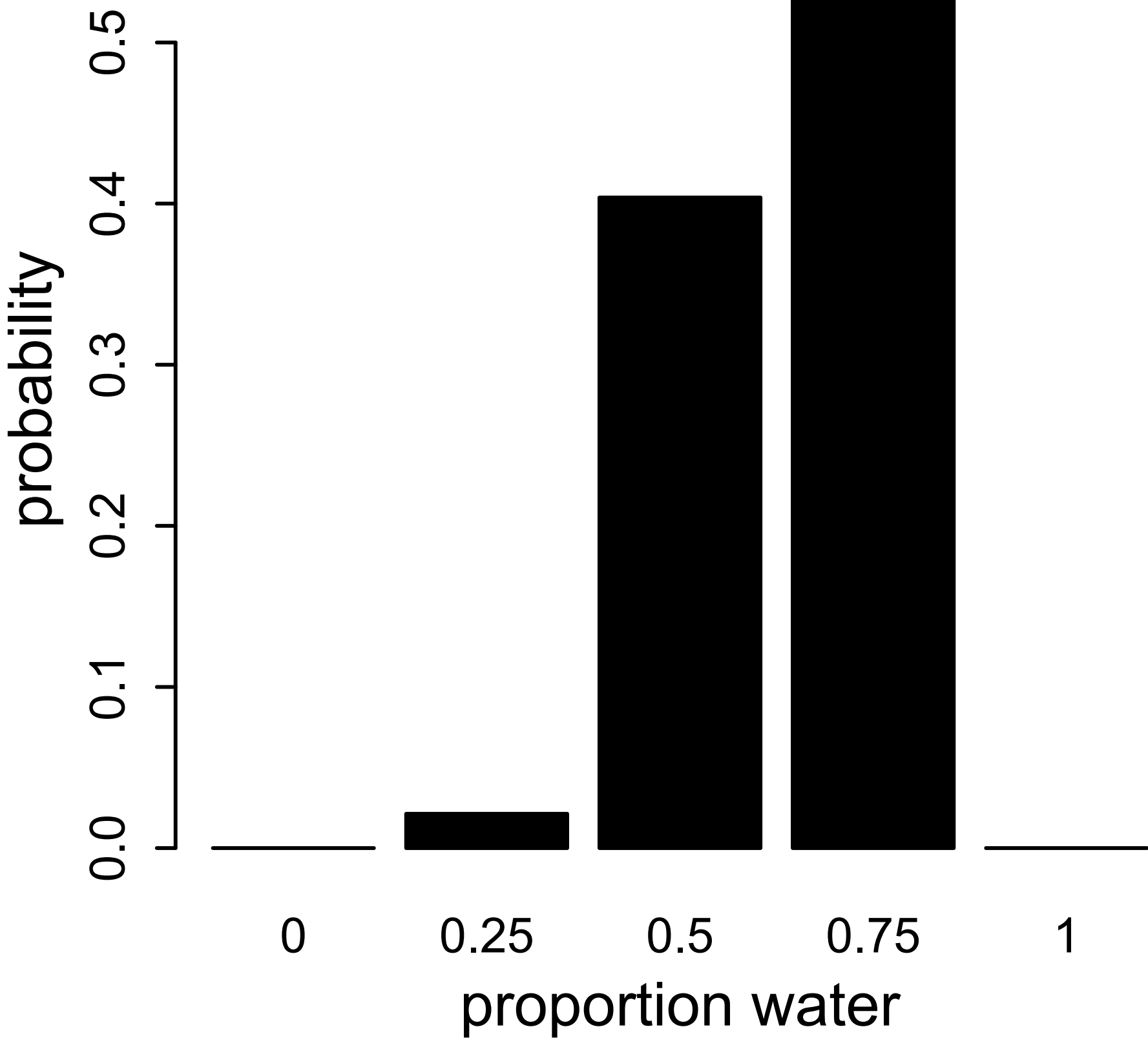
Suppose $W=20$, $L=10$. Then $p=0.5$ has

$$2^W \times 2^L = 1,073,741,824$$

ways to produce sample. Better to convert to probability.

Probability

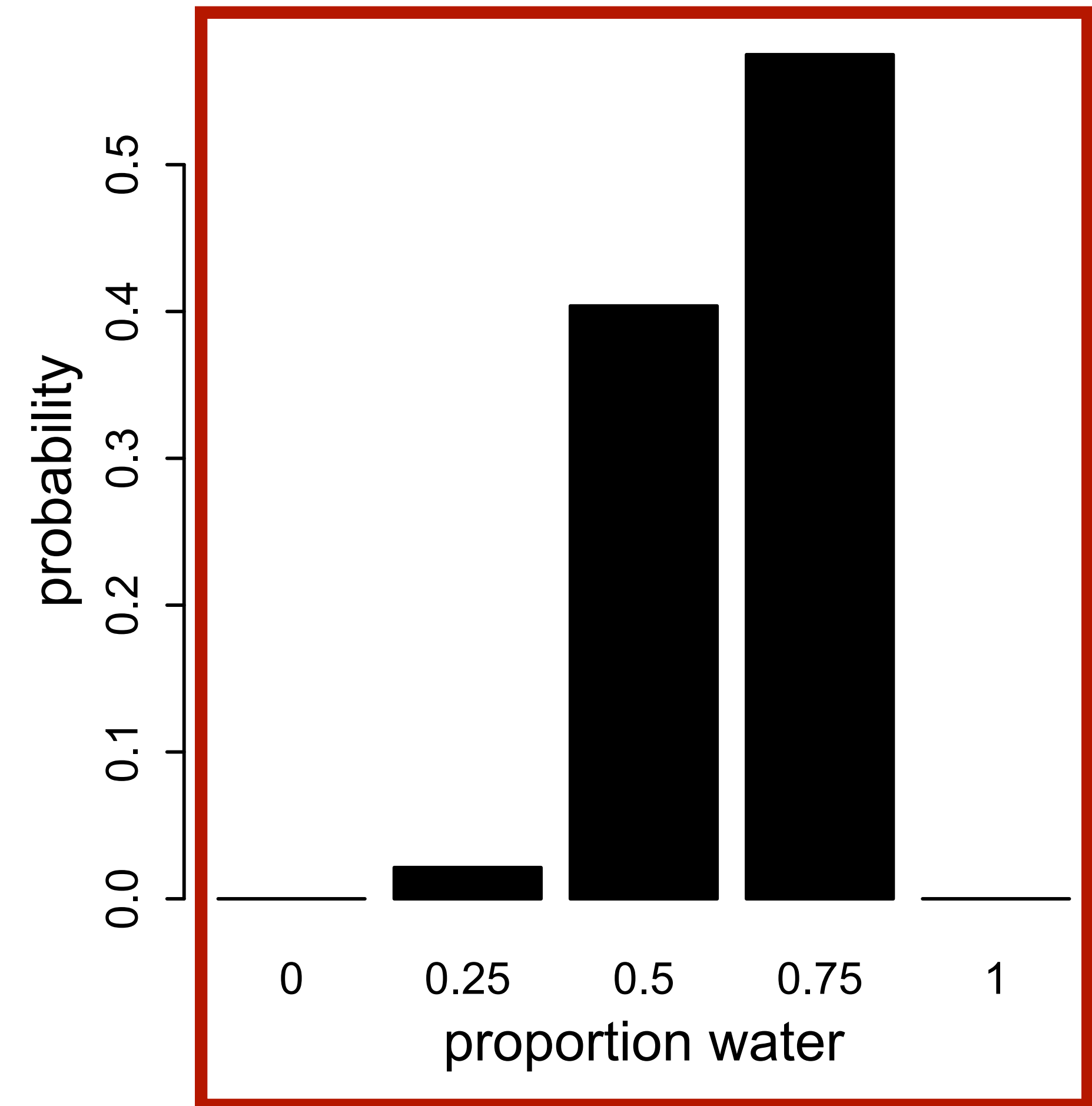
Possible proportion	Ways to produce sample	Probability of proportion
0	0	0
0.25	27	0.02
0.5	512	0.40
0.75	729	0.57
1	0	0



Probability

Posterior distribution

Possible proportion	Ways to produce sample	Probability of proportion
0	0	0
0.25	27	0.02
0.5	512	0.40
0.75	729	0.57
1	0	0

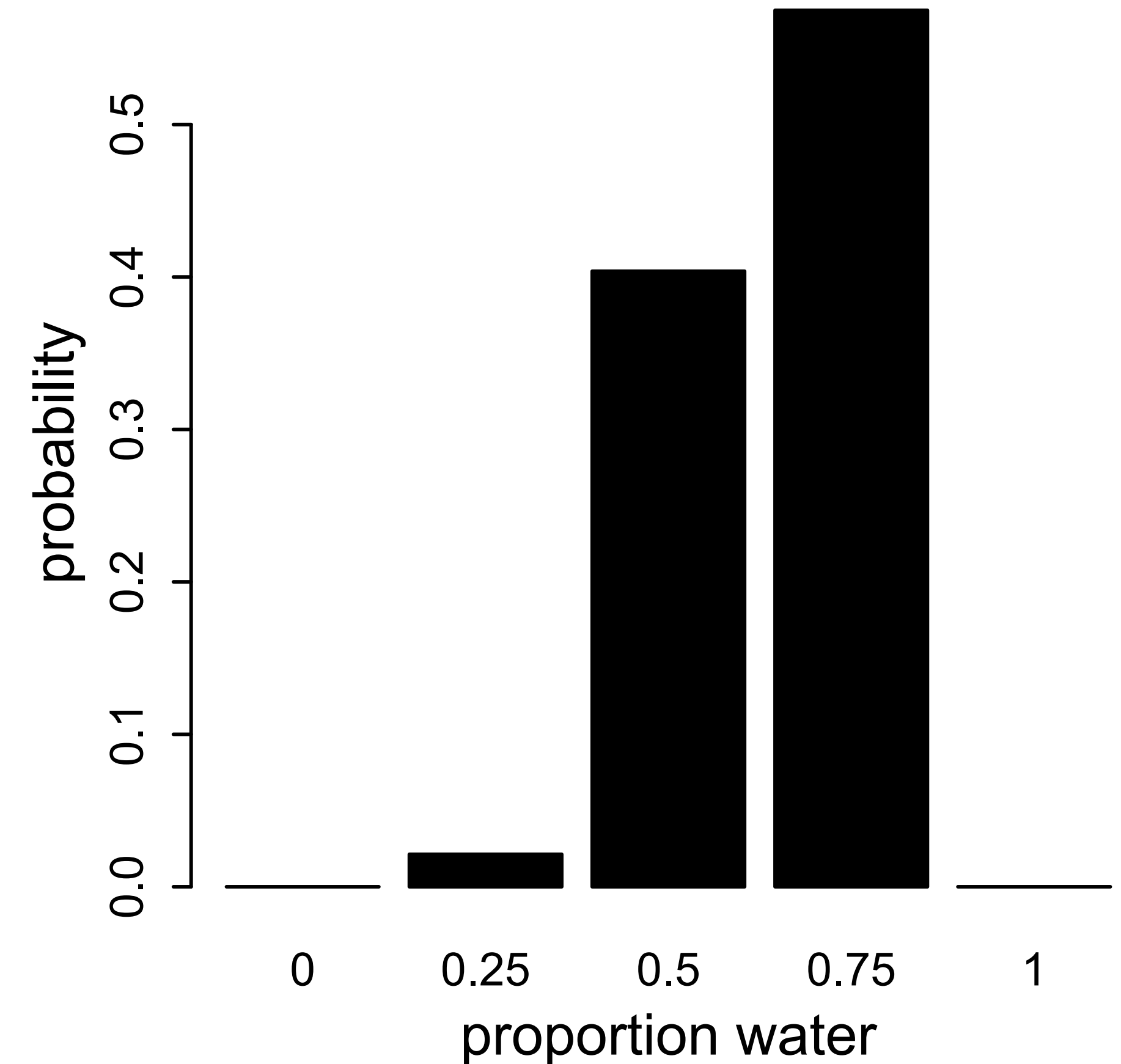


Probability

R code
2.1

```
sample <- c("W","L","W","W","W","L","W","L","W")
W <- sum(sample=="W") # number of W observed
L <- sum(sample=="L") # number of L observed
p <- c(0,0.25,0.5,0.75,1) # proportions W
ways <- sapply( p , function(q) (q*4)^W * ((1-q)*4)^L )
prob <- ways/sum(ways)
cbind( p , ways , prob )
```

	p	ways	prob
[1,]	0.00	0	0.000000000
[2,]	0.25	27	0.02129338
[3,]	0.50	512	0.40378549
[4,]	0.75	729	0.57492114
[5,]	1.00	0	0.000000000



Workflow

- (1) Define generative model of the sample
- (2) Define a specific estimand
- (3) Design a statistical way to produce estimate
- (4) Test (3) using (1)
- (5) Analyze sample, summarize



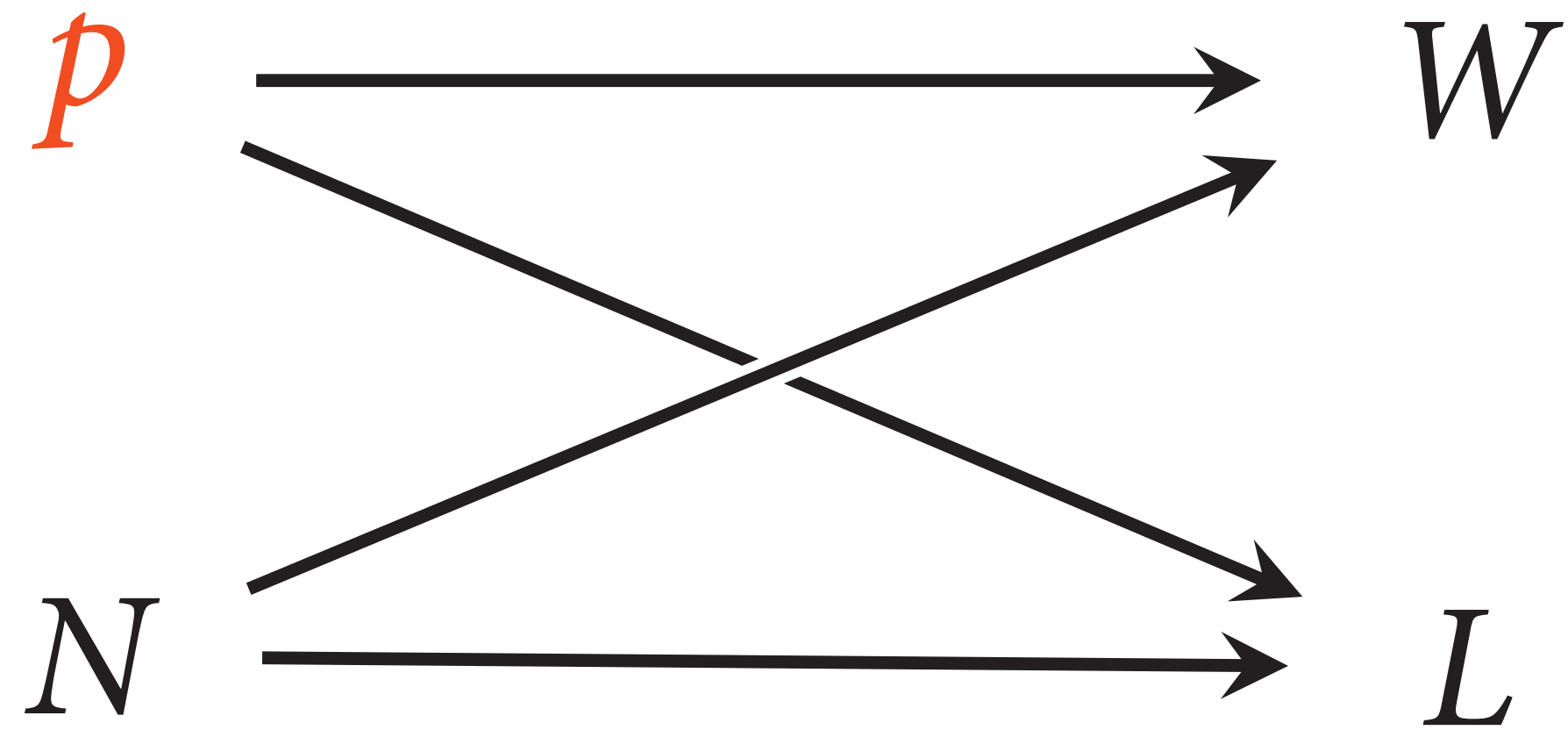
Test Before You Est(imate)

- (1) Code a generative simulation
- (2) Code an estimator
- (3) Test (2) with (1)



Extremely powerful, fun

Generative simulation



$$W, L = f(p, N)$$

```
# function to toss a globe covered p by water N times
sim_globe <- function( p=0.7 , N=9 ) {
  sample(c("W","L"),size=N,prob=c(p,1-p),replace=TRUE)
}
```

R code
2.3

R code
2.3

```
# function to toss a globe covered p by water N times  
sim_globe <- function( p=0.7 , N=9 ) {  
  sample(c("W","L"),size=N,prob=c(p,1-p),replace=TRUE)  
}
```

*Possible
observations*

*Number
of tosses*

*Probability of each
possible observation*


```
# function to toss a globe covered p by water N times
sim_globe <- function( p=0.7 , N=9 ) {
  sample(c("W","L"),size=N,prob=c(p,1-p),replace=TRUE)
}
```

R code
2.3

```
sim_globe()
```

R code
2.4

```
[1] "L" "W" "W" "W" "L" "L" "L" "W" "L"
```



```
sim_globe()
```

R code
2.4

```
[1] "L" "W" "W" "W" "L" "L" "L" "W" "L"
```

```
replicate(sim_globe(p=0.5,N=9),n=10)
```

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]
[1,]	"W"	"L"	"L"	"W"	"W"	"L"	"L"	"W"	"W"	"L"
[2,]	"W"	"L"	"W"	"L"	"W"	"L"	"L"	"W"	"L"	"L"
[3,]	"W"	"L"	"L"	"L"	"L"	"W"	"L"	"W"	"W"	"W"
[4,]	"W"	"W"	"L"	"W"	"L"	"W"	"W"	"W"	"W"	"W"
[5,]	"L"	"W"	"W"	"W"	"W"	"W"	"L"	"W"	"L"	"L"
[6,]	"L"	"W"	"L"	"L"	"W"	"L"	"W"	"W"	"W"	"W"
[7,]	"W"	"W"	"W"	"L"	"W"	"W"	"W"	"L"	"L"	"L"
[8,]	"L"	"W"	"L"	"L"	"L"	"W"	"L"	"W"	"W"	"W"
[9,]	"W"	"L"	"L"	"W"	"L"	"W"	"W"	"W"	"L"	"L"

```
sim_globe()
```

R code
2.4

```
[1] "L" "W" "W" "W" "L" "L" "L" "W" "L"
```

Test the simulation on extreme settings

R code
2.5

```
sim_globe( p=1 , N=11 )
```

```
[1] "W" "W" "W" "W" "W" "W" "W" "W" "W" "W" "W"
```

R code
2.6

```
sum( sim_globe( p=0.5 , N=1e4 ) == "W" ) / 1e4
```

```
[1] 0.505
```

IF YOU TEST

NOTHING

YOU MISS

EVERYTHING

Code the estimator

Ways for p to produce $W, L = (4p)^W \times (4-4p)^L$

```
# function to compute posterior distribution
compute_posterior <- function( the_sample , poss=c(0,0.25,0.5,0.75,1) ) {
  W <- sum(the_sample=="W") # number of W observed
  L <- sum(the_sample=="L") # number of L observed
  ways <- sapply( poss , function(q) (q*4)^W * ((1-q)*4)^L )
  post <- ways/sum(ways)
  bars <- sapply( post, function(q) make_bar(q) )
  data.frame( poss , ways , post=round(post,3) , bars )
}
```


Code the estimator

Ways for p to produce $W, L = (4p)^W \times (4-4p)^L$

```
# function to compute posterior distribution
compute_posterior <- function( the_sample , poss=c(0,0.25,0.5,0.75,1) ) {
  W <- sum(the_sample=="W") # number of W observed
  L <- sum(the_sample=="L") # number of L observed
  ways <- sapply( poss , function(q) (q*4)^W * ((1-q)*4)^L )
  post <- ways/sum(ways)
  bars <- sapply( post, function(q) make_bar(q) )
  data.frame( poss , ways , post=round(post,3) , bars )
}
```

Code the estimator

Ways for p to produce $W, L = (4p)^W \times (4-4p)^L$

```
# function to compute posterior distribution
compute_posterior <- function( the_sample , poss=c(0,0.25,0.5,0.75,1) ) {
  W <- sum(the_sample=="W") # number of W observed
  L <- sum(the_sample=="L") # number of L observed
  ways <- sapply( poss , function(q) (q*4)^W * ((1-q)*4)^L )
  post <- ways/sum(ways)
  bars <- sapply( post, function(q) make_bar(q) )
  data.frame( poss , ways , post=round(post,3) , bars )
}
```

Code the estimator

Ways for p to produce $W, L = (4p)^W \times (4-4p)^L$

```
# function to compute posterior distribution
compute_posterior <- function( the_sample , poss=c(0,0.25,0.5,0.75,1) ) {
  W <- sum(the_sample=="W") # number of W observed
  L <- sum(the_sample=="L") # number of L observed
  ways <- sapply( poss , function(q) (q*4)^W * ((1-q)*4)^L )
  post <- ways/sum(ways)
  bars <- sapply( post, function(q) make_bar(q) )
  data.frame( poss , ways , post=round(post,3) , bars )
}
```


Code the estimator

Ways for p to produce $W, L = (4p)^W \times (4-4p)^L$

```
# function to compute posterior distribution
compute_posterior <- function( the_sample , poss=c(0,0.25,0.5,0.75,1) ) {
  W <- sum(the_sample=="W") # number of W observed
  L <- sum(the_sample=="L") # number of L observed
  ways <- sapply( poss , function(q) (q*4)^W * ((1-q)*4)^L )
  post <- ways/sum(ways)
  bars <- sapply( post, function(q) make_bar(q) )
  data.frame( poss , ways , post=round(post,3) , bars )
}
```



```
compute_posterior( sim_globe() )
```

R code
2.9

```
      poss ways  post      bars
1 0.00      0 0.000
2 0.25    243 0.291 #####
3 0.50    512 0.612 #####
4 0.75     81 0.097 ##
5 1.00      0 0.000
```

- (1) Test the estimator where the answer is known
- (2) Explore different sampling designs
- (3) Develop intuition for sampling and estimation

More possibilities

4-sided globe



[0 0.25 0.5 0.75 1]

More possibilities

4-sided globe



[0 0.25 0.5 0.75 1]

10-sided globe



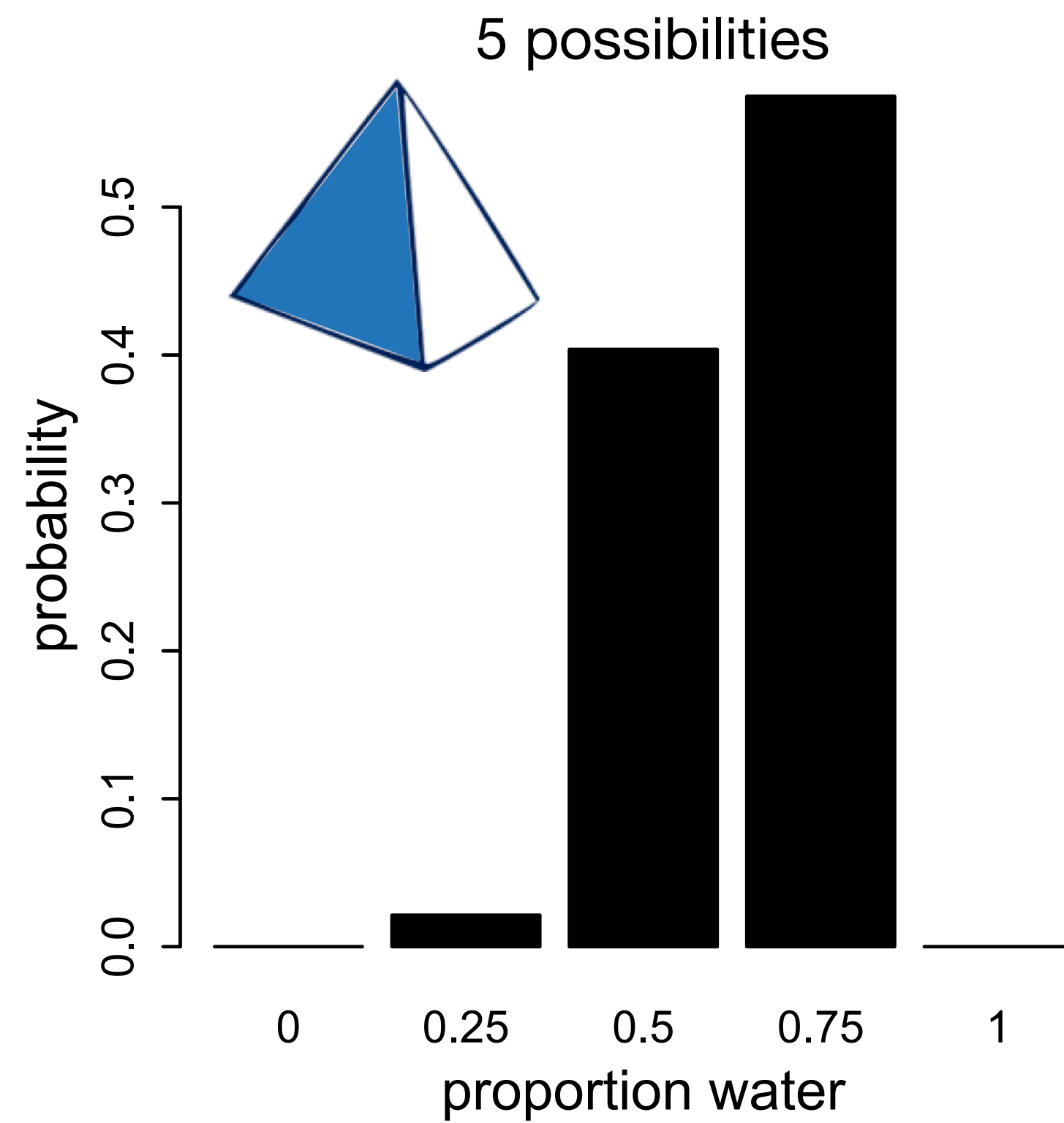
[0 0.1 0.2 0.3 0.4 0.5
0.6 0.7 0.8 0.9 1]

20-sided globe

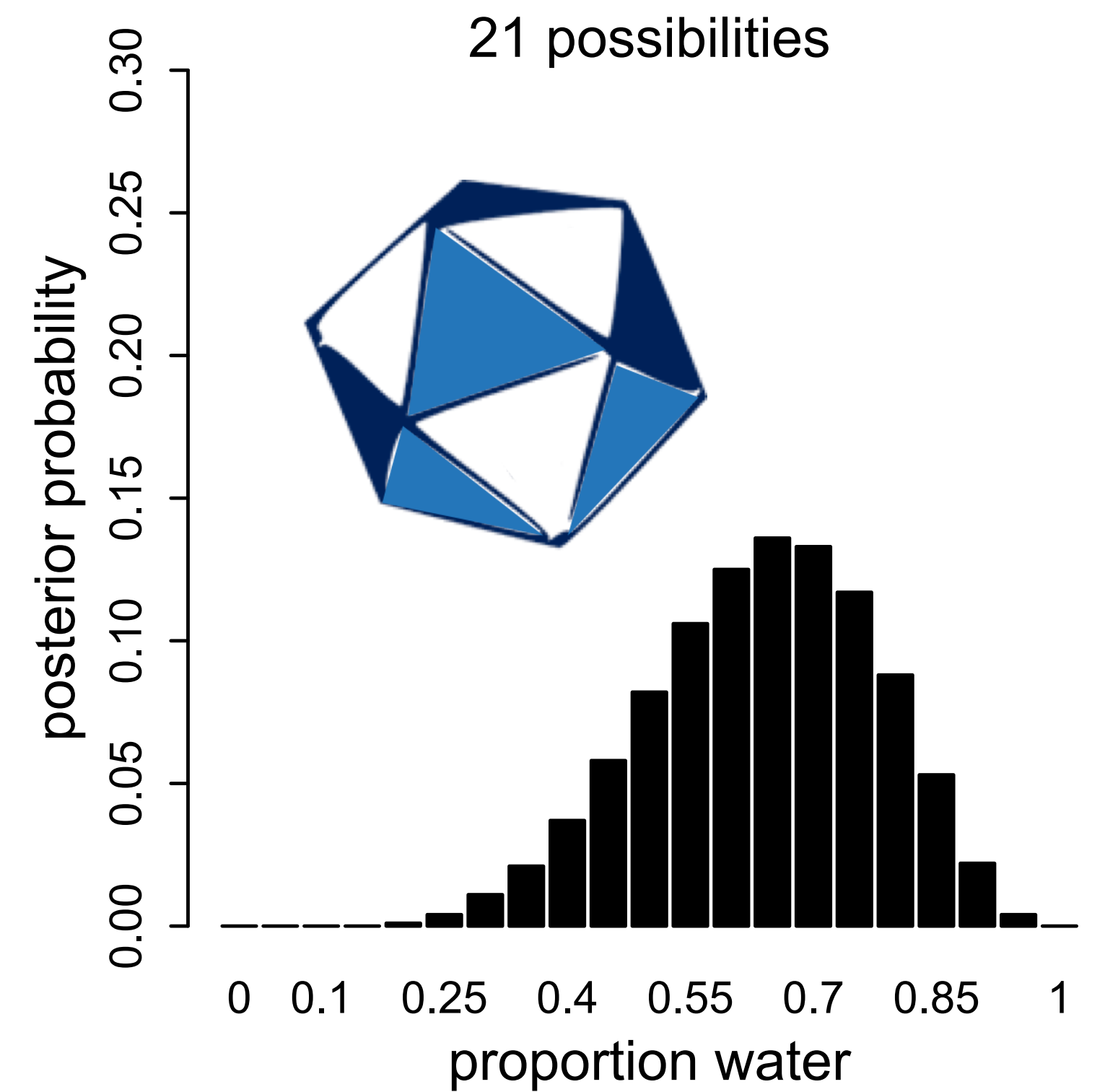
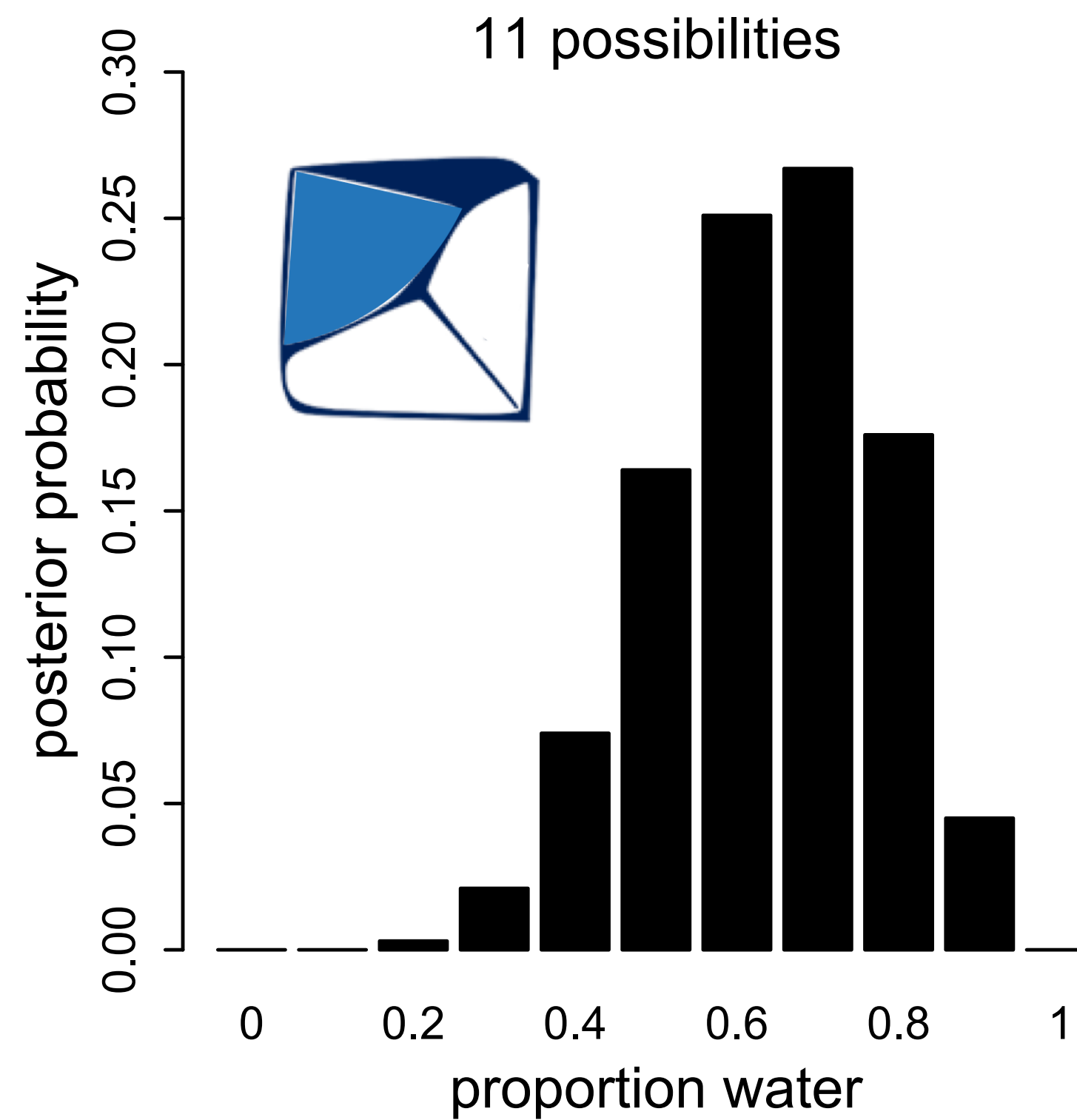
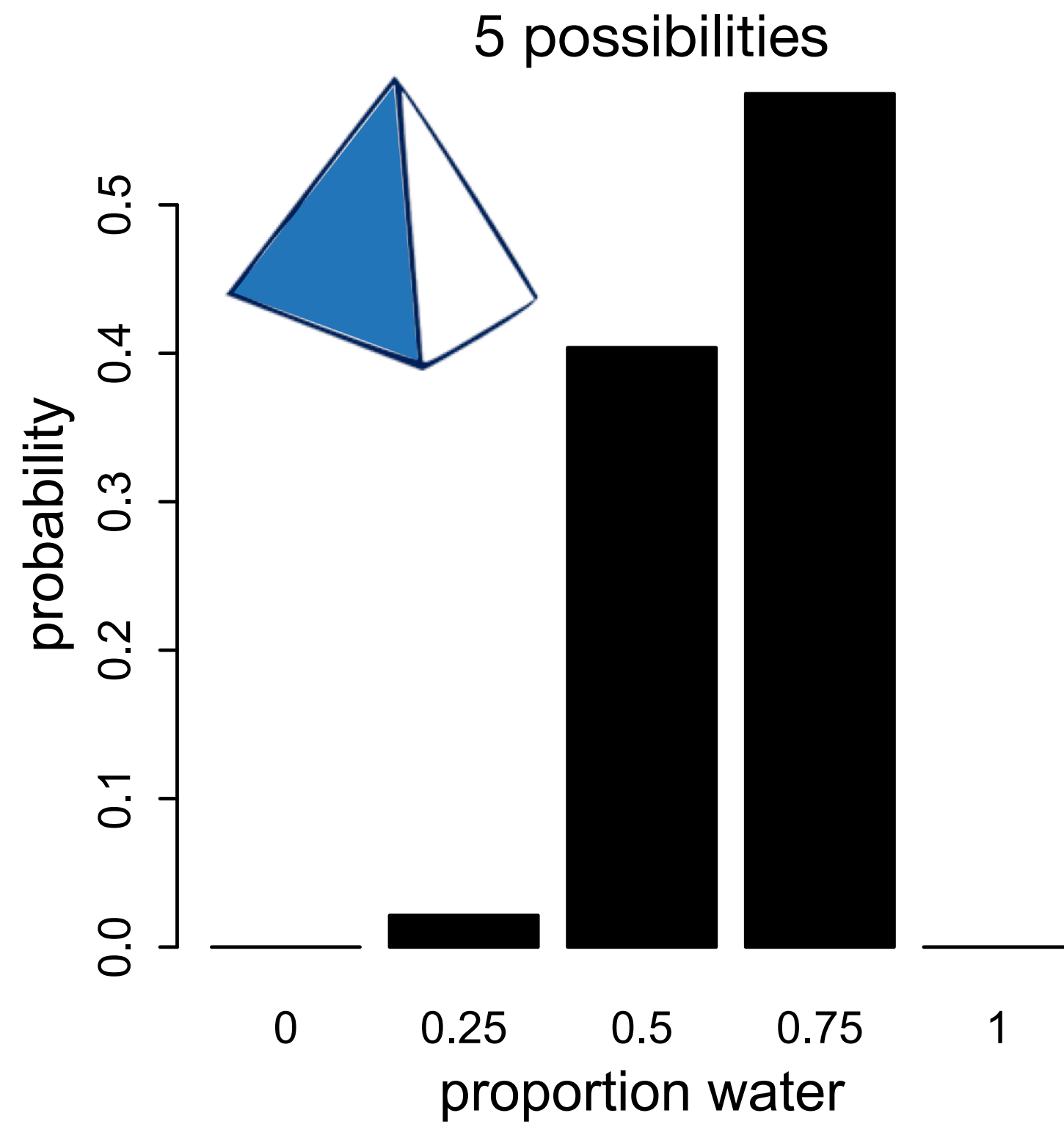


[0 0.05 0.10 0.15 0.20 0.25 0.30
0.35 0.40 0.45 0.50 0.55 0.60 0.65
0.70 0.75 0.80 0.85 0.90 0.95 1]

More possibilities



More possibilities



Infinite possibilities

The globe is a polyhedron with an infinite number of sides

The posterior probability of any “side” p is proportional to:

$$p^W (1 - p)^L$$

Infinite possibilities

The globe is a polyhedron with an infinite number of sides

The posterior probability of any “side” p is proportional to:

$$p^W (1 - p)^L$$

Only trick is normalizing to probability. After a little calculus:

$$\text{Posterior probability of } p = \frac{(W + L + 1)!}{W!L!} p^W (1 - p)^L$$

Infinite possibilities

The “Beta” distribution

Posterior probability of $p = \frac{(W + L + 1)!}{W!L!} p^W (1 - p)^L$

*Normalizing
constant*

*relative number
of ways to
observe sample*

Infinite possibilities

The “Beta” distribution

$$\text{Posterior probability of } p = \frac{(W + L + 1)!}{W!L!} p^W (1 - p)^L$$

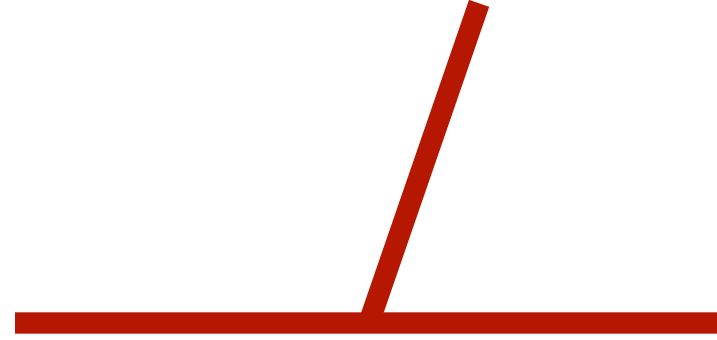
relative number of ways to observe sample

Normalizing constant = Beta function

Infinite possibilities

The “Beta” distribution

*relative number
of ways to
observe sample*



$$\text{Posterior probability of } p = \frac{(W + L + 1)!}{W!L!} p^W (1 - p)^L$$

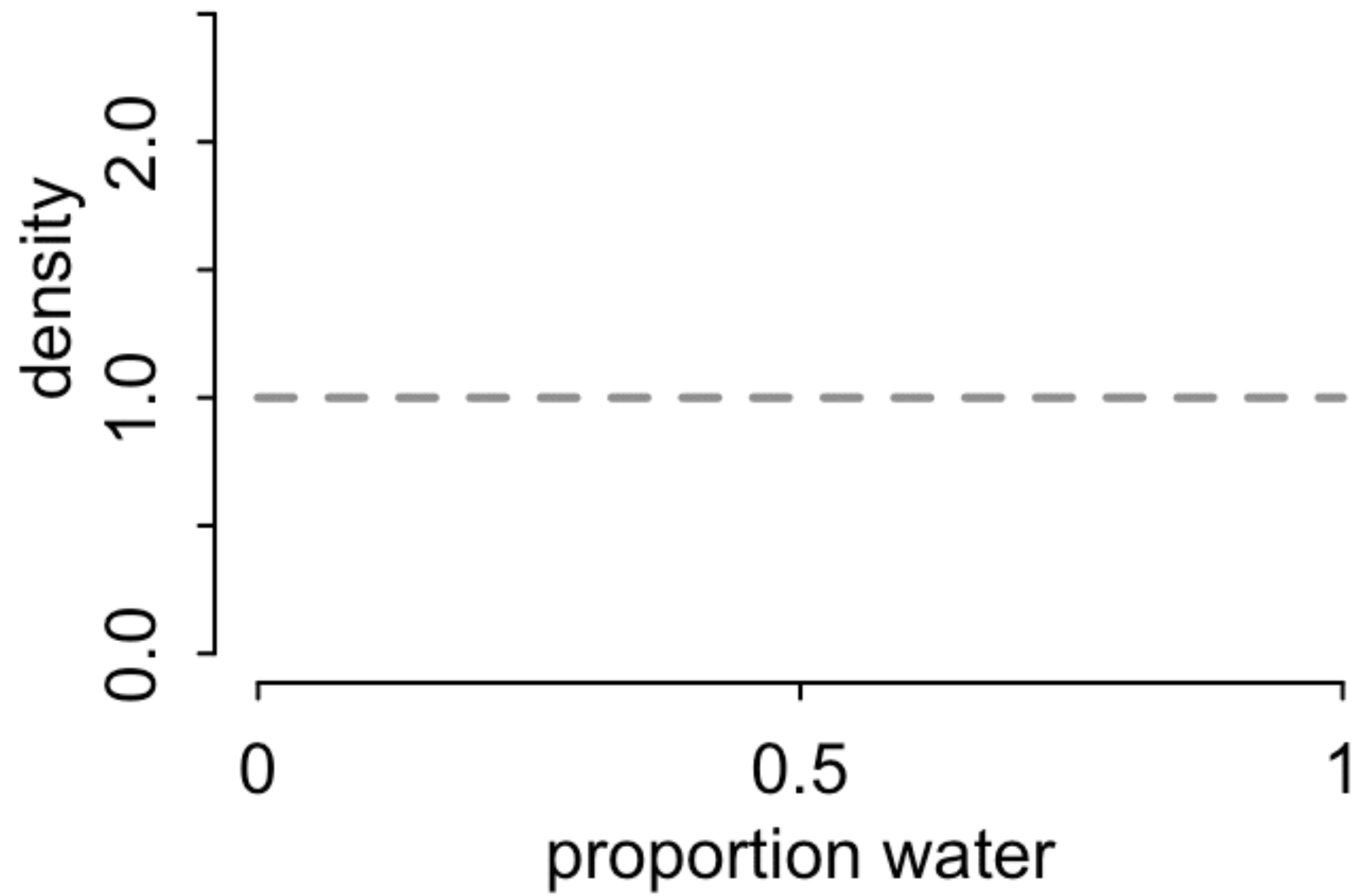
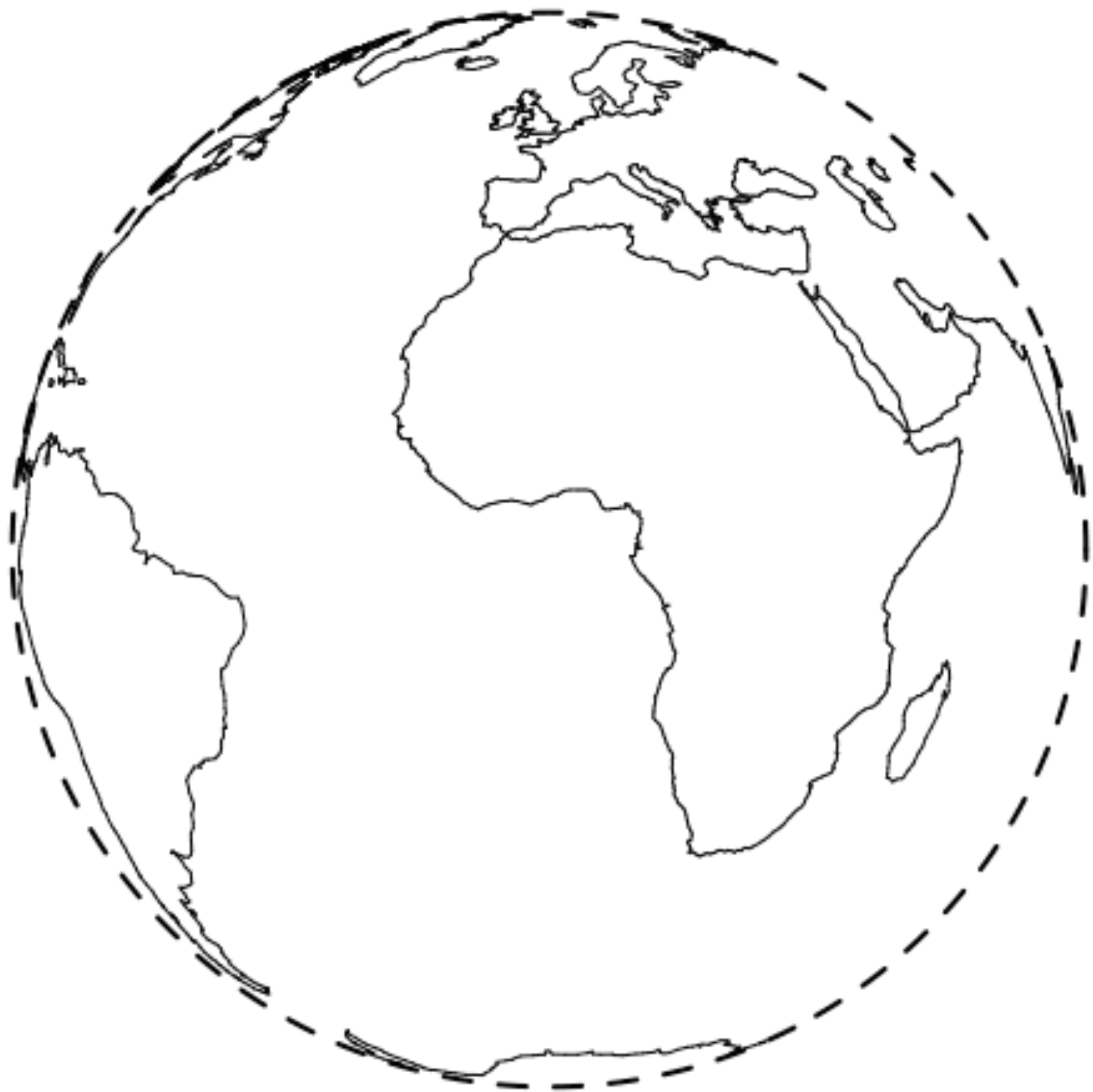
Beta distribution

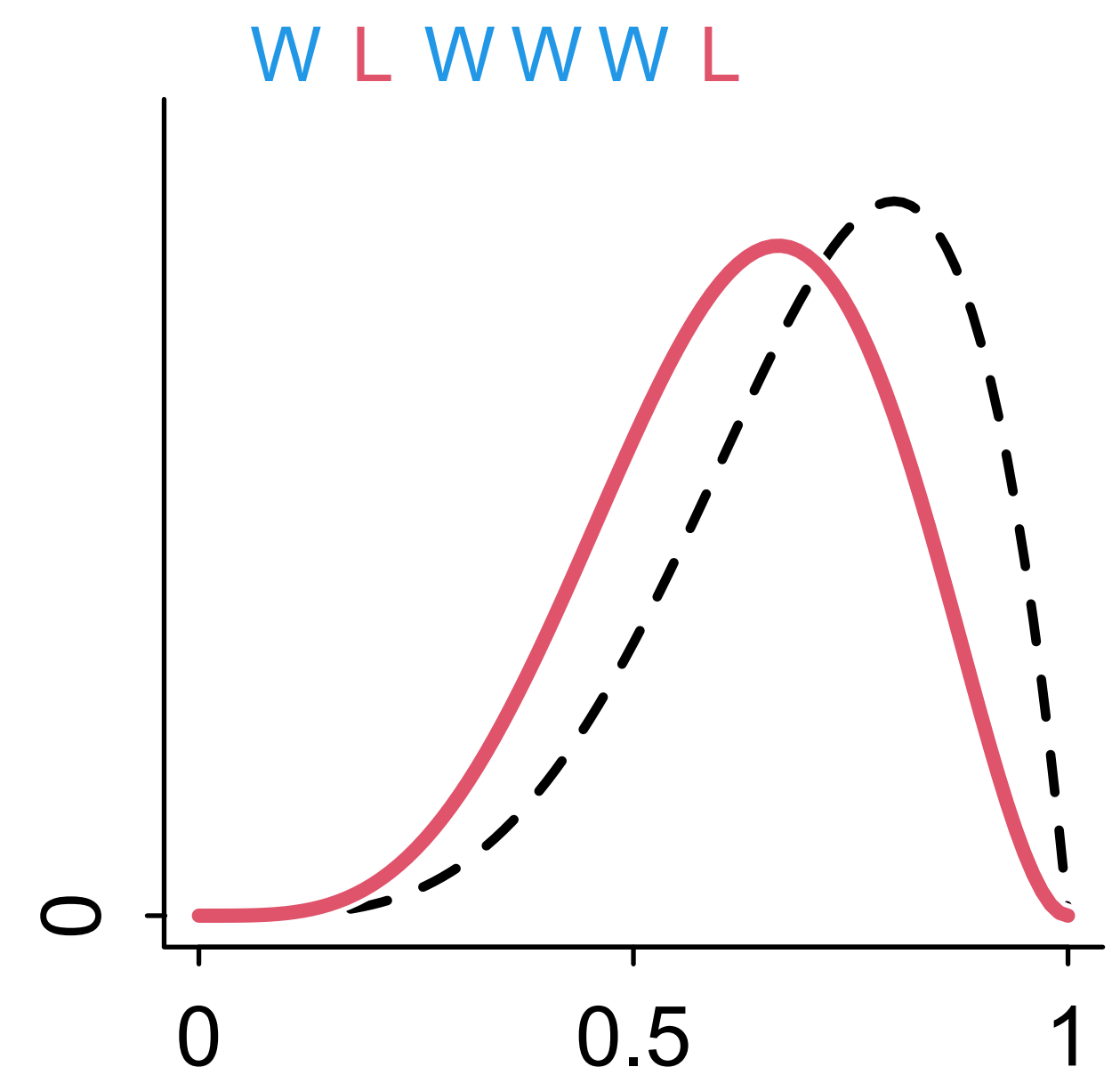
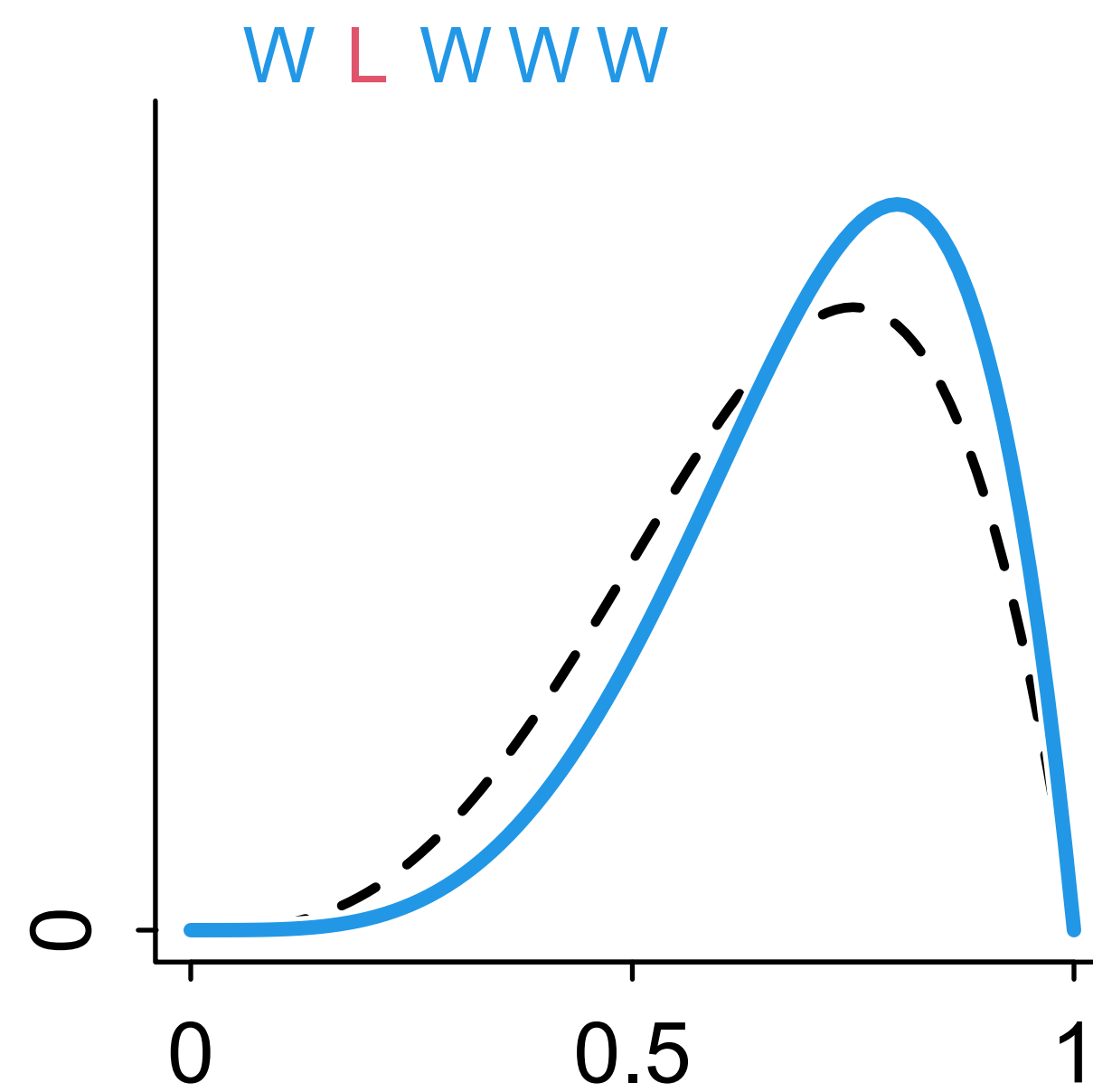
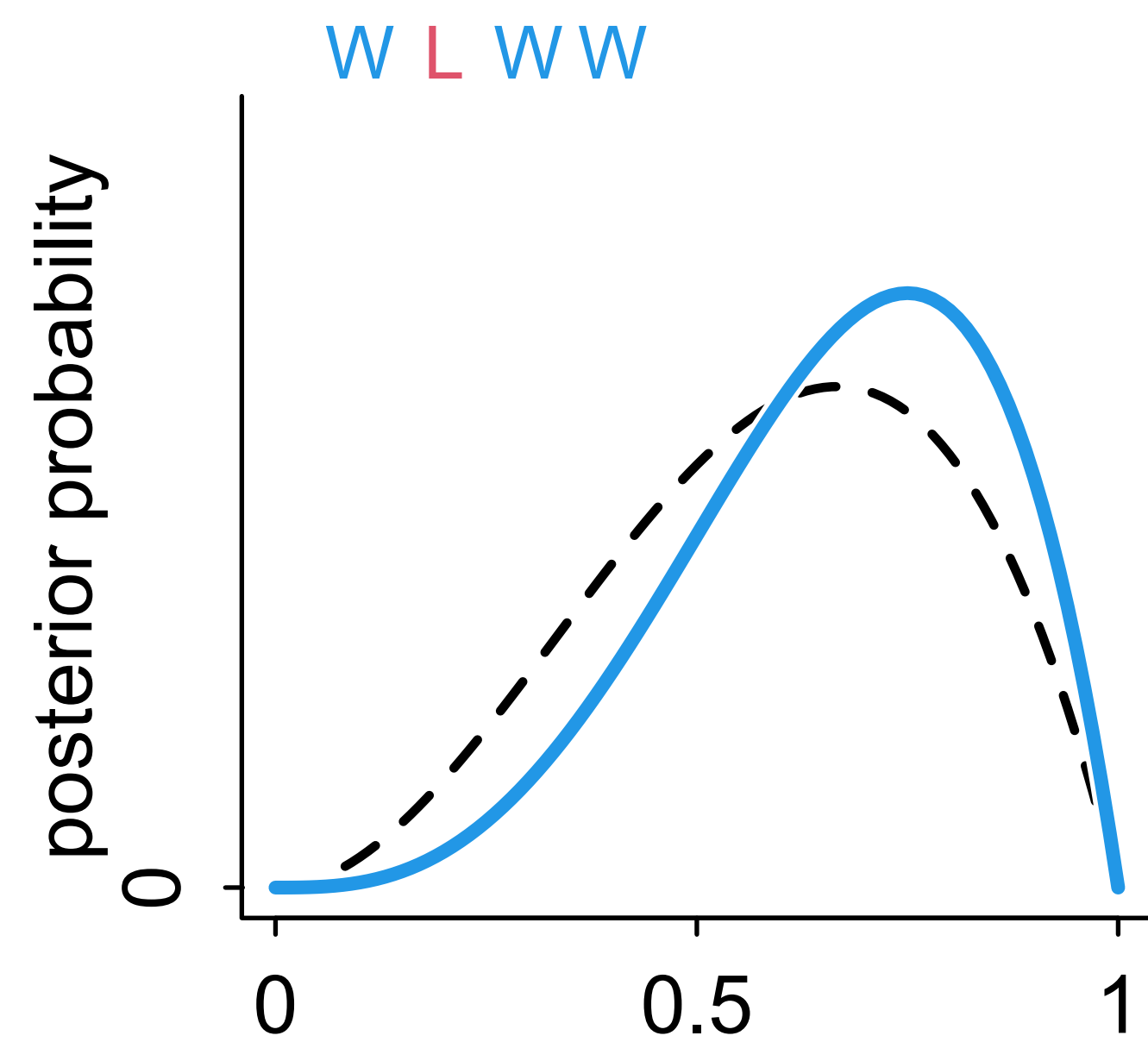
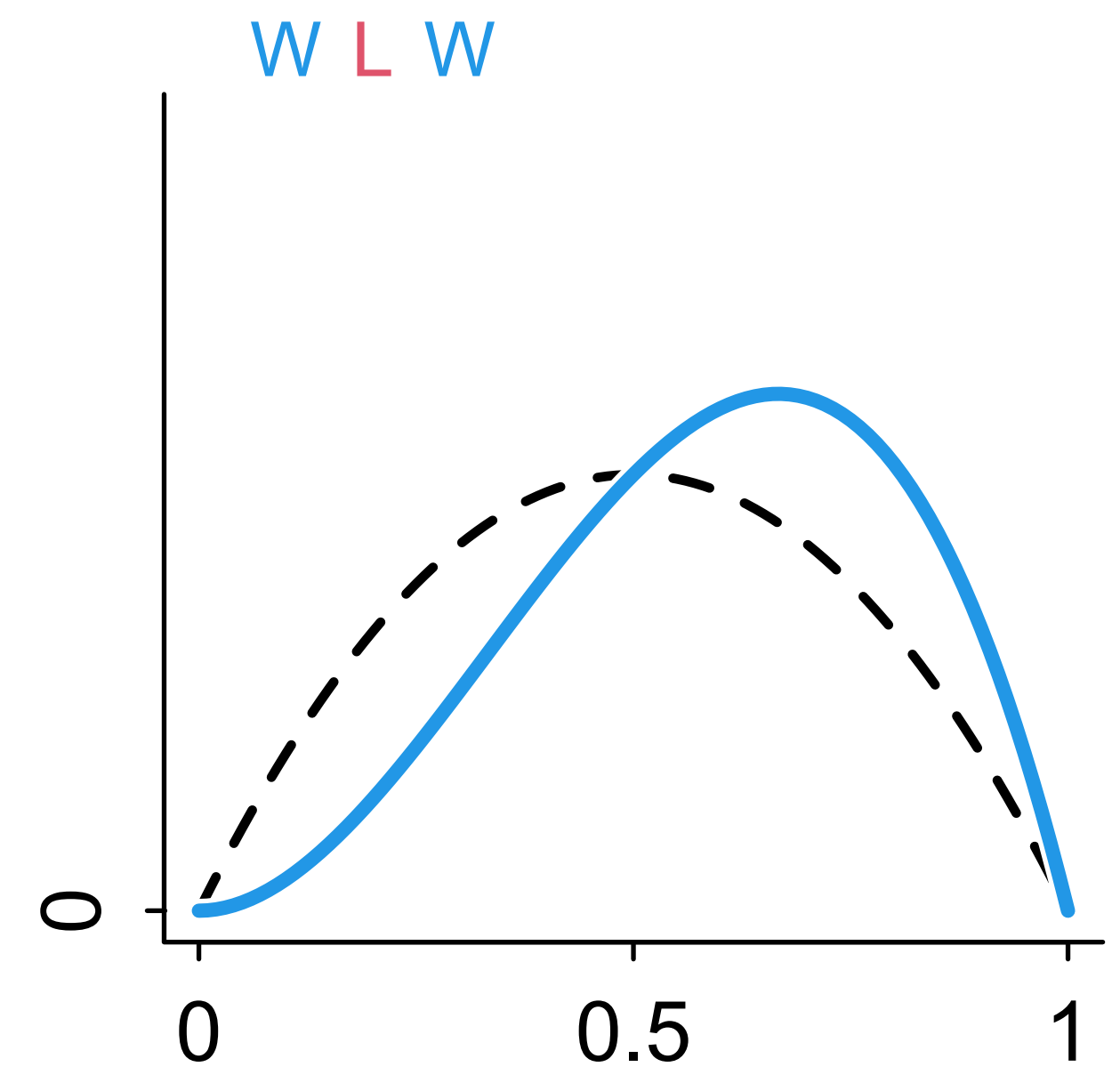
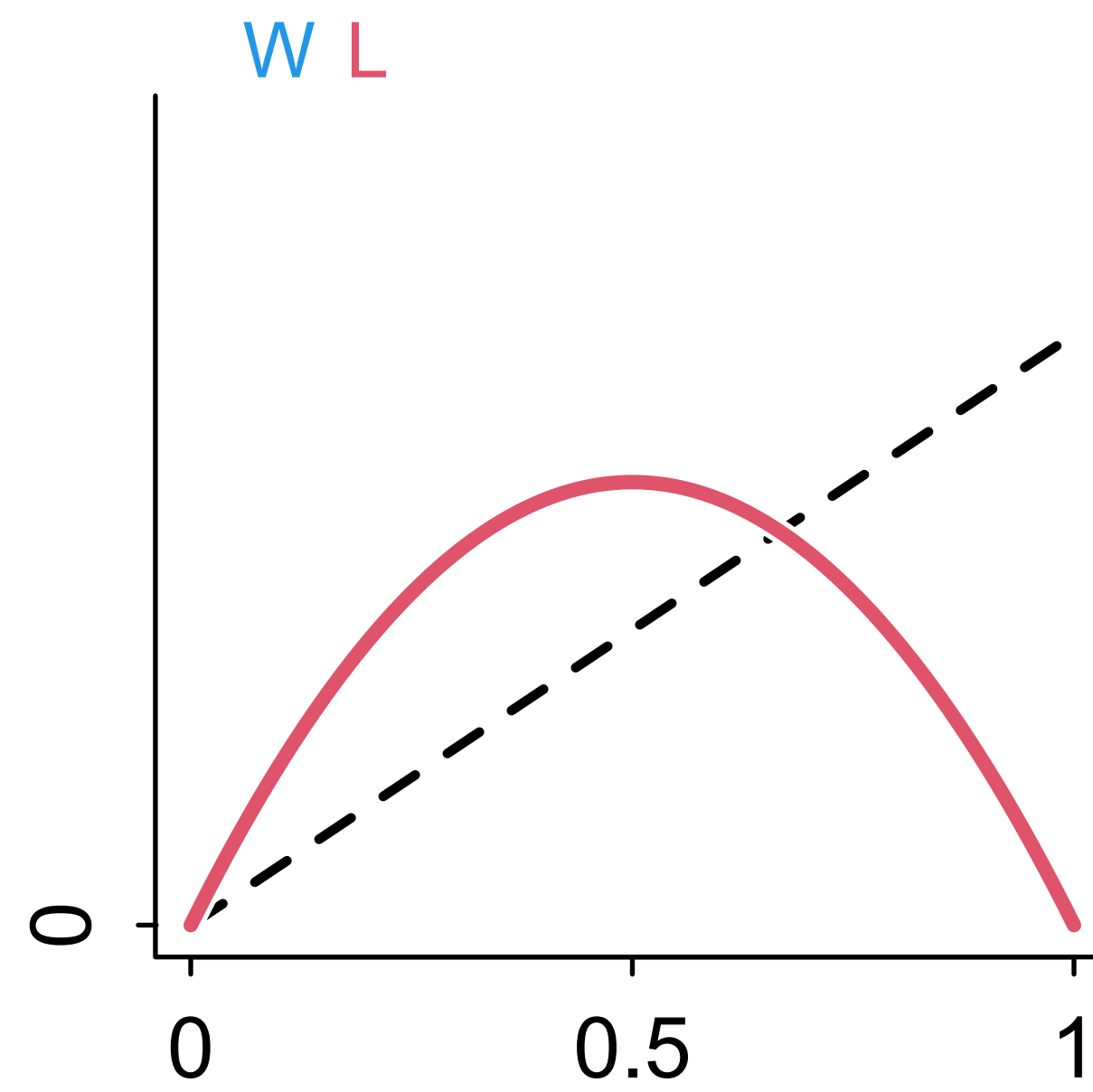
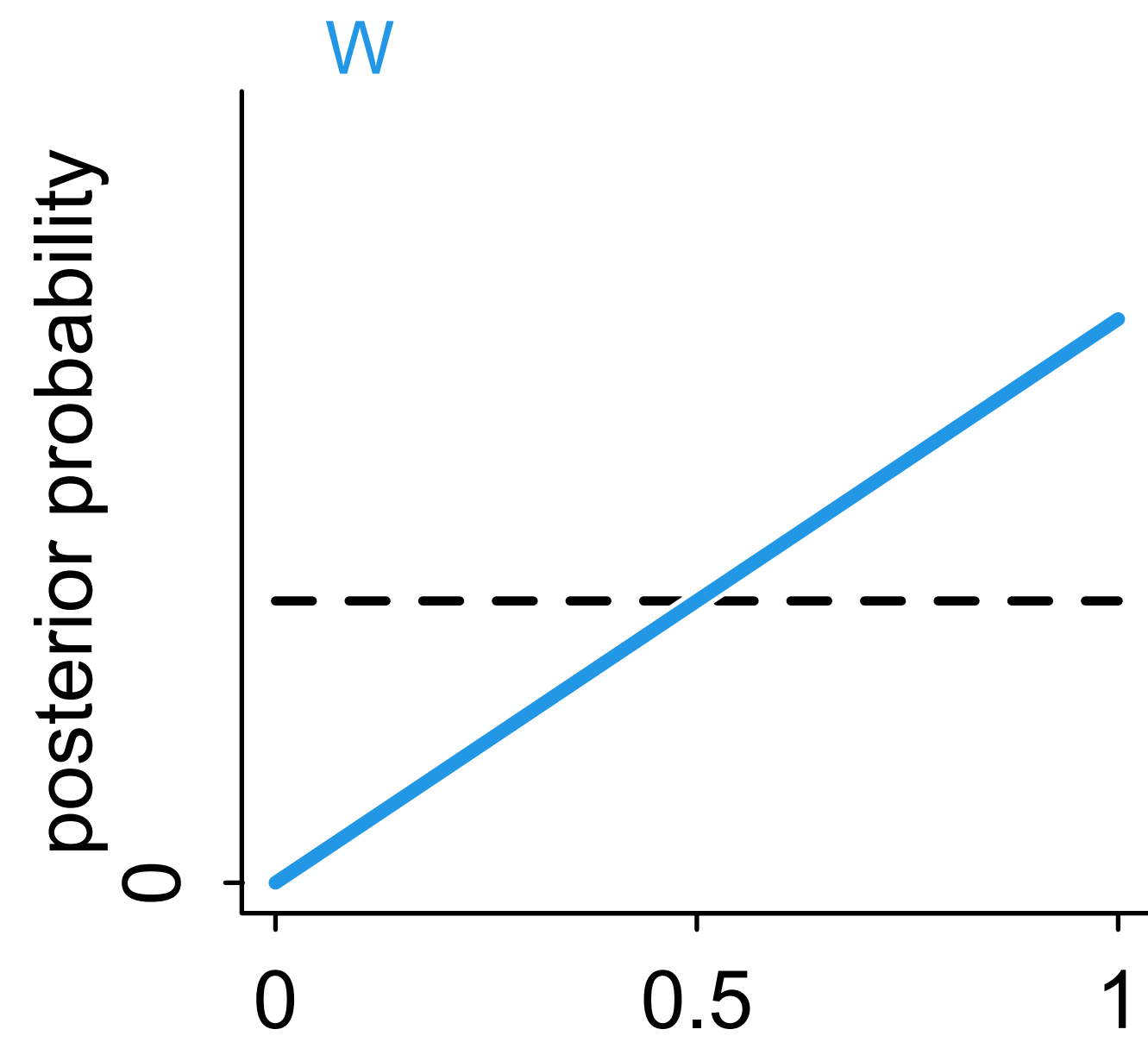
$$\text{mean} = W / (W + L)$$

$$\text{mode} = (W - 1) / (W + L - 2), \text{ for } W \text{ and } L > 1$$

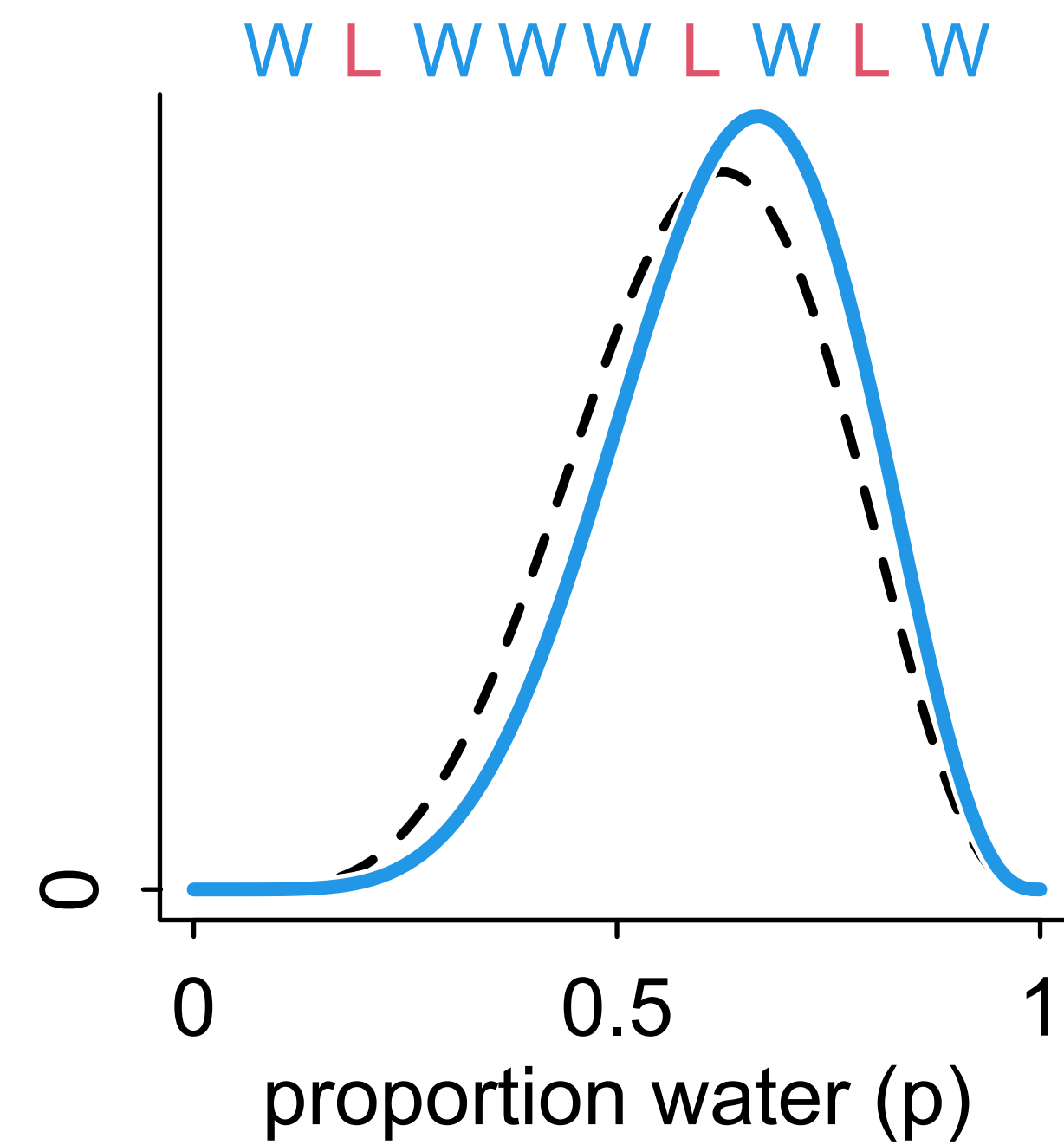
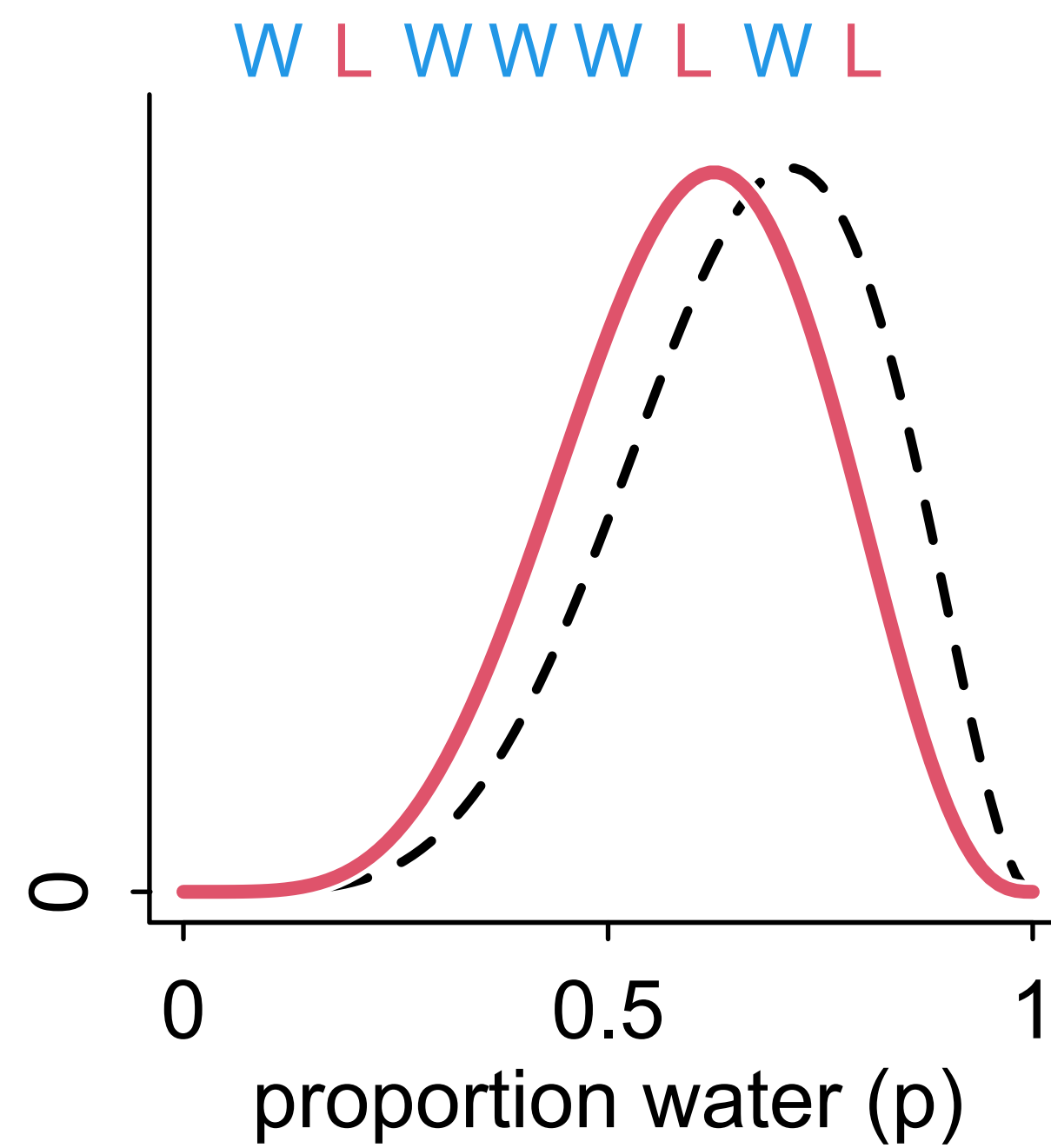
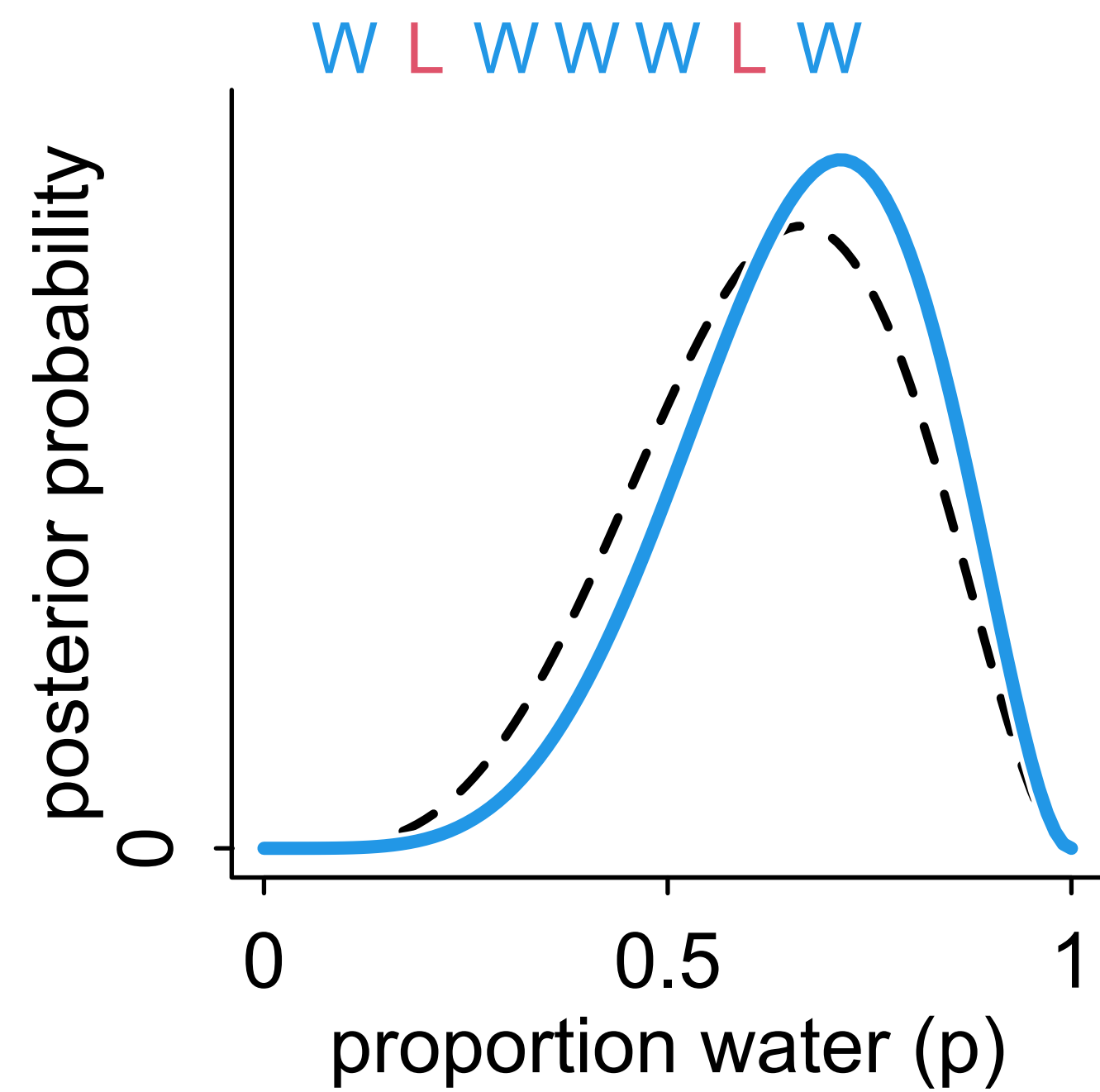
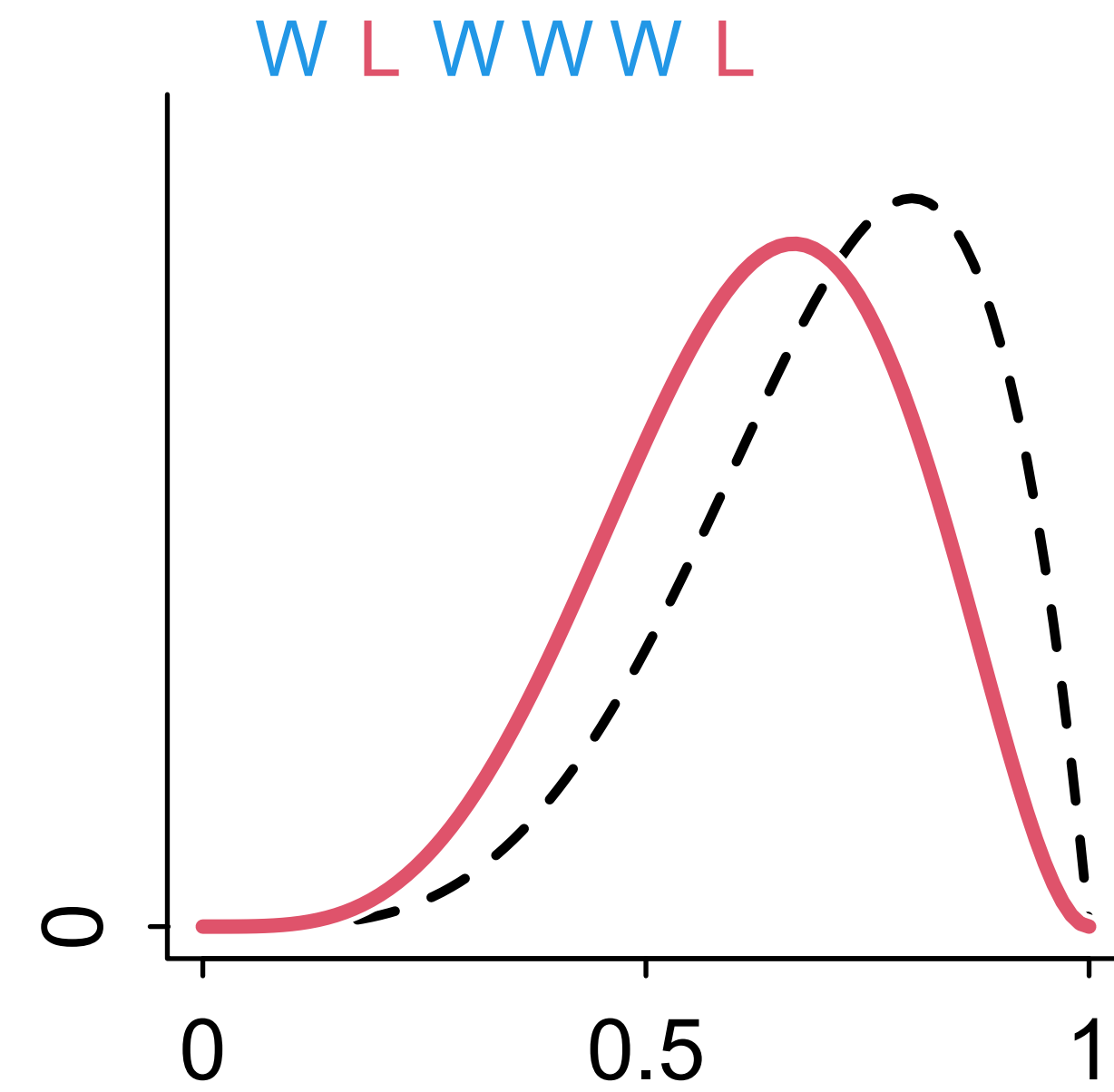
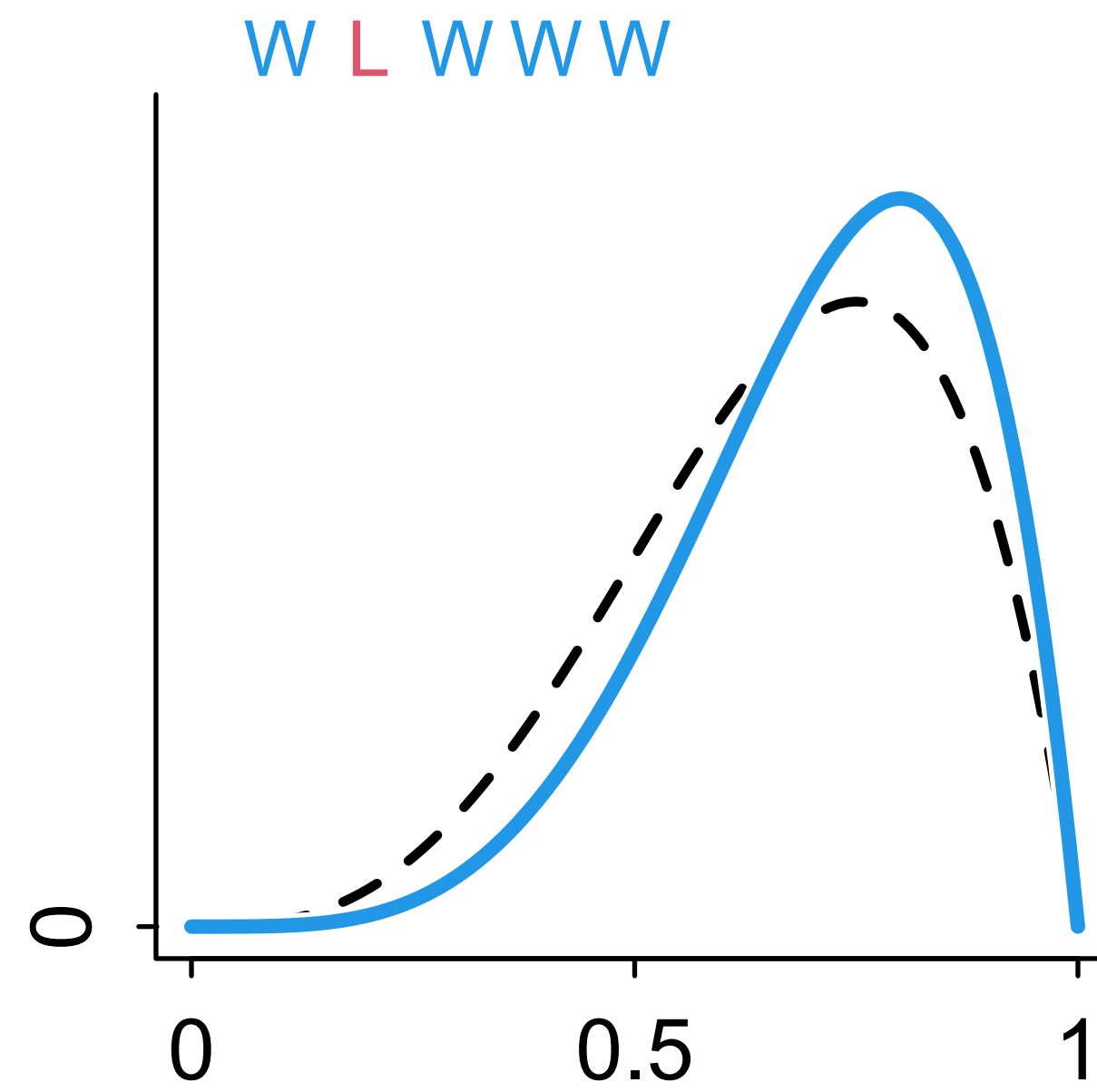
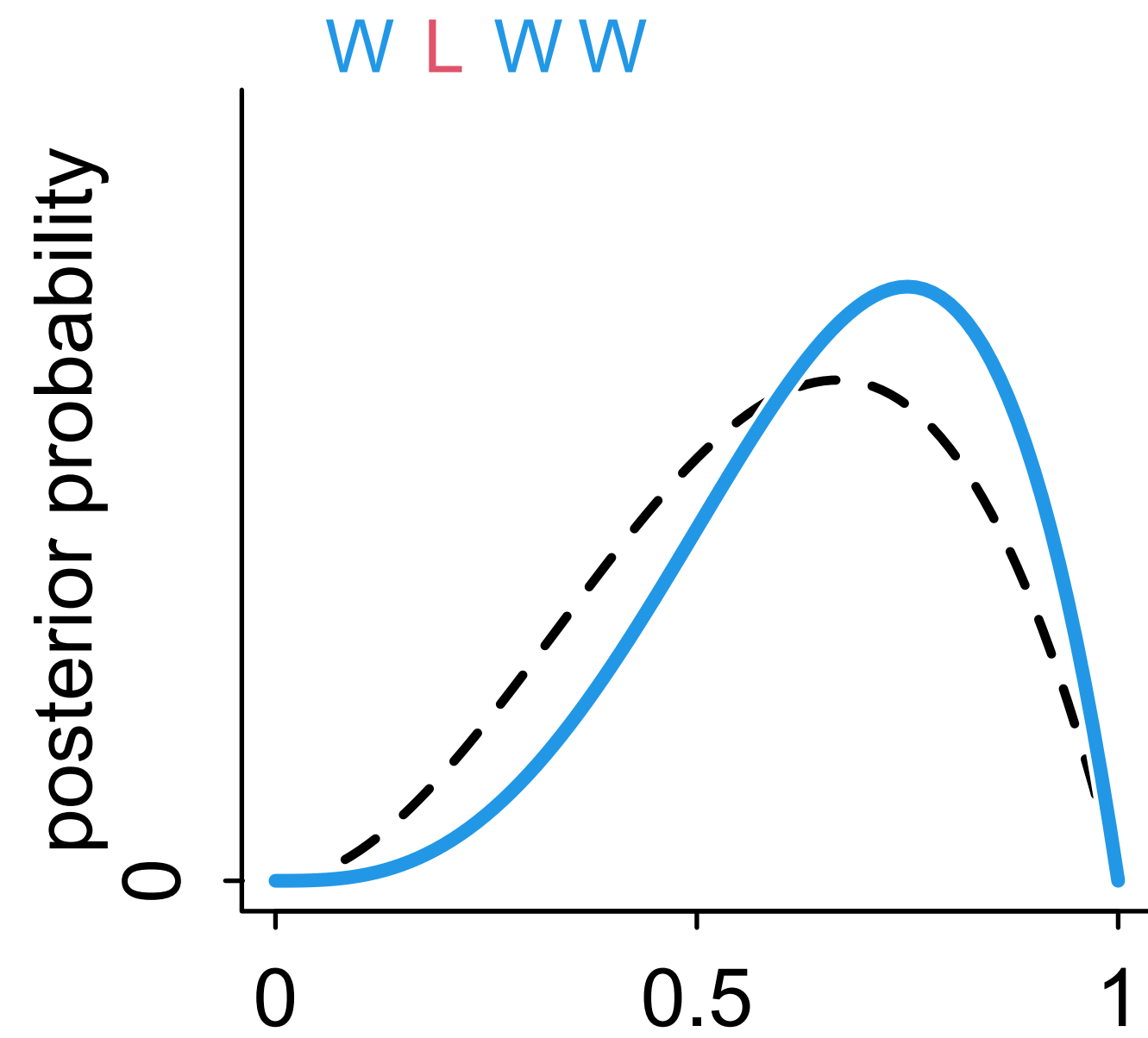
SD is inversely related to $(W + L)$

Ten tosses of the globe

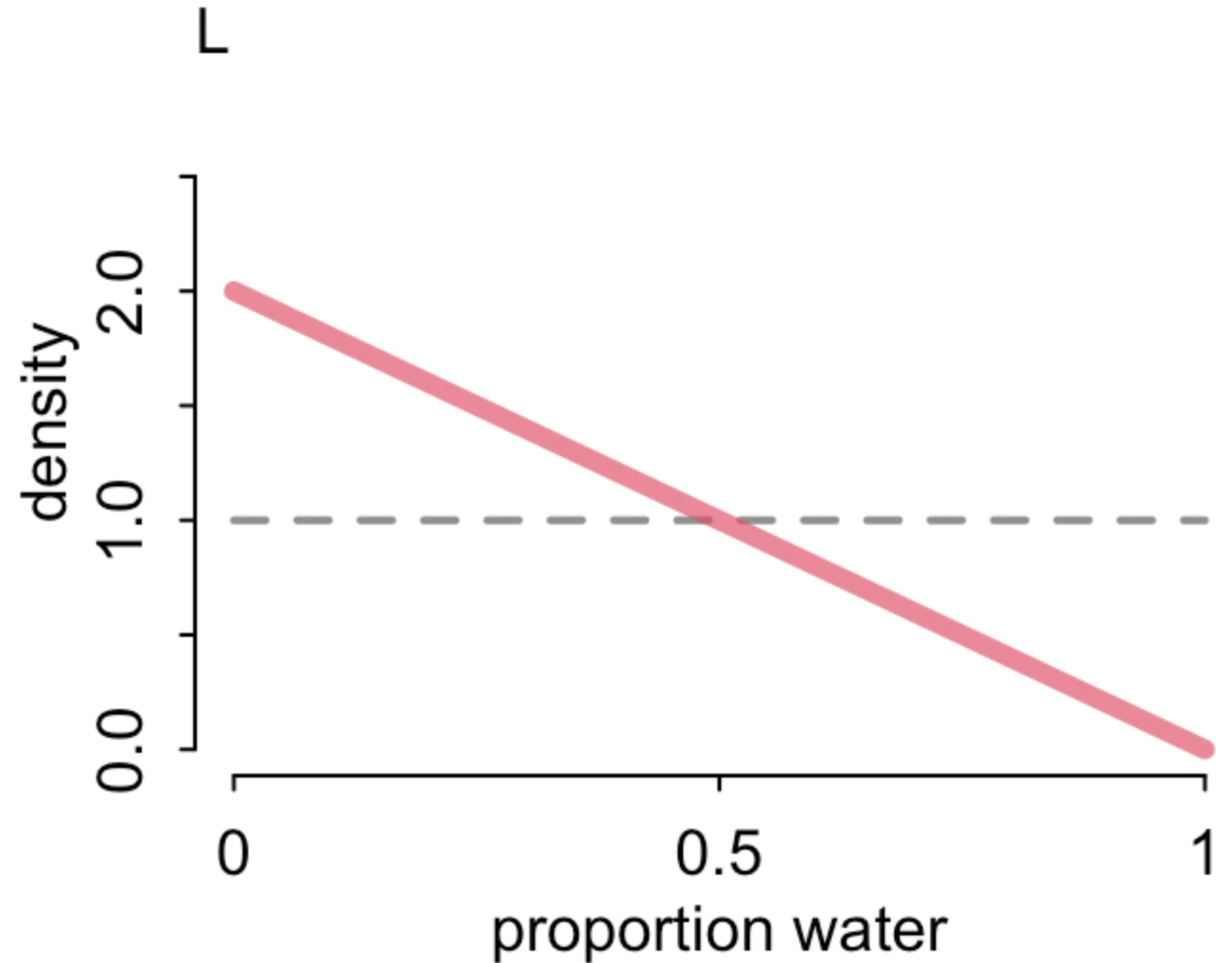
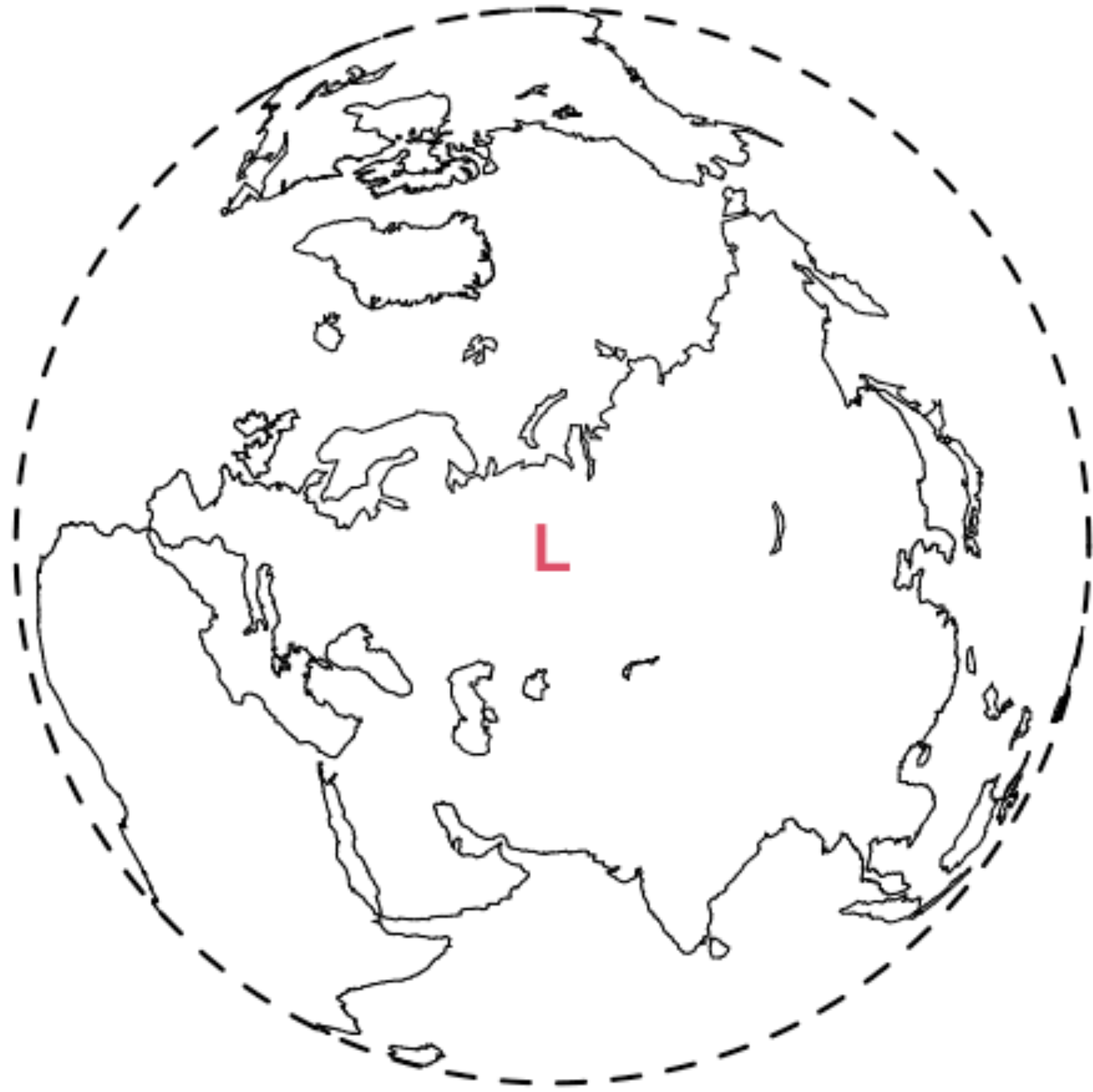




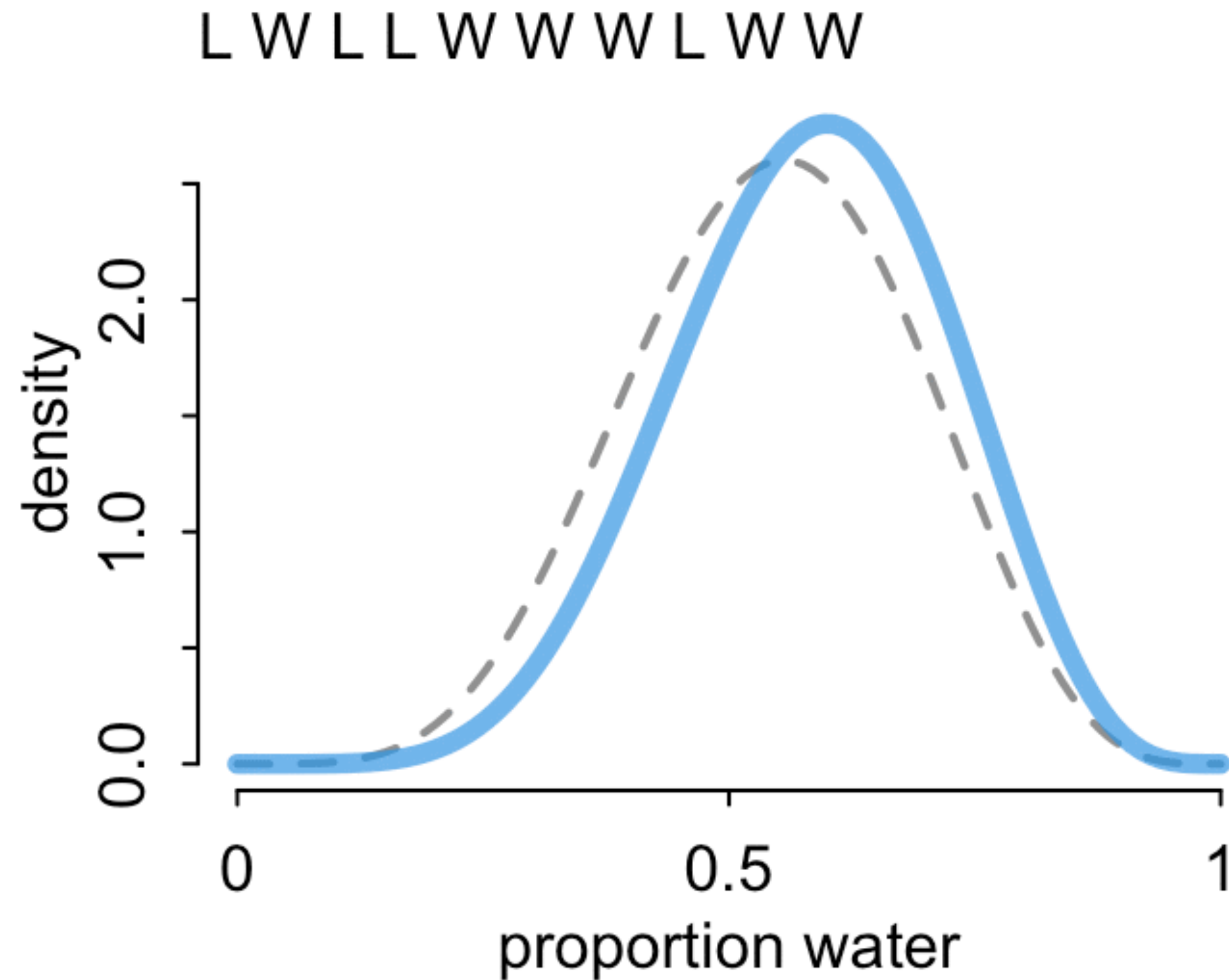
In each plot, previous plausibilities (dashed curve) are updated in light of the latest observation to produce a new set of plausibilities (solid curve).



(1) No minimum sample size



(2) Shape embodies sample size



(2) Shape embodies sample size

Beta distribution properties:

$$\text{mean} = W / (W + L)$$

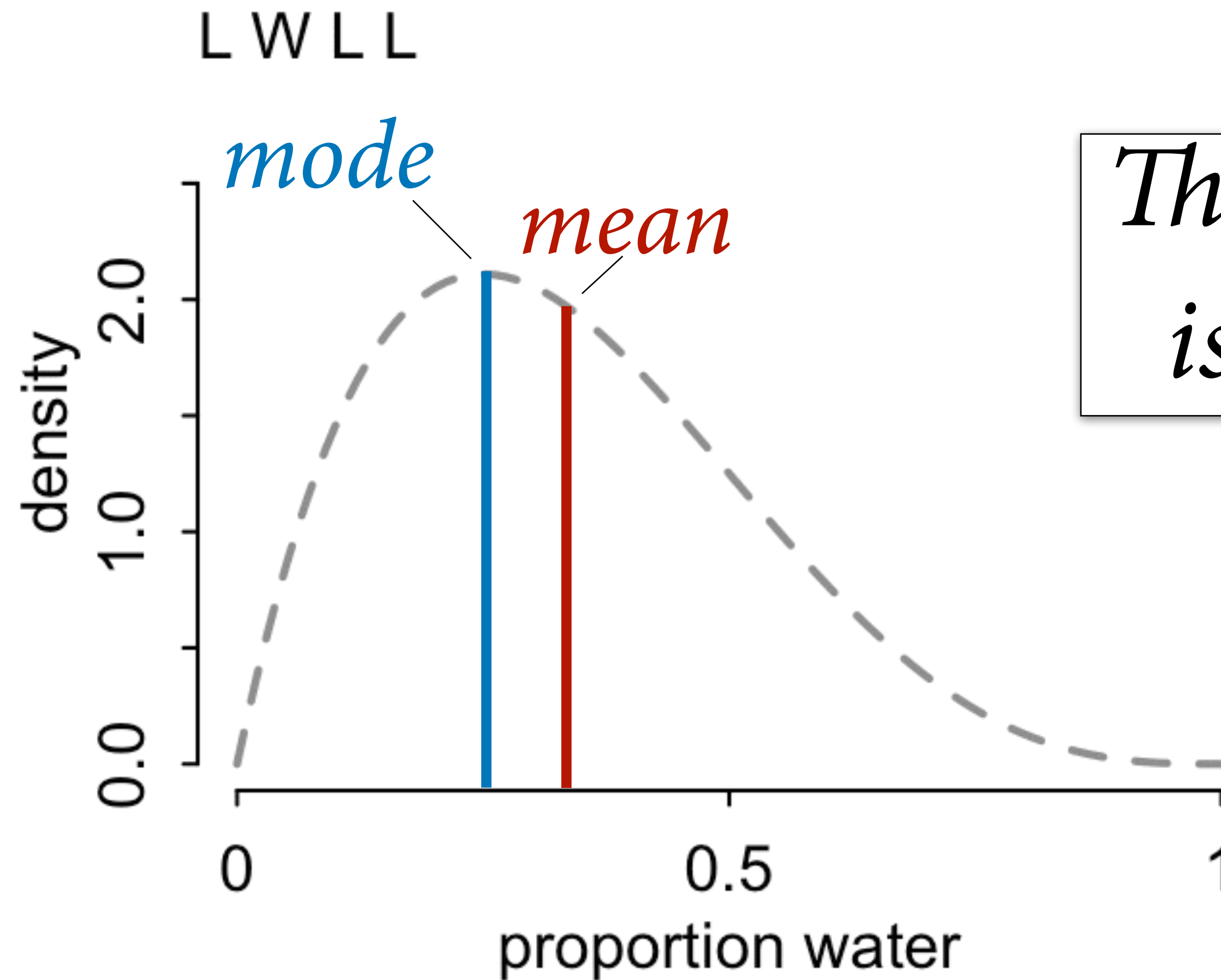
$$\text{mode} = (W - 1) / (W + L - 2), \text{ for } W \text{ and } L > 1$$

SD is inversely related to $(W + L)$

R code snippet to plot different Beta distributions

```
plot(function(x) dbeta(x, 1, 1), 0, 1)
```

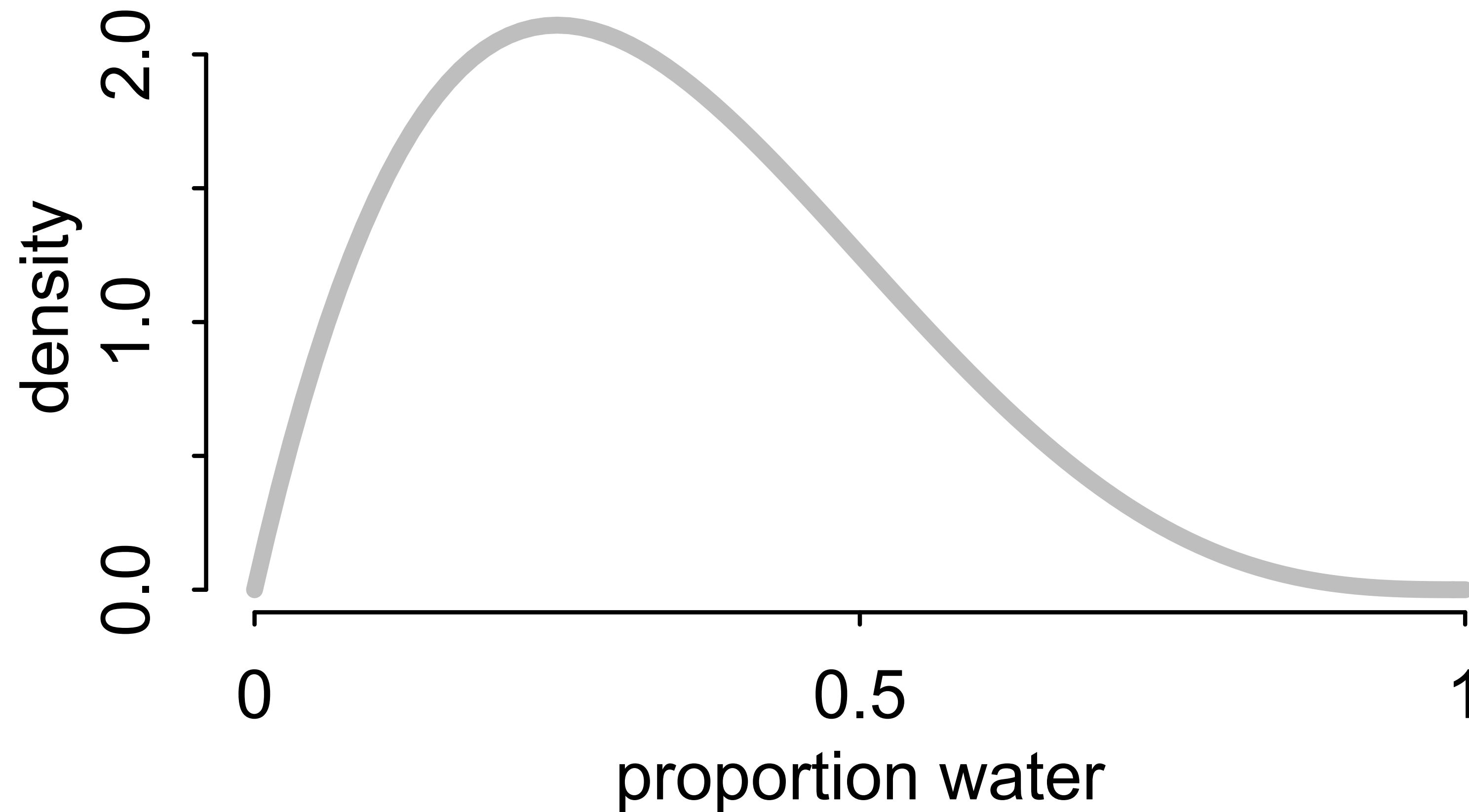

(3) No point estimate



*The distribution
is the estimate*

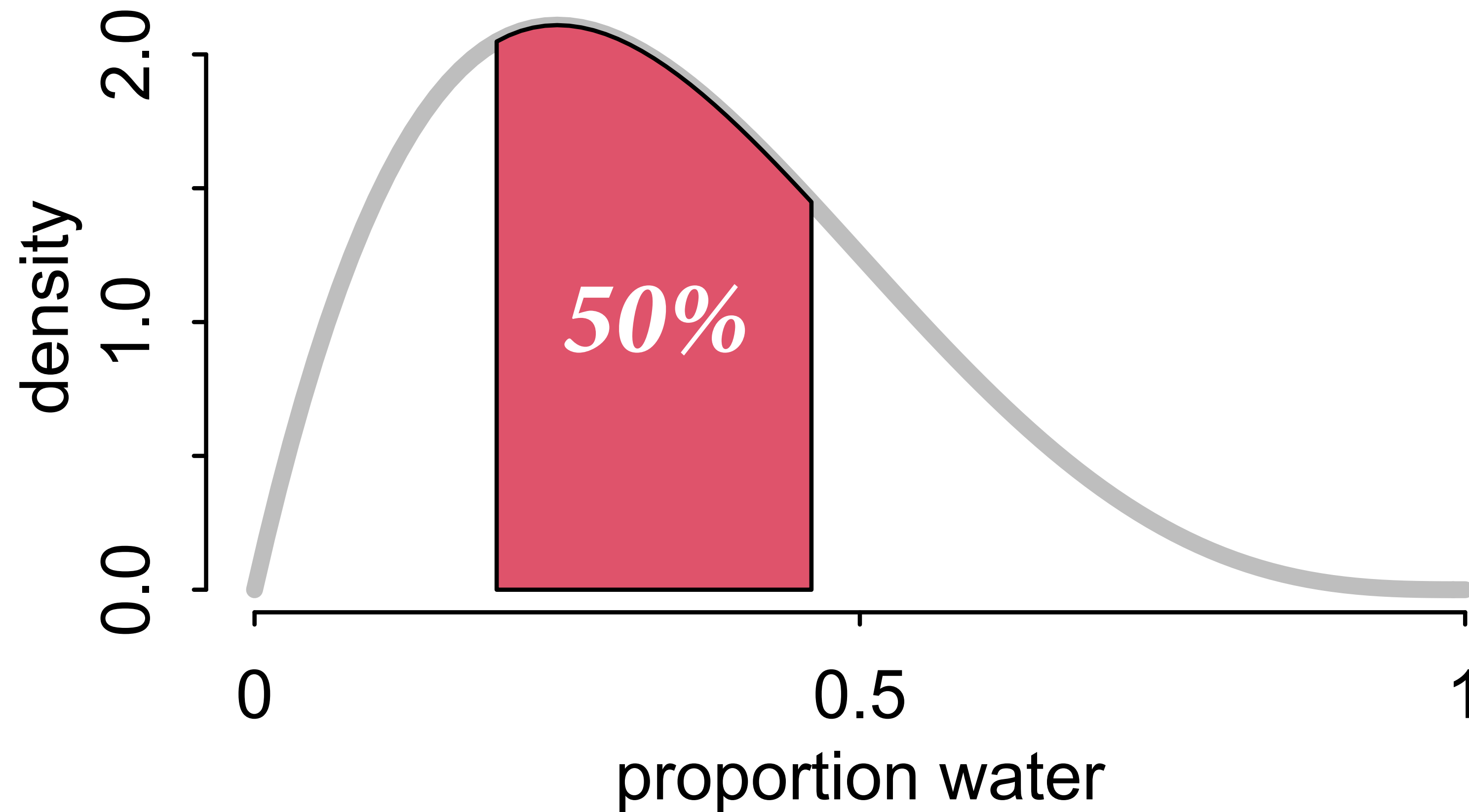
*Always use the
entire distribution*

(4) No one true interval



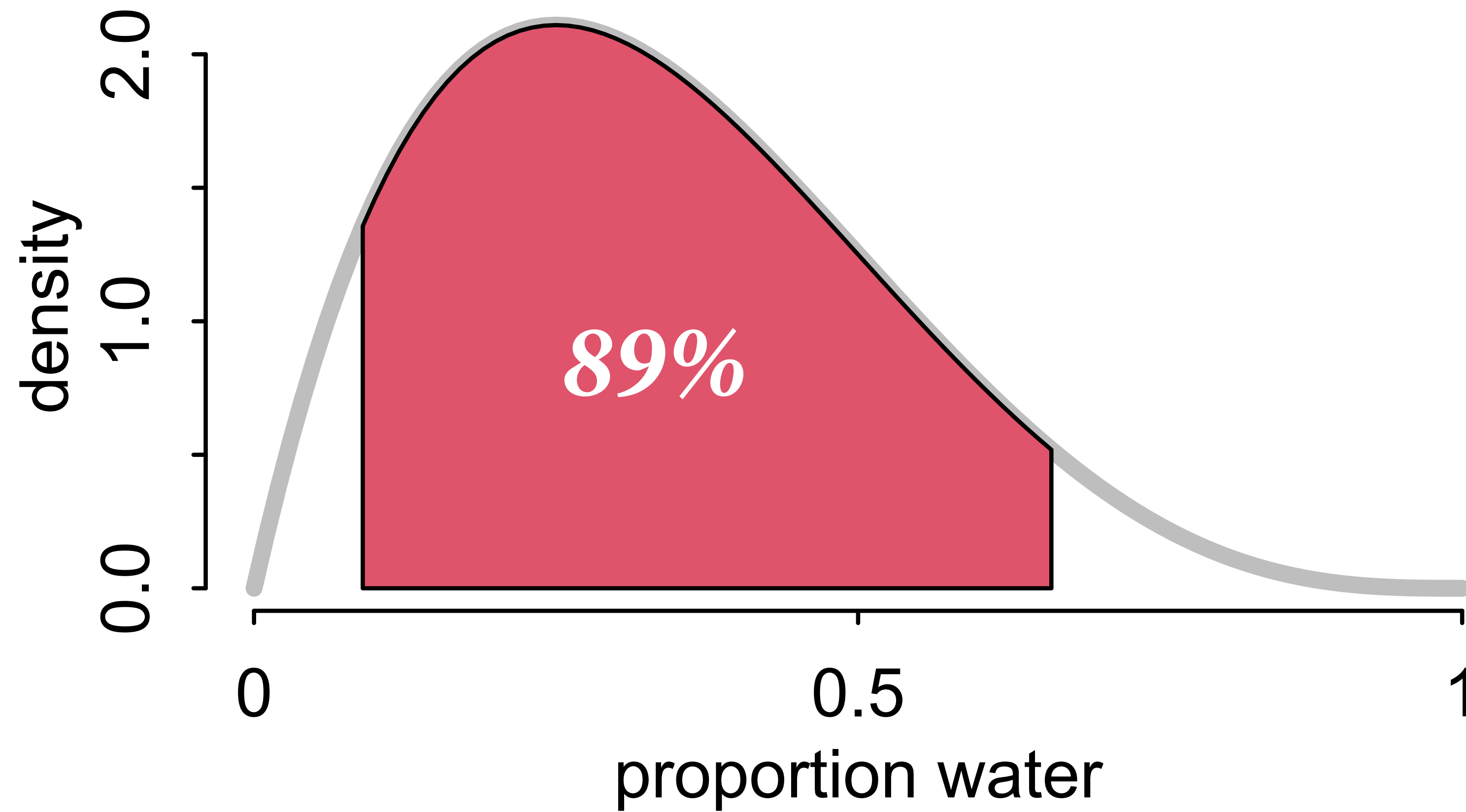
*Intervals
communicate shape
of posterior*

(4) No one true interval



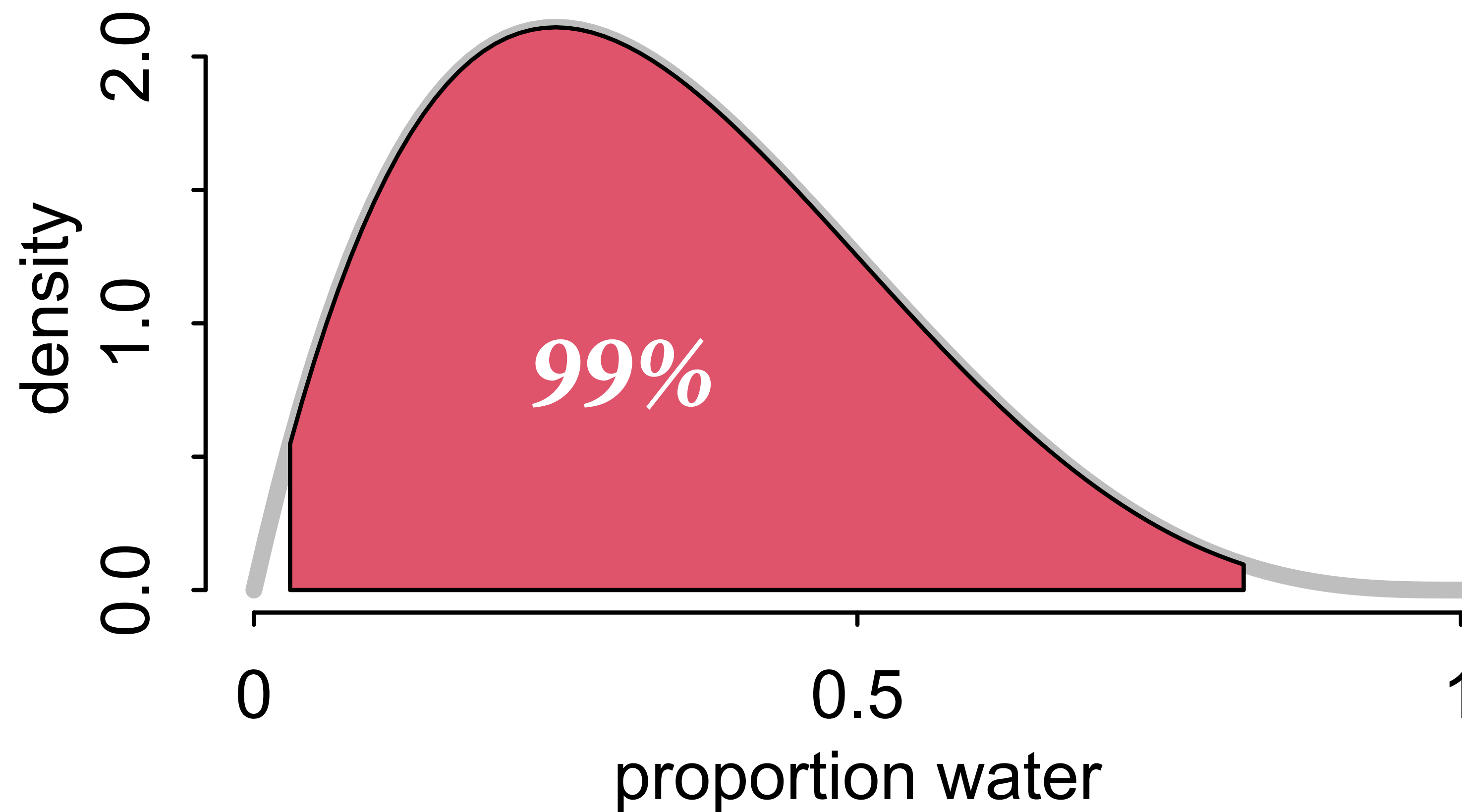
*Intervals
communicate shape
of posterior*

(4) No one true interval



*Intervals
communicate shape
of posterior*

(4) No one true interval



*Intervals
communicate shape
of posterior*

*95% is obvious
superstition. Nothing
magical happens at
the boundary.*

Letters From My Reviewers

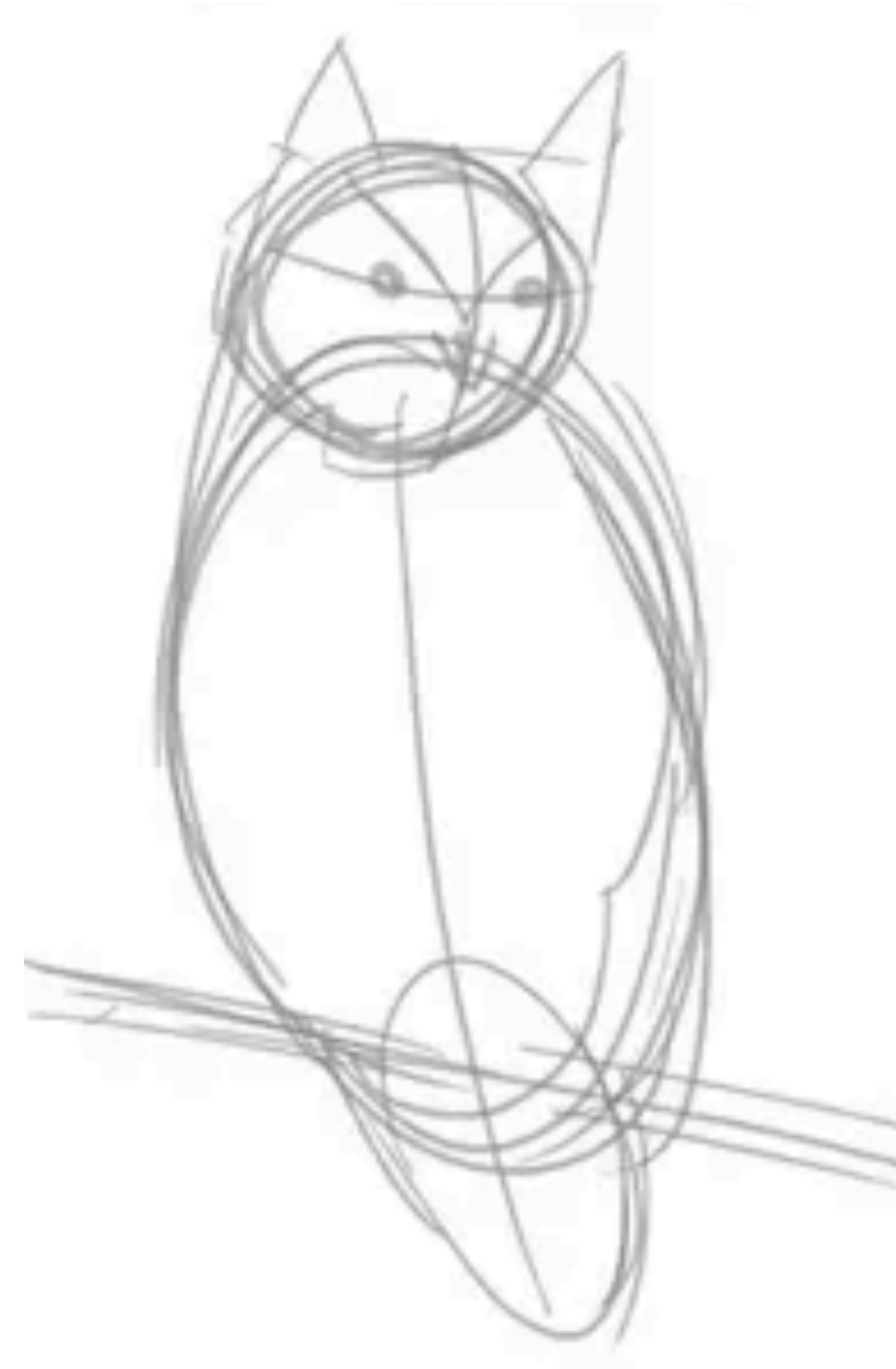
“The author uses these cute
89% intervals, but we need
to see the **95% intervals** so
we can tell whether any of
the effects are **robust**.”



*That an arbitrary interval contains an arbitrary
value is not meaningful. Use the whole distribution.*

Workflow

- (1) Define generative model of the sample
- (2) Define a specific estimand
- (3) Design a statistical way to produce estimate
- (4) Test (3) using (1)
- (5) Analyze sample, summarize



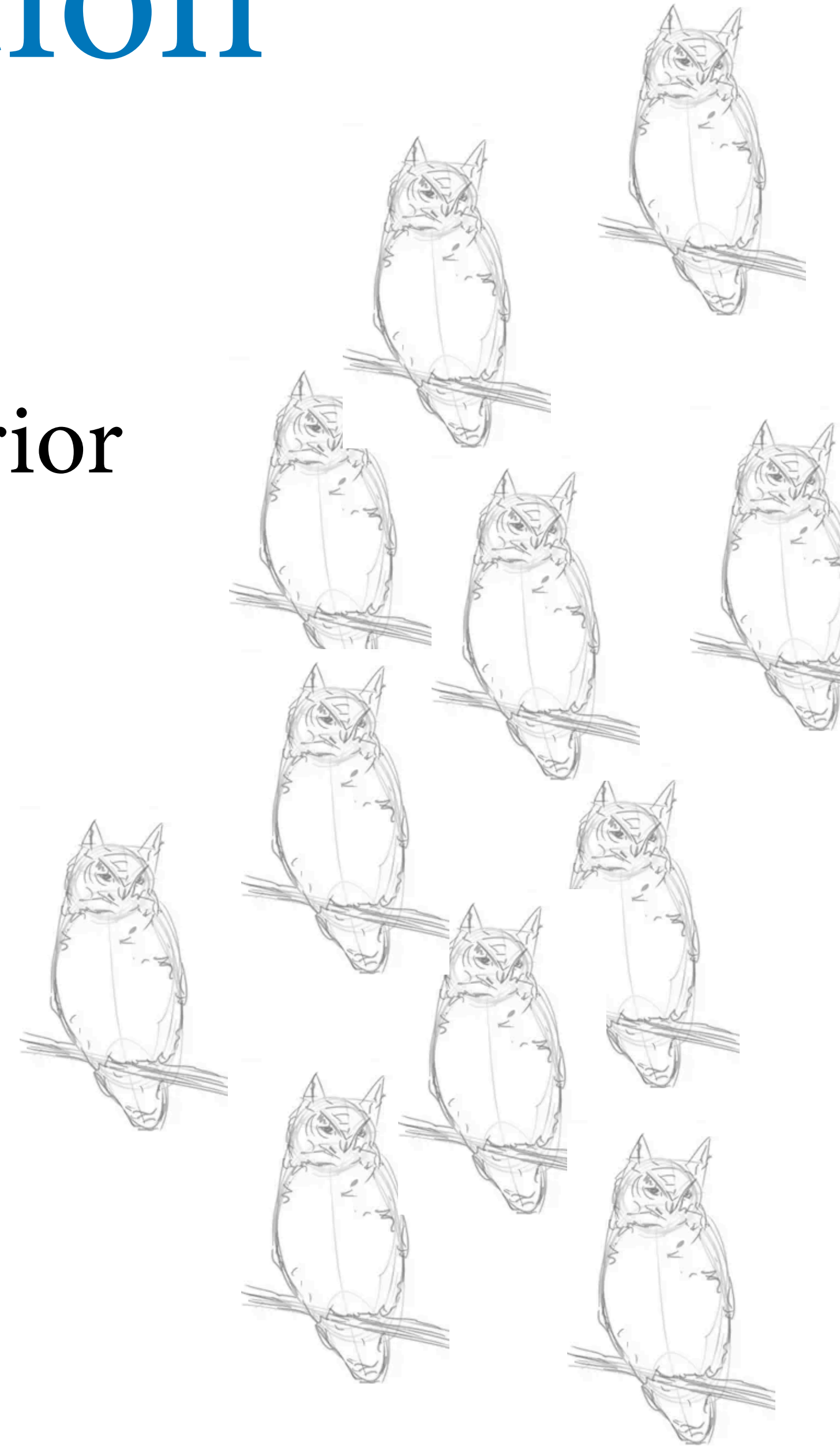
From Posterior to Prediction

Implications of model depend upon **entire** posterior

Must average any inference over entire posterior

This usually requires integral calculus

OR we can just take samples from the posterior

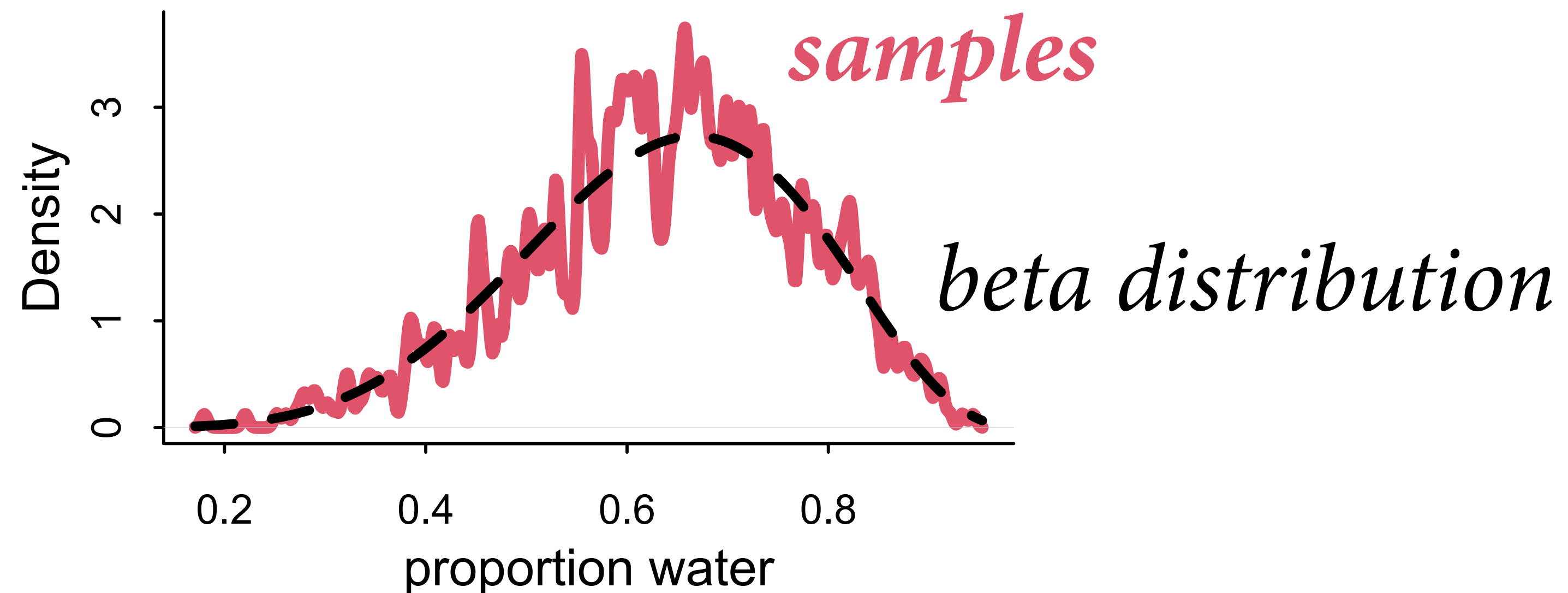


Sampling the posterior

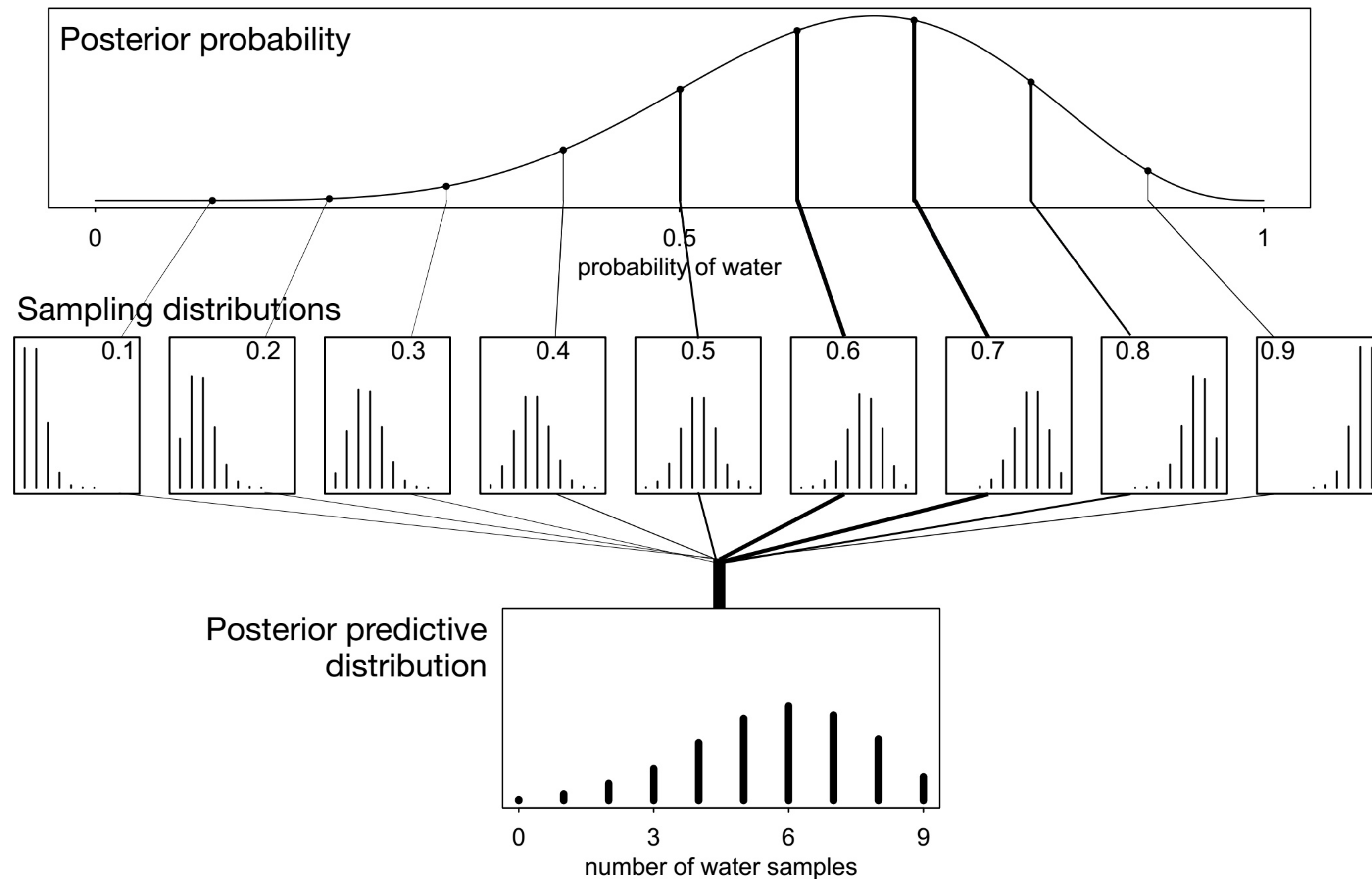
```
post_samples <- rbeta( 1e3 , 6+1 , 3+1 )
```

R code
2.19

```
dens( post_samples , lwd=4 , col=2 , xlab="proportion water" , adj=0.1 )  
curve( dbeta(x,6+1,3+1) , add=TRUE , lty=2 , lwd=3 )
```

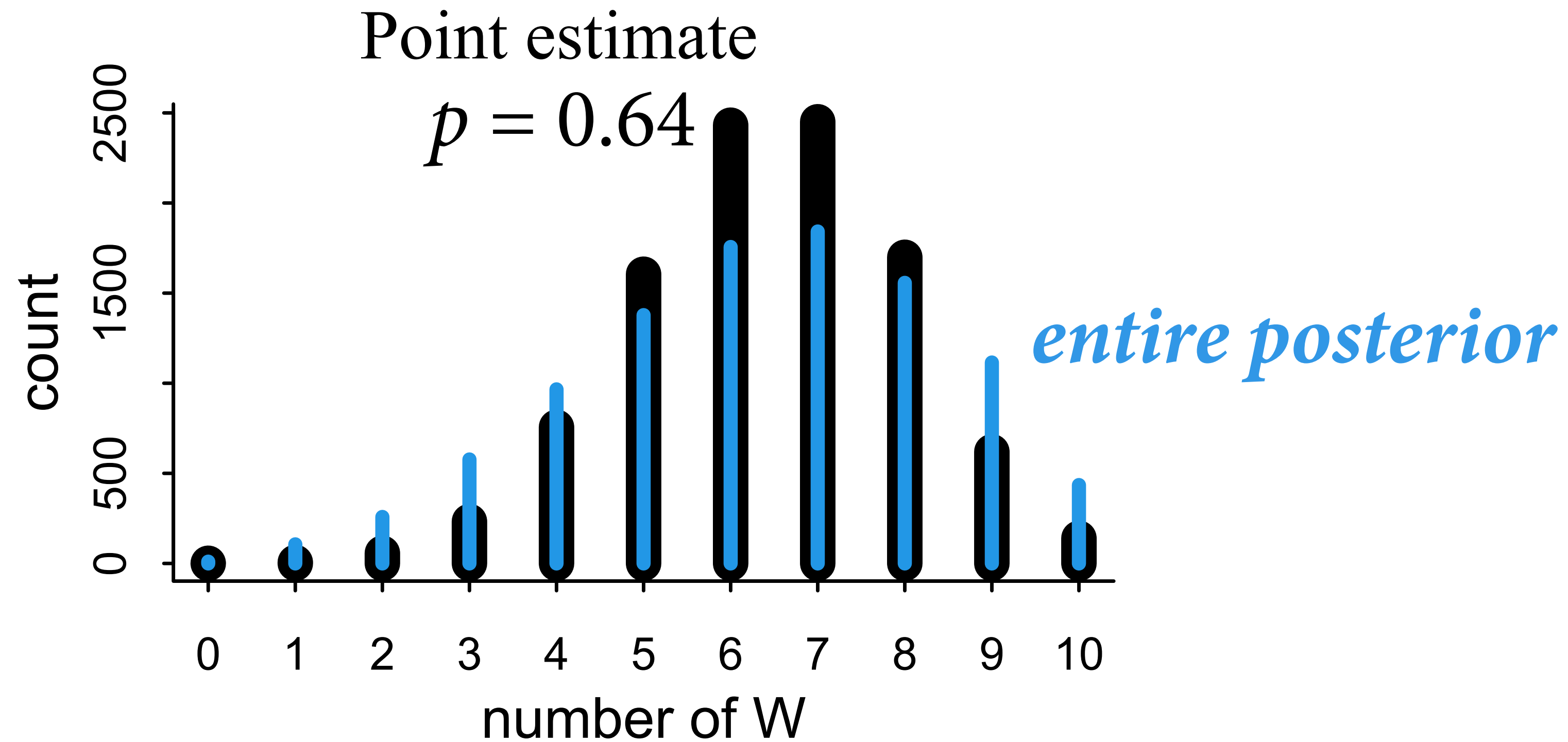


Propagating uncertainty via sampling



1. draw values of p from the posterior
2. generate W, L outcomes with each value of p
3. combine all sampling distributions into a posterior predictive distribution

```
# now simulate posterior predictive distribution
post_samples <- rbeta(1e4,6+1,3+1)
pred_post <- sapply( post_samples , function(p) sum(sim_globe(p,10)=="W") )
tab_post <- table(pred_post)
for ( i in 0:10 ) lines(c(i,i),c(0,tab_post[i+1]),lwd=4,col=4)
```

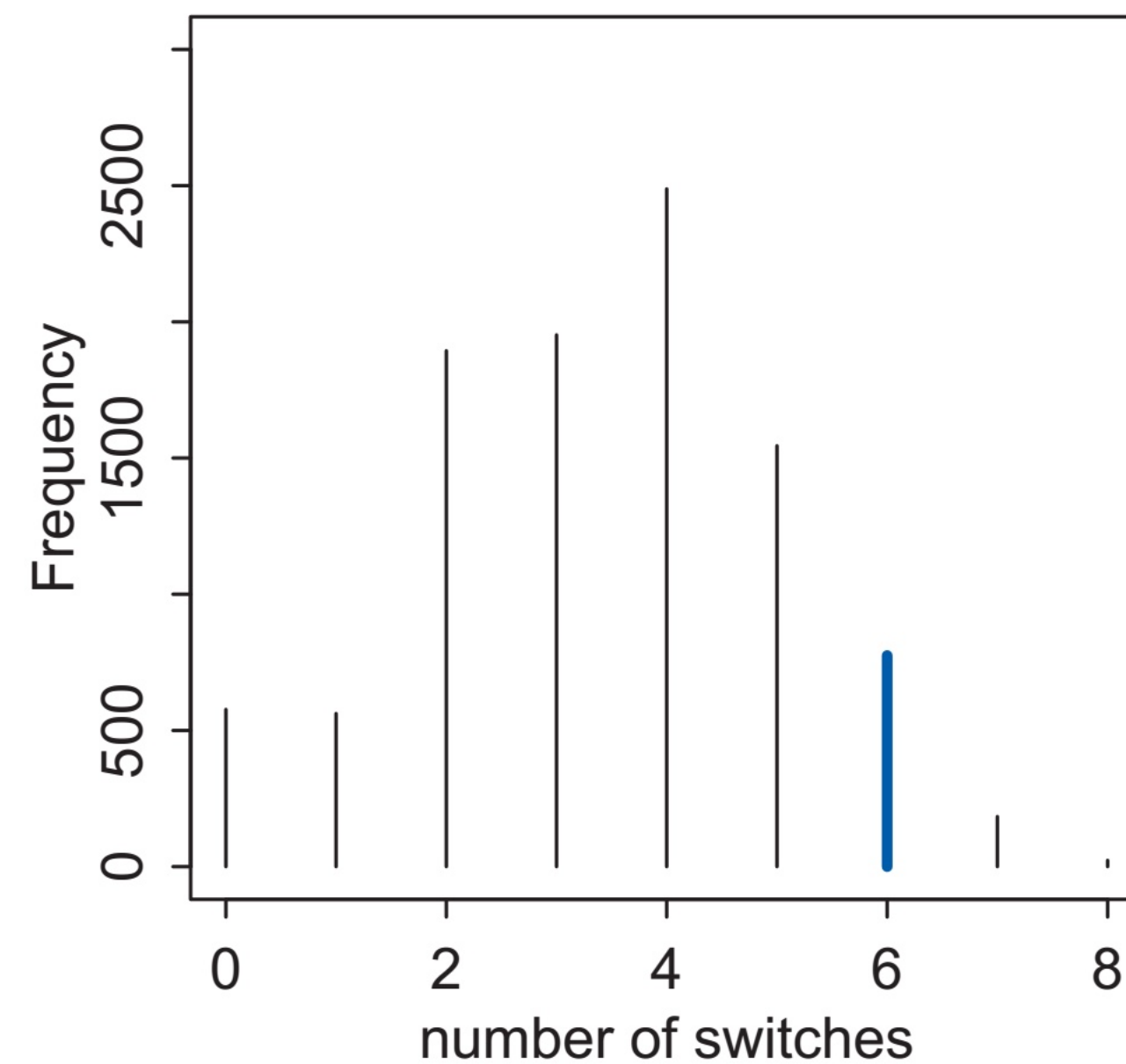
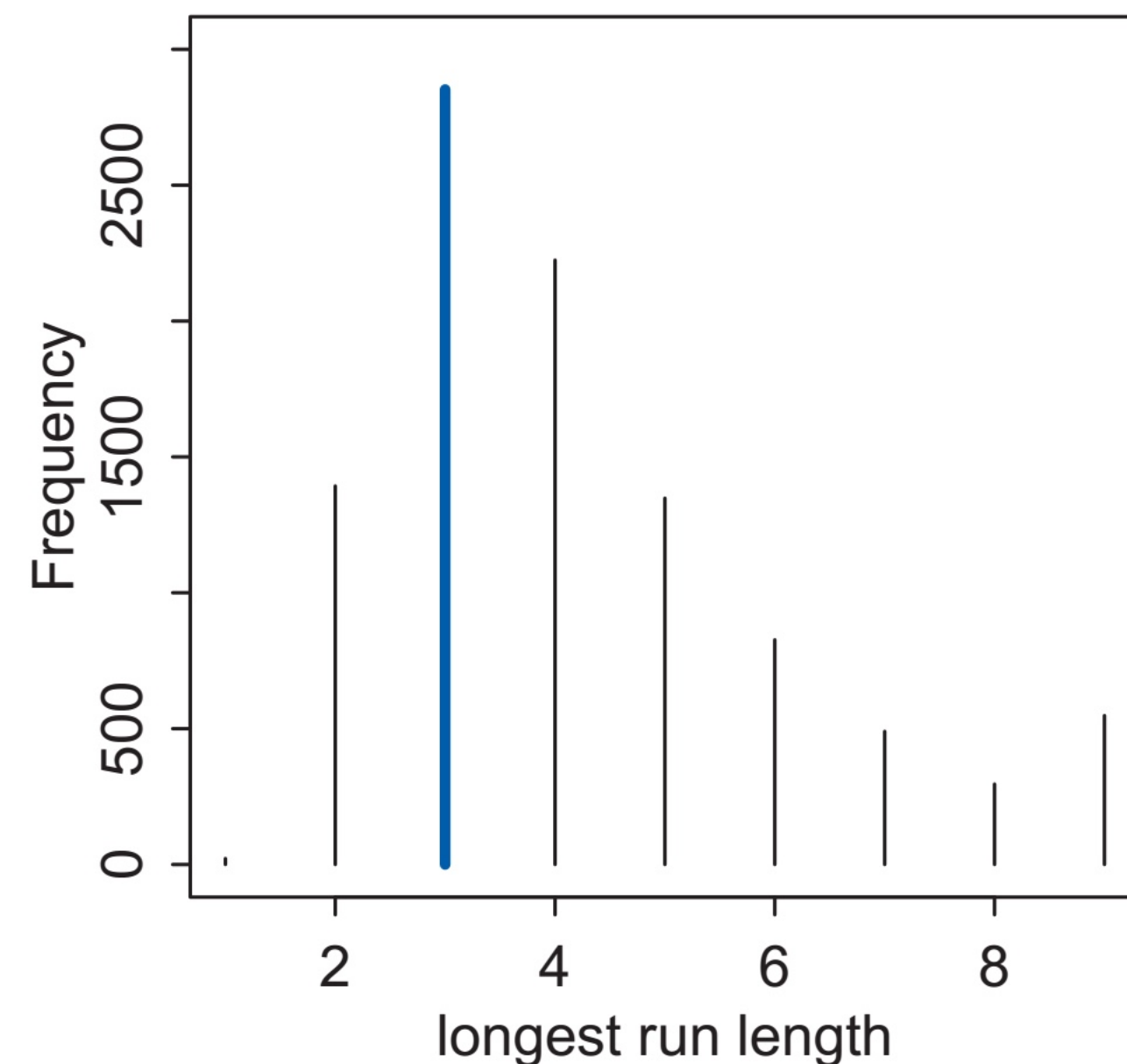


Check other features of the observed and generated data

Observed sequence of 9 tosses: W L W W W L W L W

POTENTIAL question:

Does the model generate a similar number of “runs” and switches?



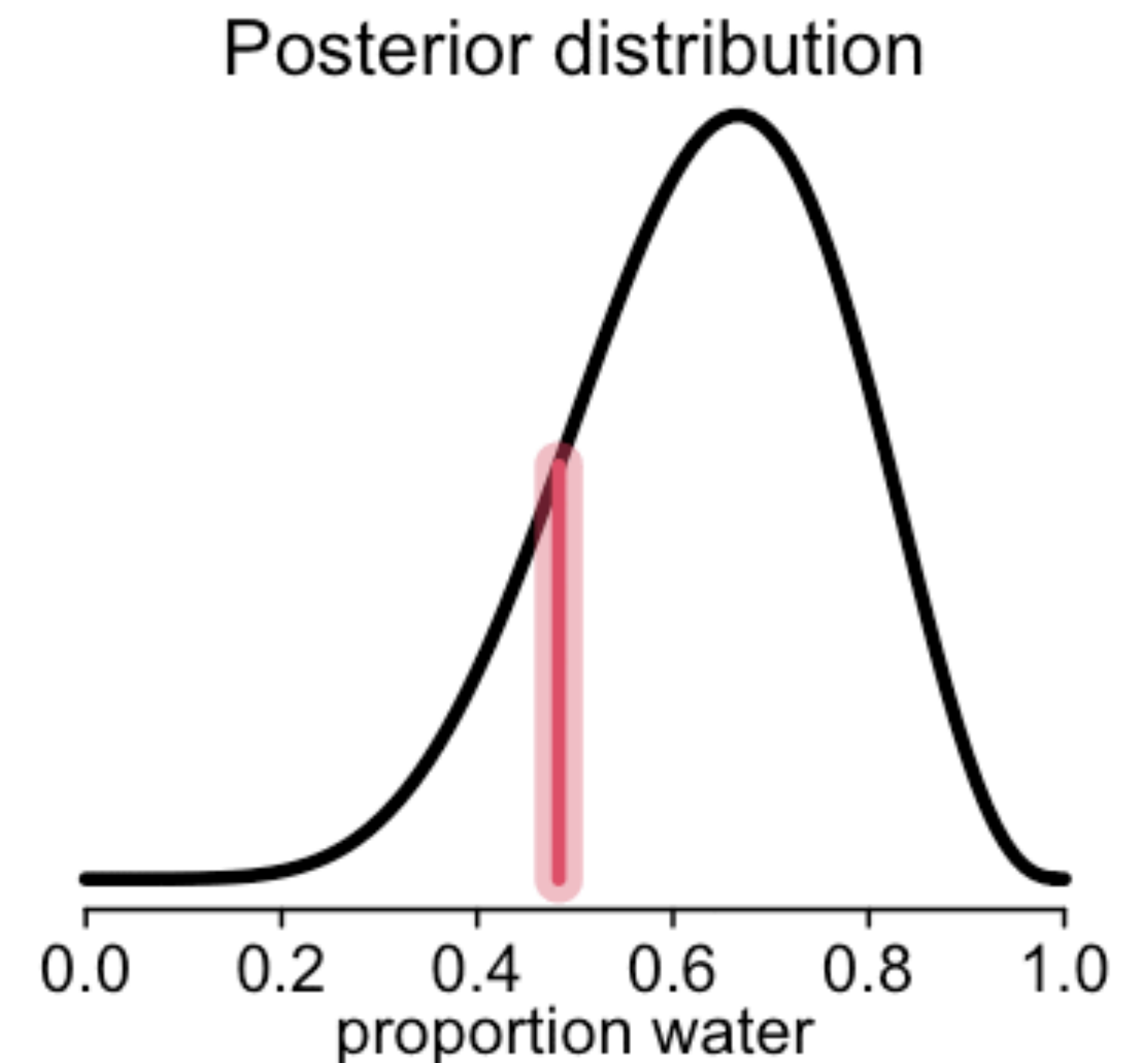
Sampling is Fun & Easy

Sample from posterior, compute desired quantity for each sample, profit

Much easier than doing integrals

Turn a **calculus problem** into a **data summary problem**

MCMC produces only samples anyway



Sampling is Handsome & Handy

Things we'll compute with sampling:

Model-based forecasts

Causal effects

Counterfactuals

Prior predictions



Bayesian data analysis

For each possible explanation of the data,

Count all the ways data can happen.

Explanations with more ways to produce the data are more plausible.

Bayesian modesty

*No guarantees except **logical***

*Probability theory is a method of logically deducing **implications of data** under assumptions that you must choose*

Any framework selling you more is hiding assumptions

